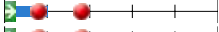
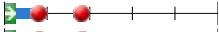
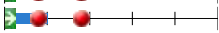
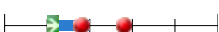
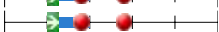
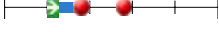
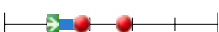
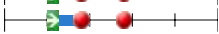
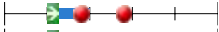
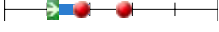


Program overview

12-Sep-2023 11:23

Year 2023/2024
 Organization Civil Engineering and Geosciences
 Education Master Civil Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
CIE Master 2023	CIE Master 2023 (New Master Programme)		
CIE Year 1 2023			
CIE Faculty Core Module (MUDE)			
CEGM1000	Modelling, Uncertainty and Data for Engineers	12	
CIE Programme Base: Mechanics and Interdisciplinary Perspectives			
CIEM0000	Interdisciplinary Mechanics and Design for CE	9	
CIE Track Construction Materials (CM)			
CIEM1000	Base: Fundamentals of Construction Materials	15	
CIEM1110	Measuring and Modelling Construction Material Behaviour	9	
CIEM1210	Construction Materials Research	15	
CIEM1220	Design and Engineering of Construction Materials	15	
CIE Track Geotechnical Engineering (GE)			
CIEM2000	Base: Geotechnical Engineering	15	
CIEM2110	Geotechnical Modelling	9	
CIEM2210	Geotechnical Structures	15	
CIEM2220	Advanced Soil Mechanics	15	
CIEM2230	Delta Geotechnics	15	
CIE Track Hydraulic Engineering (HE)			
CIEM3000	Base: Hydraulic Engineering Fundamentals and Environments Part I	15	
CIEM3110	Hydraulic Engineering Fundamentals and Environments Part II	9	
CIEM3210	Coastal Engineering	15	
CIEM3220	River Engineering	15	
CIEM3230	Advanced Design of Ports and Waterways Systems and Interventions	15	
CIE Track Hydraulic & Offshore Structures (HOS)			
CIEM4000	Base: Hydraulic and Offshore Structures	15	
CIEM4110	Hydraulic Structures	9	
CIEM4120	Offshore Structures	9	
CIEM4210	Marine Renewables	15	
CIEM4220	Dams, Dikes and Breakwaters	15	
CIEM4230	Floating and Submerged Structures	15	
CIE Track Structural Engineering (SE)			
CIEM5000	Base: Structural Engineering	15	
CIEM5110	Structural Mechanics and Dynamics	9	
CIEM5120	Design of Structural Components	9	
CIEM5130	Design of Civil Structures and Infrastructures	9	
CIEM5210	Applied Mechanics of Structures	15	
CIEM5220	Applied Dynamics of Structures	15	
CIEM5230	Concrete Structures	15	
CIEM5240	Steel and Composite Structures	15	
CIEM5250	Building Engineering	15	
CIEM5260	Transportation Infrastructures	15	
CIE Track Traffic & Transport Engineering (TTE)			
CIEM6000	Base: Traffic and Transport Engineering	15	
CIEM6110	Methods in Traffic and Transport Engineering	9	
CIEM6210	Transport Networks and Systems	15	
CIEM6220	Road Traffic Systems	15	
CIEM6230	Public Transport and Railway Systems	15	
CIEM6240	Road and Railway Engineering	15	
CIE Year 2 2023			
CIE Electives Construction Materials (CM)			
CIEM1301	Advanced Computation Mechanics	5	
CIEM1302	Forensic Construction Materials Engineering	5	
CIEM1303	Upscaling Techniques in Construction Materials Design and Engineering	5	
CIEM1304	Glass Science and Engineering	5	
CIE Electives Geotechnical Engineering (GE)			
CIEM2301	Offshore Geotechnical Engineering	5	
CIEM2302	Trenchless Technologies	5	
CIEM2303	Rock Mechanics Applications	5	
CIEM2304	Environmental and Energy Geotechnics	5	
CIE Electives Hydraulic Engineering (HE)			
CIEM3301	Building with Nature	5	

CIEM3302	Dredging for Sustainable Infrastructure	5	
CIEM3303	Advanced Modelling of Turbulent Flows and Transport	5	
CIEM3304	Physical Oceanography	5	
CIE Electives Hydraulic & Offshore Structures (HOS)			
CIEM4301	Onshore Hydropower	5	
CIEM4302	Cold Regions Engineering	5	
CIEM4303	Flood Risk	5	
CIEM4304	Hydraulics Fieldwork	5	
CIE Electives Structural Engineering (SE)			
CIEM5301	Shell Structures	5	
CIEM5304	CO2 Neutral Structures	5	
CIEM5305	Fire Safety Design	5	
CIEM5306	Assessment of Existing Concrete Structures	5	
CIEM5307	Forensic Structural Engineering	5	
CIEM5308	Parametric Design and Digital Fabrication	5	
CIEM5309	Advanced Building Physics	5	
CIEM5310	Contact and Interface Mechanics for Engineering Structures	5	
CIEM5311	Loading and Life Time Prediction of Transportation Infrastructure	5	
CIEM5312	Emerging Technologies for Transportation Infrastructure	5	
CIE Electives Traffic & Transport Engineering (TTE)			
CIEM6301	Railway Traffic Management	5	
CIEM6302	Advanced Data Science for Traffic and Transportation Engineering	5	
CIEM6303	Transitions, Sustainability & Innovation	5	
CIEM6304	eXtended Reality for Civil Engineering	5	
CIE Projects			
CEGM3000	Multidisciplinary Project (MDP)	15	
CIEM0120	Research Internship	10	
TUD4040	Joint Interdisciplinary Project	15	
CIE Cross overs choose one			
CEGM2000	Suspensions, Sludges and Soils	10	
CEGM2001	Sustainable Cities	10	
CEGM2002	Engineering for Global Development	10	
CEGM2003	Data Science and Artificial Intelligence for Engineers	10	
CEGM2004	Noise and Vibration: Generation, Propagation and Effect on Humans and Environment	10	
CEGM2005	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE)	10	
CEGM2006	Subsurface Storage for Energy, Water and Climate Applications	10	
CEGM2007	Resilient Deltas	10	
CEGM2008	Monitoring of Structural Health and Geohazards	10	
CIEM0210	Engineering Management Systems	10	
CIE Master Thesis preparation			
CIEM0400	Master Thesis preparation	5	
CIE Master Thesis			
CIEM0500	Master Thesis	30	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Master 2023

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Year 1 2023

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Faculty Core Module (MUDE)

CEGM1000	Modelling, Uncertainty and Data for Engineers	12
Responsible Instructor	Dr.ir. R.C. Lanzafame	
Module Manager	Dr.ir. R.C. Lanzafame	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr.ir. A.A. Verhagen	
Instructor	Dr. A. Amiri Simkoei	
Instructor	F.P. van der Meer	
Instructor	D. Mehta	
Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. R. Taormina	
Instructor	A. Jagadeesh	
Instructor	Dr. P.M.N. Mares Nasarre	
Instructor	T.R. van Woudenberg	
Instructor	J.C. Pittman	
Contact Hours / Week x/x/x/x	4HC, 4W, 4PJ	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Modelling, Uncertainty and Data for Engineers (MUDE) is a module shared by students in all three programs within the Civil Engineering and Geosciences Faculty (Civil Engineering, Environmental Engineering, Applied Earth Sciences). In a typical week you will be introduced to new theory, practice it on your own through a series of exercises and work together with teachers and classmates in a medium-sized classroom setting. At the end of each week you will apply these concepts in a project along with a group of fellow students. You will also have weekly coding assignments to build your programming skills in a way that will help you apply the methods learned in MUDE using real data sets and models, as well as prepare you for your future MSC thesis work.	
Study Goals	After successfully completing the MUDE module a student is able to 1. Apply a diverse set of models and analysis techniques to solve engineering problems through a combination of mathematical formulation and implementation in computer code. 2. Process and analyze data from different sources and formats to gain insight on the underlying systems and incorporate in modeling approaches. 3. Mathematically formulate, solve and discuss estimation, risk, reliability and optimization problems for systems to support decision making. 4. Analyze and reflect on the methodology, performance, and results of modeling approaches and analysis techniques, accounting for uncertainty. 5. Implement and carry out analysis using Python programming language following specific coding standards and best practices while facilitating collaboration. 7. Communicate and interpret results of analyses with appropriate documentation, metrics and visualization techniques.	
Education Method	The weekly format is: - Mon: lecture - Tue: open question hours (attendance optional) - Wed: classroom workshops (teacher-led activities) - Fri: supervised group projects (bi-weekly assignments) Practice material is provided in the form of a digital textbook with analytic and programming exercises and is the primary focus of the Mon-Wed activities. These concepts are applied on Friday during the project sessions. Weekly coding assignments build fundamental programming skills that are also applied in the Friday project sessions.	
Assessment	- Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) The exams are paper-based. Project portfolio includes 7 bi-weekly projects (80%) as well as weekly coding assignments (20%). You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.	
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis, Python programming	
Academic Skills	Analysis, coding, communication, visualization, team skills	
Literature & Study Materials	An online digital textbook consisting of text, practice material and videos, as well as related materials (e.g., Jupyter Notebooks).	
Judgement	- Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) The exams are paper-based. Project portfolio includes 7 bi-weekly projects (80%) as well as periodic coding assignments (20%). You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.	
Permitted Materials during Exam	Calculator, pen/pencil. All formulas will be provided by the instructors.	
Collegerama	Yes No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Programme Base: Mechanics and Interdisciplinary Perspectives

CIEM0000	Interdisciplinary Mechanics and Design for CE	9
Responsible Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Ir. R. Binnekamp	
Instructor	Dr.ir. R.J. Labeur	
Instructor	Dr.ir. K.N. van Dalen	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. G.A. van Nederveen	
Instructor	Prof.dr.ir. L.A. Tavasszy	
Instructor	Prof.dr.ir. A.R.M. Wolfert	
Instructor	Prof.dr. P.J. Vardon	
Instructor	L.M. Leclercq	
Instructor	J.A. Arriaga Garcia	
Contact Hours / Week x/x/x/x	6 (lecture room)/ 4 (instruction room)/ 2 (assignment support)/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Unit 1 (4EC): Provides an overview of how the surrounding social-technical-natural-economic environment challenges CE designs and teaches students how to take this into account in design by applying principles of Systems Engineering in an Open Design Learning approach. Unit 2 (5EC): Provides students with basic concepts, equations and mathematical tools of continuum mechanics relevant to all disciplines of civil engineering, teaches them to reduce the basic problems to simpler engineering forms (i.e., Laplace-type, wave-type, diffusion-type and 29 transport-type equations), equips them with a unified and coherent framework for analysis of the obtained PDEs to analyse the system behaviour, and enables them to recognize the differences and similarities between the various mechanical problems across civil engineering.	
Study Goals	On completion of this module, the student will be able to [LO1] Know how to develop Civil Engineering Systems (CES) within the broader social, economic, natural and built environment [LO2] use the basic concepts of continuum mechanics to define a mathematical model for a CE problem. [LO3] reduce the model to a simpler (engineering) one, explain the corresponding assumptions and classify the obtained result. [LO4] solve the reduced model/problem using the appropriate technique, interpret the behaviour and its relevance to CE practice.	
Education Method	Lectures and assignments	
Assessment	- Group assignment (44%) - Written exam (56%)	
Expected prior Knowledge	Basic knowledge of mechanics and mathematics. Prior experience with programming (preferably using Python)	
Academic Skills	Unit 1: Critical and interdisciplinary design thinking, problem-based learning and solving, continuous investigation, information collection and synthesis, communication and responsiveness, time management and anticipation. Unit 2: application of theory to new situations using the relevant equations, explaining differences and similarities between problems and their physical behaviours encountered in the various domains of CE.	
Literature & Study Materials	Unit 1: Reader. Unit 2: lecture pots	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Unit 2: Closed book exam. Formula sheet will be provided	
Collegerama	Yes	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Construction Materials (CM)
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CIEM1000	Base: Fundamentals of Construction Materials	15
Responsible Instructor	O. Copuroglu	
Instructor	Dr. F. Di Maio	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Dr. X. Liu	
Instructor	Dr.ir. F.A. Veer	
Instructor	G. Ye	
Instructor	O. Copuroglu	
Instructor	Dr.ir. D.A. Koleva	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Prof.dr.ir. P.C. Louter	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	F. Messali	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. M. Pavlovic	
Instructor	A.T. Gebremariam	
Instructor	Dr. Y. Mosleh	
Instructor	Dr. M. Fotouhi	
Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	x/10/8/x	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The field of construction materials explores the chemical roots of mechanical behaviour. Therefore, the fundamentals of material chemistry and mechanics form the base of the CM-track. In the first part of the track base (during Q2), two units will be offered; Science of Construction Materials (5EC) and Fracture Mechanics(4EC), which provide the students with composition, manufacturing, processing, and mechanical behaviour aspects of construction materials. These fundamental subjects will form a sound basis for the construction materials selection (2.5EC) and technology (3.5EC) in the second part of the track base (during Q3). The details of the module content are as follows: MODULE: FUNDAMENTALS OF CONSTRUCTION MATERIALS Module coordinator: Dr. O. Copuroglu Unit 1: Science of Construction Materials (Q2, 5EC) Unit coordinator: Dr. O. Copuroglu U1: Introduction; Chemistry of Construction Materials; Phases and Microstructures U1: Thermodynamic concepts; First and Second Law of Thermodynamics U1: Earth Materials; Aggregates U1: Portland Cement U1: Electrochemistry U1: Glass; Metals; FRC U1: Wood and Timber; Bituminous Materials U1: Polymers and Organic Additives; Recycled Construction Materials; Sustainability Aspects Unit 2: Fracture Mechanics (Q2, 4EC) Unit coordinator: Dr. F. Messali U2: Introduction: Theory of Elasticity U2: Theory of Griffith; Linear Elastic Fracture Mechanics U2: Linear Elastic Fracture Mechanics U2: Mixed Mode Fracture U2: Elastic/Plastic Fracture Mechanics U2: Quasi-Brittle Failure; The Cohesive Zone Model U2: Fracture in Timber; Environmental Influences on Fracture U2: Fracture Mechanics in Steel; Implementation of the Prediction Methods in Real Case Situations Unit 3: Construction Materials Selection (Q3, 2.5EC) Unit Coordinator: Dr. ir. F. Veer U3: Engineering Materials in the Build Environment 1800-2050; Properties of Engineering Materials, the Logic of the Ashby Diagram U3: Criteria for Materials Selection U3: Materials Selection based on Optimized Performance U3: The Shape Factor as a parameter and Integrated Materials U3: Optimizing using Multiple Criteria, Combined structural and Eco-Impact Optimization U3: Finding if a design concept works U3: Finding if a concept material is useful. Designing concept materials using the synthesizer U3: Ethics of materials selection.	

	Unit 4: Construction Materials Technology (Q3, 3.5EC) Unit Coordinator: Dr. G. Ye U4: Introduction to Concrete Technology; Microstructural of Concrete; Introduction to Asphalt Technology U4: Pore Structure in Concrete.; Fresh concrete properties; Aggregates properties U4: Early Age Concrete Properties; Basic Properties of Asphalt cement U4: Volume Stability: Shrinkages; Creep and Relaxation; Admixtures for (asphalt) concrete U4: Transport and Concrete Durability. Concrete Mix Design. Asphalt concrete mix design U4: Concrete Construction Practices; Curing and Quality Control of Concrete U4: Concrete Recycling. Concrete Sustainability. Reuse and Recycling in maintenance of Asphalt pavements U4: Special Concrete Types and Applications; Special asphalt concrete types and applications.
Study Goals	On completion of this module, the student will be able to 1. Explain the chemical roots of mechanical behaviour of construction materials (1, 4-7) 2. Categorize construction materials based on their composition, engineering properties and function (1, 8) 3. Design and predict engineering properties of construction materials based on composition and production methods (1-7) 4. Quantify sustainability of construction materials. (1, 5) 5. Analyse, model and predict failure of construction materials (1) 6. Apply fracture mechanics principles to develop materials more resistant to cracking (1,7) 7. Select construction materials based on functional requirements, economic, environmental and ethical considerations (3, 4) 8. Design mixture of concrete, asphalt and other composite materials for application-related requirements (1, 2, 3, 6) (with link to final attainments of CM track in parentheses)
Education Method	Lectures, in class interactive experiment/demo, lecture with in-class exercises, lab practical, lectures on theory and interactive lectures on case studies
Assessment	You complete the module successfully when passing the following summative assessments: Unit 1: Science of construction materials (BASE-1) written exam (33%) Unit 2: Fracture mechanics (BASE-2) written exam (27%) Unit 3: Materials selection in civil engineering (BASE-3) written exam (17%) Unit 4: Construction materials technology (BASE-4) written exam (23%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Expected prior Knowledge	Prerequisites to be able to start the full module BSc Civil Engineering or Applied Science (or equivalent). Faculty base and CE programme base.
Academic Skills	Fundamental Civil Engineering skills required to enter the CE MSc. program
Literature & Study Materials	Lecture notes of the units will be provided via Brightspace. Unit 2: Textbook (CT Sun and Z-H Lin, Fracture Mechanics, 2012, or alternative) complemented with handouts on selected topics Unit 3: Materials selection in mechanical design, M. F. Ashby, 2016
Judgement	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Unit 1: Formula sheet and calculator. Unit 2: Formula sheet and calculator. Unit 3: Laptop and the course book (by Ashby) Unit 4: Formula sheet and calculator
Collegerama	The unit coordinators will inform the students in case they may use additional materials during the exam. No

CIEM1110	Measuring and Modelling Construction Material Behaviour	9
Responsible Instructor	F.P. van der Meer	
Instructor	Ir. C. Kasbergen	
Instructor	Ir. P.A. de Vries	
Instructor	G. Ye	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	O. Copuroglu	
Instructor	Dr.ir. D.A. Koleva	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	R. Esposito	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. Y. Mosleh	
Instructor	Dr. M. Fotouhi	
Contact Hours / Week x/x/x/x	12/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Use of materials for structural applications requires quantification of the material performance. Such quantification is related to two scientific disciplines: experimental characterization and numerical modelling. Safe engineering with construction materials requires that material behaviour has been experimentally tested under controlled conditions and that the performance of the material under expected conditions can be predicted with simulations. In materials science, the two disciplines meet when numerical simulations are used to interpret laboratory experiments and when experiments are used to calibrate and validate models. This module provides students in the construction materials track with a solid basis in both disciplines of measuring and modelling. Students will learn about using advanced experimental techniques that can be used in the laboratory and on-site to monitor and characterise construction material performance. Furthermore, they will be trained to interpret measured data and determine consequences for their short and long-term performance in the specific application. This module will also prepare students for the development and application of multi- scale and multi-physics modelling techniques to study the behaviour of construction materials and to explain deterioration and damage accumulation.	
Study Goals	On completion of this module, the student will be able to 1. Set up a computational model for simulation of the behaviour of materials 2. Perform linear and nonlinear finite element analysis for mechanical and transport problems 3. Calibrate and validate computational models with experimental observations 4. Set up a testing plan for characterisation of materials to define input parameters for structural analysis 5. Analyse experimental measurements to characterise the performance of construction materials 6. Advise structural engineers on characterisation and modelling strategies for describing material behaviour 7. Quantify uncertainty in models and measurements	
Education Method	Videos, interactive lectures, computer lab sessions, lectures, lab experiments.	
Assessment	Assessment of the modules is done with three examinations: - Written exam 1: Numerical modelling (31%) - Written exam 2: Experimental characterization (39%) - Presentation based on combined assignments (30%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all examinations.	
Academic Skills	BSc Civil Engineering or Applied Science (or equivalent). MUDE and CE programme base.	
Literature & Study Materials	Analytical thinking, coding, data collection and processing, reporting.	
Judgement	Lecture notes will be provided via Brightspace.	
Permitted Materials during Exam	See Assessment	
Collegerama	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
	None	
	No	

CIEM1210	Construction Materials Research	15
Responsible Instructor	Dr.ir. B. Šavija	
Instructor	Ir. C. Kasbergen	
Instructor	Prof.dr.ir. L.J. Sluijs	
Instructor	G. Ye	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	O. Copuroglu	
Instructor	Dr.ir. D.A. Koleva	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	M. Lukovi	
Instructor	Dr. K. Anupam	
Instructor	F. Messali	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. M. Fotouhi	
Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	18/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Module B1 is designed to offer students the opportunity to develop indepth scientific insight into the behaviour of construction materials under various environmental and loading conditions. Specifically, students will learn about degradation mechanisms and durability (5EC), imposed deformations (5EC), and constitutive modelling (5EC).	
Study Goals	On completion of this module, the student will be able to 1. Define durability concepts of construction materials, describe material properties and their effect on durability. (1, 2, 7) 2. List and explain physical, chemical and biological degradation mechanisms of construction materials. (2) 3. Perform service life design of structures based on material properties. (2) 4. Set-up repair and maintenance strategies for degraded structures (7) 5. Analyse and predict the influence of eigenstresses in materials on the behaviour of structural elements and components (1, 2) 6. Explain how environmental effects and different gradients on material level and structural level impose deformations and stresses (1, 2) 7. Formulate constitutive models to describe the linear/nonlinear response of materials and implement them in FEM codes (1, 7) 8. Select and apply constitutive models for various physicochemical material behaviours. (1) (with link to final attainments of CM track in parentheses)	
Education Method	Lectures, practicals with hands on exercises, workshop, case studies, computer lab sessions.	
Assessment	You complete the module successfully when passing the following 4 summative assessments: B1-1 Durability: Written exam (24%) B1-1 Durability: Report (10%) B1-2 Imposed: Oral exam (33%) B1-3 Constitutive: Oral exam (33%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Prerequisites to be able to start the full module BSc Civil Engineering or Applied Science (or equivalent). Faculty base and CE programme base.	
Academic Skills	Knowledge acquired in track core	
Literature & Study Materials	Lecture notes will be provided via Brightspace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	None	
Collegerama	No	

CIEM1220	Design and Engineering of Construction Materials	15
Responsible Instructor	A. Varveri	
Instructor	Dr. F. Di Maio	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Dr. X. Liu	
Instructor	G. Ye	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	O. Copuroglu	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Dr. H.M. Jonkers	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. M. Ottele	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	Dr.ir. J.A. Pascoe	
Instructor	A.T. Gebremariam	
Instructor	Dr. Y. Mosleh	
Instructor	Dr. M. Fotouhi	
Contact Hours / Week x/x/x/x	14/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Module B2 focuses on the design, performance and engineering of smart materials, self-healing materials, composite materials, bio-based materials, and emerging new materials. Innovative technologies such as 3D printing will receive special attention as well as recycling and renewable construction materials. The module is organized in three units: (1) The first module unit B2-1 on Recycling and renewable construction materials considers the design and engineering of construction materials from the perspective of recycling and sustainability. (2) The second module unit B2-2 on Advanced construction materials and smart technologies focuses on design and performance of new high-performance or multifunctional materials including new manufacturing technologies. (3) The third module unit B2-3 on Bio-based/hybrid/composite materials in construction technology addresses the design and engineering performance of bio-based and polymer-based composite and hybrid materials.	
Study Goals	On completion of this module, the student will be able to 1. Advise on the use of construction materials and production processes for reliable and sustainable structural design (1, 3, 8) 2. Design potential routes for recycling and reusing of EoL (End-of-Life) construction materials 3. Analyse the environmental impact of construction materials in their complete life cycle (4) 4. Develop strategies for sustainable use of materials in construction (4,7) 5. Facilitate the use of new, high performance and sustainable materials (e.g. UHPFRC, SHCC, geopolymer, bio-based materials, epoxy-modified bitumen) in structural engineering (5) 6. Evaluate current trends in the development of smart and sustainable construction materials (5) 7. Explain the opportunities and challenges of additive manufacturing technologies (such as 3D printing) in construction (3) 8. Analyse the short- and long-term behaviour of polymer and bio-based composite materials (1,5,7)	
Education Method	Interactive lectures, lab hands-on practicals, lab demonstrations, industrial visit, assignments	
Assessment	Students complete the module successfully when passing the following four (4) summative assessments: B2-1 Recycling: Lab report (17%) and written exam (17%) B2-2 Smart materials: Report portfolio (33%) B2-3 Bio-base & Composite: Written exam (33%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Construction Materials track base module.	
Academic Skills	Collection and synthesis of information, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate with their colleagues, produce reports and present their work, so organizational, interpersonal, writing and presentation skills will be also developed.	
Literature & Study Materials	Lecture notes, slides and handouts will be provided in Brightspace.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Geotechnical Engineering (GE)
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CIEM2000	Base: Geotechnical Engineering	15
Responsible Instructor	Dr. K.G. Gavin	
Instructor	Dr.ir. R.B.J. Brinkgreve	
Instructor	Prof.dr.ir. C. Jommi	
Instructor	S. Muraro	
Instructor	M. Korf	
Responsible for assignments	M.A. Cabrera	
Contact Hours / Week x/x/x/x	Contact hours: Q2: 4+6 hr per week Q3: 6 hr per week	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The Geotechnical Engineering Track Base Module includes three units, Theory and modelling of geo-processes (3 ECTS), Testing and modelling of soil behavior (6 ECTS), and Foundations and Excavations (6 ECTS). Students will learn how to describe mechanical behavior (stress-strain response) of soils and rocks based on laboratory testing and data analysis. The concepts of constitutive modelling will be introduced, and the applications are discussed. The observed soil behavior from standard geotechnical laboratory tests will be presented and students will learn to analyze and interpret the test data to translate them into mathematical formulations and model parameters needed for engineering design. Formulation, assessment, and evaluation of the soil coupled hydro-mechanical response in engineering applications will be covered. The students will use the derived soil parameters in common design approaches to predict the behaviour of foundations and excavations. Case studies where the actual response of structure was measured will be used to promote reflective learning. Moreover, the interaction between the subsurface and geotechnical structures as well as the impact of the environment on the design, construction and performance of geotechnical structures, and vice versa will be presented. Topics such as communicating with stakeholders from all relevant backgrounds about geotechnical problems and solutions will be presented and discussed. In Track Base, the learning line of Monitoring and Data Science is covered within all three units. The basics of topics related to the monitoring of geotechnical structures will be introduced in Foundations and excavations. Additionally, all three units provide opportunities for elaborating the data, and development of models for predicting soil and geotechnical structures behavior. Such elaboration is guided throughout the module for students to learn how to process/visualize/present information for clients and stakeholders. This module will add to the learning line on Ethics related to design of geotechnical structures considering issues such as usability/safety of geotechnical systems, evaluation, and assessment of the interaction between the subsurface and geotechnical structures as well as effective Communication with stakeholders.	
Study Goals	On completion of this module, the student will be able to: C-LO1: Describe the mechanical behavior (stress-strain response) of soils and rocks based on laboratory testing and data analysis C-LO2: Recognize and apply the concept of constitutive modelling C-LO3: Analysis and interpret laboratory tests and derive and apply mathematical formulations for engineering design C-LO4: Evaluate and assess the interaction between the subsurface and geotechnical structures C-LO5: Recognize, and determine the impact of the environment on the design, construction, and performance of geotechnical structures, and vice versa C-LO6: Communicate the results with experts from other disciplines and other stakeholders C-LO7: Assess the impact of hydro-mechanical coupling on the soil response	
Education Method	Lectures, individual and group assignments, laboratory practicals and site visits. See unit descriptions on Brightspace for details.	
Assessment	The module includes in total 4 summative assessments. Three of them are unit-based. The 4th summative assessment is an overarching group assignment, which covers LOs and aspects of all three units. The student groups need to elaborate on the disciplinary connections among the different units in the module. Formative assessment moments are organized within each unit of the module. The distribution of marks is: Theory and modelling of geo-processes - Written Exam (15%) Testing and modelling of soil behaviour - Written Exam (30%) Foundations and excavations - Oral Exam (30%) GE Track Base - Group Assignment (25%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Attendance of Faculty and Program Base Modules	
Academic Skills	Critical thinking; application of theory to new situations using the relevant methods; scientific writing	
Literature & Study Materials	Lecture Notes, Readers, Scientific papers, Lecture slides	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Communicated in advance by responsible teacher.	
Collegerama	No	

CIEM2110	Geotechnical Modelling	9
Responsible Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Prof.dr. P.J. Vardon	
Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Dr.ir. A.A.M. Dieudonné	
Contact Hours / Week x/x/x/x	0/0/14/0 4 (Unit 1) + 10 (Unit 2) = 14 contact hours (variable per week) in Q3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The module covers two complementary processes in geotechnical engineering: Engineering Geology and Numerical Modelling. A geological perspective is a prerequisite of a complete geotechnical analysis. Once this interpretation of the geology in an engineering context is established in the Engineering Geology unit, the resulting ground model provides the starting point for geotechnical analyses and predictions by numerical models. The Numerical Modelling unit covers the theory and application of these models to come to quantitative predictions, as well as the interpretation of the results in the geotechnical context, because after all a number is not an answer. Parts of this module follow the learning line on Data and Monitoring by covering 1) the manipulation of both sparse and dense data, and both soft and hard data to highlight ground related risks and opportunities, 2) the interpretation of numerical analysis results in the context of data- and model uncertainty, 3) uncertainty propagation in (numerical) models. Parts of this module follow the learning line on Ethics by considering ethical aspects in feasibility studies conducted in Engineering Geology. This allows students to apply moral frameworks to the allowable actions as a CE Engineer. The Numerical modelling unit covers modelling techniques which are at the heart of various value sensitive methods for CE designs to be understood in the learning line on Ethics. Keywords: Engineering geology; desk study; soil and rock properties; conceptual ground and ground water flow modelling; anticipated ground conditions; geo-hazards and geotechnical issues; feasibility study; finite element method; numerical models; engineering solutions; sensitivity analysis;	
Study Goals	On completion of this module, you should be able to: A-LO1: Acquire conceptual (hydro)geological knowledge on project site; A-LO2: Integrate (hydro-)geological information and knowledge into ground models and estimate resulting uncertainty for a variety of geotechnical applications A-LO3: Formulate, explain and solve the (equations of the) finite element method in the context of numerical modelling of geotechnical applications; A-LO4: Design and execute numerical simulations for geotechnical applications; A-LO5: Critically evaluate the results and the limitations of model solutions for geological and geotechnical problems, considering data and model uncertainty; A-LO6: Communicate results of geo-engineering investigation and numerical simulations to stakeholders in terms of geo-risks and opportunities, in the context of a responsible use of the subsurface for multi-purposes.	
Education Method	Lectures, tutorial sessions, question hours, exercises, assignments, oral presentation, peer and self-evaluation, coursework, field excursions	
Assessment	Numerical modelling in Geotechnical Engineering - Written Exam (4 ECTS) Engineering Geology - Written Exam (5 ECTS) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Attendance of MUDE + Program core + track base units Q2	
Academic Skills	Critical and interdisciplinary thinking, problem-based learning and solving, continuous investigation, information collection and synthesis, communication, technical writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations. Making inferences of physical situations by analysing observations/data.	
Literature & Study Materials	Lecture slides, scientific papers, software packages, accompanying lecture notes Selected chapters out of textbooks: 1) Programming the Finite Element Method, 5th edition - Smith, Griffiths & Margets - 2014 John Wiley & Sons, Ltd 2) Waltham: Foundation of Engineering Geology, 3) Fetter: Applied hydrogeology, and 4) ICE manual of geotechnical engineering, Volume I, Geotechnical Engineering Principles, Problematic Soils and Site Investigation	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Unit 1: 1 A4 sheet of handwritten notes Unit 2: all permitted, including laptop with internet connection disabled.	
Collegerama	No	

CIEM2210	Geotechnical Structures	15
Responsible Instructor	Dr. G. Giardina	
Instructor	Dr.ir. W. Broere	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Dr.ir. C. Zwanenburg	
Instructor	Dr. G. Giardina	
Instructor	Prof.dr. M.A. Hicks	
Instructor	Prof.dr.ir. C. Jommi	
Instructor	S. Muraro	
Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Drs.ir. R.E.P. de Nijs	
Instructor	Dr.ir. A.A.M. Dieudonné	
Contact Hours / Week x/x/x/x	6,6,6+4 (18 hours of lectures, 4 hours practicals)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module covers the design, monitoring and assessment of geotechnical structures. It addresses the measurement, characterization and numerical modelling of spatial variability, the evaluation of risk and uncertainty in geotechnical engineering, the interaction between soil layers and the geotechnical structure, and the impact of geotechnical structures on other structures in an urban environment. Applications include the design of geotechnical structural elements (e.g., tunnel lining, soil reinforcement). The learning line on Monitoring and Data Science is included through the analysis of conventional and innovative geotechnical monitoring, and the evaluation of data from site investigation and physical modelling (Site Investigation modelling, Risk and variability in geotechnical engineering). Parts of this module follow the learning line on Ethics by considering ethical aspects in geo-risk management. In the Site Investigation and monitoring unit, students will gain understanding into approaches used to reduce risk in geotechnical engineering. Ethical aspects associated with human factors and risk communication in geo-engineering will be also covered.	
Study Goals	On completion of this module, the student will be able to: B1-LO1: describe the nature, impact and characterisation of soil spatial variability. B1-LO2: explain the relationship between soil spatial variability and geotechnical design and risk assessment. B1-LO3: evaluate advantages and shortcomings of different alternative choices for the design of infrastructure. B1-LO4: evaluate the design and construction methods for different types of infrastructure. B1-LO5: identify suitable site investigation techniques and critically analyse the results of a site investigation survey. B1-LO6: analyse geotechnical problems based on physical and numerical modelling techniques.	
Education Method	Lectures, individual and group assignments, laboratory practicals and site visits. See unit descriptions on Brightspace for details.	
Assessment	The module includes in total 4 summative assessments, transversal to the LOs of the module. 3 of them are unit-based, with an attention for connecting the assessment of LOs that are common to more than one unit. Formative assessment moments are organised within each unit of the module. They include an individual assignment (30%), a report (30%) and an oral exam (30%). The 4th summative assessment, included in the unit Geotechnical Design, is a group design assessment which addresses disciplinary connections among the different units in the module. (10%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Attendance of Track Base and Module A of GE track	
Academic Skills	Critical thinking; interdisciplinary thinking; problem-based learning and solving; application of theory to new situations using the relevant equations; time management; making inferences of physical situations by analysing observations/data.	
Literature & Study Materials	See Brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Assignments may be carried out by using all the teaching materials provided during the course and further background material as needed. The report on the group assignment can be used as reference, no further materials are needed for the oral exam.	
Collegegerama	No	

CIEM2220	Advanced Soil Mechanics	15
Responsible Instructor	M.A. Cabrera	
Instructor	Dr.ir. R.B.J. Brinkgreve	
Instructor	Dr.ir. C. Zwanenburg	
Instructor	Prof.dr. M.A. Hicks	
Instructor	S. Muraro	
Instructor	F. Pisanò	
Instructor	M.A. Cabrera	
Contact Hours / Week x/x/x/x	Contact hours: 16 hours lectures + 4 hours lab per week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module includes three units, namely Mechanics of Deltaic Soils, Experimental Soil Mechanics, and Soil Dynamics. Students will dive into non-standard aspects of Soil Mechanics, both in terms of analysis/modelling techniques and geotechnical applications. From a methodological standpoint, the module covers advanced topics regarding the testing and constitutive modelling of soils, up to the solution of boundary value problems through both physical and numerical modelling techniques. Relevant applications include the analysis of geotechnical problems in soft deltaic soils, and/or in the presence of cyclic/dynamic loading conditions. In this module the learning line on Monitoring and Data Science is enriched with further content. In particular, the unit on Experimental Soil Mechanics addresses topics related to the monitoring of physical geo-models during small-scale centrifuge experiments, including the selection of the most suitable sensors for real-time measurements. Overall, the three units provide opportunities for elaborating the data, often of big size, returned by physical and numerical models. Such elaboration is guided throughout the module for students to learn how to process/visualise/present information for clients and stakeholders. Ethical aspects associated with Advanced Soil Mechanics subjects are discussed throughout the module. Given the emphasis on experimental and numerical analyses as essential input to geotechnical design, important implications are pointed out regarding the impact of (i) selection of testing methods, (ii) identification of soil model parameters, (iii) validation of physical/numerical models on the usability/safety of geotechnical systems. Particularly, conflicts of interest, and their potentially catastrophic consequences, between (financial benefit of) service providers and (satisfaction/safety of) clients is discussed throughout the three units from a mechanics-oriented perspective.	
Study Goals	On completion of this module, the student will be able to: B2-LO1: analyse the role of advanced, time-dependent features of soils' hydro-mechanical behaviour B2-LO2: identify and integrate testing and constitutive modelling strategies for soils subjected to environmental and man-induced perturbations B2-LO3: select and perform advanced analyses of geotechnical problems based on most appropriate physical/numerical modelling techniques B2-LO4: critically reflect on benefits and limitations of advanced physical and numerical modelling strategies in geotechnical engineering B2-LO5: effectively report in writing the outcome of advanced experimental and computational analyses in relation to geotechnical engineering applications	
Education Method	MECHANICS OF DELTAIC SOILS Introduction, overview and issues with deltaic soils Week 1 (6 hrs): Lectures with active teaching including practical examples, quizzes and discussions. Instr.: M. Hicks, C. Zwanenburg Time-dependent behaviour, creep; Assignment 1a Week 2 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: R. Brinkgreve, C. Zwanenburg Stress path-dependent shear strength, anisotropy, fabric, bonding; Assignment 1b Week 3 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: R. Brinkgreve, C. Zwanenburg Issues related to loose soils, static liquefaction; Assignment 1c Week 4 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: M. Hicks, R. Brinkgreve Advanced numerical modelling of large deformations; engineering and design aspects of geotechnical projects; introduction of case study Week 5 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: M. Hicks, C. Zwanenburg, R. Brinkgreve Start numerical modelling of case study Week 6 (6 hrs): Practical example, problem-based learning through numerical simulation Instr.: C. Zwanenburg, R. Brinkgreve Elaboration of case study and results; Assignment 1d Week 7 (6 hrs): Practical example, problem-based learning through numerical simulation Instr.: C. Zwanenburg, R. Brinkgreve EXPERIMENTAL SOIL MECHANICS Introduction & soil classification week 1 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, laboratory activities Instr.: M. Cabrera, L. Flessati	

Direct simple shear and ring shear test
 Week 2 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies,
 Instr.: , L. Flessati

Small strain stiffness
 Week 3 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies
 Instr.: , L. Flessati

Physical modelling
 Week 4 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises
 Instr.: M. Cabrera

Physical modelling
 Week 5 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, visit to geotechnical companies
 Instr.: M. Cabrera

Physical modelling
 Week 6 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies
 Instructor(s): M. Cabrera

Physical modelling
 Week 7 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies
 Instr.: M. Cabrera

SOIL DYNAMICS

Waves in Soils
 Weeks 1-2 (12 hrs): lectures with active teaching,
 Instr.: F. Pisano

Man-induced and environmental soil vibrations
 Week 3 (6 hrs): lectures with active teaching
 Instr.: F. Pisano

From linear to non-linear cyclic/dynamic soil behaviour
 Week 4 (5 hrs): lectures with active teaching
 Instr.: F. Pisano

Seismic site response
 Week 5 (6 hrs): lectures with active teaching
 Instr.: R. Brinkgreve, F. Pisano

Computational soil dynamics
 Weeks 6-7 (12 hrs): lectures with active teaching
 Instr.: R. Brinkgreve, F. Pisano

Assessment

The module includes in total 4 summative assessments, which are all transversal to the LOs of the module. 3 of them are individual and more pronouncedly unit-based, with an attention for connecting the assessment of those LOs that are common to more than one unit. The 4th summative assessment is an overarching assignment, during which students will be challenged to elaborate on the disciplinary connections among the different units in the module. The overarching assignment will build around advanced geotechnical applications preliminarily suggested by the teachers. Formative assessment moments will be organized within each unit of the module.

Module assessment will be overall distributed as follows:

3 unit assignments - 90% of the total (3x30%)
 1 overarching assignment - 10% of the total

A successful participation in the module, requires a minimum evaluation of 5.8 in all units.

Expected prior Knowledge Academic Skills

Attendance of Track Base and Module A in the Geotechnical Engineering Track

Critical thinking; problem-based learning and solving; application of theory to new situations using the relevant equations; making inferences of physical situations by analyzing observations/data; scientific writing

Literature & Study Materials

MECHANICS OF DELTAIC SOILS

tbd

EXPERIMENTAL SOIL MECHANICS

- Das, B.M. (2010). Geotechnical engineering handbook. J. Ross publishing. (selected chapters)
- Head, K.H. (1980). Manual of soil laboratory testing. London: Pentech press. (selected chapters)
- Taylor, R.N. (2018). Geotechnical centrifuge technology. CRC Press. (selected chapters)
- Wood, D. M. (2017). Geotechnical modelling. CRC press. (selected chapters)

Further suggested readings (research papers) will be provided during the course.

SOIL DYNAMICS

- Lecture slides

	- Kramer, S.L. (1996). Geotechnical Earthquake Engineering. Prentice-Hall. - Towhata, I. (2008). Geotechnical Earthquake Engineering. Springer Science & Business Media. - van Dalen, K.N. (2015). Body Waves in an Elastic Continuum & Rayleigh Waves at the Free Surface. Lecture notes - Zienkiewicz, O.C., Chan, A.H.C., Pastor, M., Schrefler, B.A., & Shiomi, T. (1999). Computational Geomechanics -- with special reference to Earthquake Engineering. Wiley - Holscher, P. (2016). Soil Dynamics in Urban Areas. Lecture notes (Chapters 9-10) (provided through Brightspace)
Judgement	Further suggested readings (research papers) will be provided during the course. See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Collegerama	Take-home assignments may be carried out by using all the teaching materials provided during the course. No further materials needed for the final oral exam. No

CIEM2230	Delta Geotechnics	15
Responsible Instructor	Dr.ir. W. Broere	
Instructor	Dr.ir. R.B.J. Brinkgreve	
Instructor	Dr.ir. W. Broere	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Dr.ir. C. Zwanenburg	
Instructor	Dr. G. Giardina	
Instructor	Prof.dr. M.A. Hicks	
Instructor	Prof.dr.ir. C. Jommi	
Instructor	S. Muraro	
Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Drs.ir. R.E.P. de Nijs	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	M.A. Cabrera	
Contact Hours / Week x/x/x/x	x/x/Q4/x 6,6,6+4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Geotechnical Engineering in delta areas, in the Netherlands and worldwide, must face challenges associated to soft, and possibly unstable, soils. Such challenges must inevitably be tackled at different levels, i.e., from fundamental understanding and simulation of deltaic soil behaviour, to their in-situ characterisation for geotechnical design. More specifically, the module covers the design, monitoring and assessment of geotechnical structures. It addresses the mechanics of deltaic soils, the interaction between soil layers and the geotechnical structure, and the impact of geotechnical structures on other structures in an urban environment. Applications include the design of geotechnical structural elements (e.g., tunnel lining, soil reinforcement). The learning line on Monitoring and Data Science is included through the analysis of conventional and innovative geotechnical monitoring, and the evaluation of data from site investigation and physical modelling (Site Investigation & Monitoring). Parts of this module follow the learning line on Ethics by considering ethical aspects in geo-risk management. In the Site Investigation & Monitoring unit, students will gain understanding into approaches used to reduce risk in geotechnical engineering. Ethical aspects associated with human factors and risk communication in geo-engineering will be also covered.	
Study Goals	On completion of this module, the student will be able to: B3-LO1: describe testing and constitutive modelling strategies for soils subjected to environmental and man-induced perturbations. B3- LO2: explain the role of advanced, time-related features of soils' hydro-mechanical behaviour. B3- LO3: evaluate advantages and shortcomings of different alternative choices for the design of infrastructure. B3- LO4: evaluate the design and construction methods for different types of infrastructure. B3- LO5: compare the various methods for site investigation. B3- LO6: analyse geotechnical problems based on physical and numerical modelling techniques	
Education Method	Lectures, individual and group assignments, laboratory practicals and site visits. See unit descriptions on Brightspace for details.	
Assessment	The module includes in total 4 summative assessments, transversal to the LOs of the module. 3 of them are unit-based, with an attention for connecting the assessment of LOs that are common to more than one unit. Formative assessment moments are organised within each unit of the module. They include an individual assignment (30%), a report (30%) and an oral exam (30%) The 4th summative assessment, included in the unit Geotechnical Design, is a group design assessment which addresses disciplinary connections among the different units in the module. (10%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Attendance of Track Base and Module A of GE track	
Academic Skills	Critical thinking; interdisciplinary thinking; problem-based learning and solving; application of theory to new situations using the relevant equations; time management; making inferences of physical situations by analysing observations/data.	
Literature & Study Materials	See Brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Assignments may be carried out by using all the teaching materials provided during the course and further background material as needed. The report on the group assignment can be used as reference, no further materials are needed for the oral exam.	
Colleggerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Hydraulic Engineering (HE)

CIEM3000	Base: Hydraulic Engineering Fundamentals and Environments Part I	15
Responsible Instructor	Dr.ir. B.C. van Prooijen	
Instructor	Dr.ir. B.C. van Prooijen	
Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Prof.dr.ir. W.S.J. Uijtewaald	
Instructor	Dr.ir. B. Hofland	
Instructor	Dr.ir. E. Mosselman	
Instructor	Dr.ir. M. Zijlema	
Instructor	Dr.ir. W.M. Kranenburg	
Instructor	Dr.ir. S. de Vries	
Instructor	Dr.ir. A. Blom	
Instructor	Dr. M. Hrachowitz	
Instructor	Dr. M.F.S. Tissier	
Instructor	Dr. R.M.J. Schielen	
Instructor	C. Piccoli	
Instructor	Dr. L.M. Stancanelli	
Instructor	J.A. Arriaga Garcia	
Contact Hours / Week x/x/x/x	x/16/6-8/x	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The Hydraulic Engineering Extended Track Core of 24 EC consists of two parts. Part I (15 EC) contains three units: Modelling Flow, Transport and Bed Protection, MFTBP (6EC); Ocean Waves (5EC); and River Systems and Interventions (4EC). The units MFTBP and Ocean Waves provide the fundamental knowledge and skills. The unit River Systems and Interventions covers together Part II the Hydraulic Engineering Environments or systems. The unit MFTBP covers: (1) an in-depth analysis of the hydrodynamic and transport processes; (2) the theory and techniques to numerically solve the hydrodynamic and transport equations; (3) the theory and tools to describe and calculate the hydraulic loads and bed protection. The unit Ocean Waves covers the dynamics of wind-generated waves in the open ocean and in coastal waters. This unit River Systems and Interventions covers: (1) the characteristics of a river system,(2) the morphodynamic behaviour of these systems (including the response to human interventions) based on the underlying hydrodynamic and sediment transport processes, and (3) typical river engineering problems and methods design interventions.	
Course Contents Continuation	MFTBP: The unit is organised around three main topics: Fundamentals of free surface flows and transport processes: The first block treats the mathematical description of unsteady flow with free surface in open channels based on the principles of conservation of mass and momentum, including long waves, tidal basins and flood waves in rivers. Further to this, transport of dissolved matter due to advection and diffusion is covered in depth. Specifically, the underlying transport processes on the basis of which shear stress, bed roughness, logarithmic velocity profile, dispersion, turbulent diffusivity and mixing are described, but also the associated two-dimensional (horizontal) and three-dimensional (vertical) flow structures, are treated. Computational modelling of flow and transport In the second block elementary notions and techniques from numerical analysis are addressed for the approximation of solutions to partial differential equations (PDEs), with particular attention to the numerical accuracy and stability. This block is also an introduction to finite difference methods for constructing numerical solution of parabolic and hyperbolic PDEs. These equations describe a large variety of transport processes and wave propagation phenomena arising in hydraulic engineering. Emphasis is laid upon the assessment of the introduced uncertainty while using different numerical schemes in terms of the numerical diffusion and dispersion, along with measures to prevent wiggles. Lastly, a class of numerical methods is introduced for solving one-dimensional shallow water equations, principally aiming at the simulation of unsteady free surface flows in seas, estuaries, rivers, channels and lakes. Of crucial importance is the knowledge concerning the abilities and limitations of the commonly used practices related to approximating the equations of long waves. Loads and protection In this third block hydraulic loads and protections are introduced. Bed protections, filter layers, stability, porous flows are introduced. Also a 5-step design principle is introduced. Fundamental principles of physics are combined with data based (hydraulic) engineering principles and rules of thumb to facilitate engineering decisions on loads and protection. The Ocean Waves Unit focuses in the first part on the description of regular (monochromatic) waves before extending the results to realistic wind-generated wave fields that consist of wave components of varying frequencies and directions. It starts with linear wave theory which gives the interrelation between physical characteristics as the surface motion, the wave-induced pressure in the water and the motion of water particles. Commonly used nonlinear wave theories are then briefly introduced. We then discuss how the main properties of regular waves change when they propagate over varying depths and how they are affected by currents. In the second part of the Unit, we transition to irregular and directionally spread wave fields. We discuss how such wave fields can be measured and subsequently characterized, both in the time- and frequency-domain. We finally discuss how the wave field develops in the open ocean and coastal waters, and how an energy balance can be used to predict spectral wave evolution. The unit River Systems and Interventions covers three main topics: (1) The characteristics of a river system. Here we address river systems, river response to change and associated time scales, catchment hydrology, rainfall-runoff, and climate aspects. (2) The hydraulic and morphodynamic behavior of these systems. Here we address flow properties (normal flow, backwater types, flood waves, flow structures), sediment transport, and initial, transient and equilibrium morphodynamic response.	

	(3) River interventions. Here we address river-related problems, (inter)national river management, flood safety, navigation and other river functions, as well as river interventions.
Study Goals	On completion of the Hydraulic Engineering Extended Track Core (Part I and Part II combined), the student will be able to: 1. Explain the physical principles behind flow, transport, turbulence, mixing, sediment transport, surface waves and hydraulic loads. 2. Derive analytical/numerical models to assess typical hydraulic engineering problems based on physical principles combined with data. 3. Design solutions for typical hydraulic engineering problems regarding scour, siltation, contamination, sediment transport, bed stability. 4. Explain the characteristics of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems. 5. Predict and assess the behaviour of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems in response to natural and anthropogenic change taking uncertainties into account. 6. Identify the range of functions of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems.
Study Goals continuation	The specific Learning Objectives for the unit MFTBP are: LO 01: Explain the physical principles of free surface flow, turbulence, mixing and transport in open channels and the associated hydraulic loads; LO 02: Assess and select appropriate numerical techniques for solving the equations for transport and free surface flows. LO 03: Derive analytical/numerical models to assess typical hydraulic engineering problems regarding based on physical principles. LO 04: Devise suitable computing methods for the physically consistent solution to flow and transport equations; LO 05: Apply mathematical and numerical methods in combination with data to predict and assess flow and transport phenomena in hydraulic engineering; LO 06: Design solutions for typical hydraulic engineering problems regarding scour, river contamination, and bed stability The specific Learning Objectives for the unit Waves are: LO01: Describe the different observation techniques of wind-generated waves and their limitations LO 02: Explain the rationale behind the definition and computation of the wave spectrum and calculate spectral wave characteristics LO 03: Characterize wind-generated waves in a statistical framework LO 04: Explain the physical processes driving wave transformation in oceanic and coastal waters and evaluate which processes dominate in a given situation LO 05: Calculate the evolution of wind-generated waves in oceanic and coastal waters . The specific Learning Objectives for the unit River Systems and Interventions are: LO 01: Explain the role of the river system in the hydrological cycle, the controls of a river system, time scales of change of the controls versus time scale of system response (catchment to reach scale), and phases (initial, transient, quasi-equilibrium, equilibrium) of river response LO 02: derive and apply simplified forms of the one-dimensional shallow-water equations (i.e., normal flow, flood waves, and backwater curves); LO 03: predict and assess (initial, equilibrium, and transient) river response to natural and anthropogenic change of controls LO 04: identify and assess the range of river functions (e.g. flood safety, navigation, nature restoration, freshwater supply, ecology, port operability, water quality, and construction in water), system-related problems (from catchment to reach scale) in national and international river management, and possible solutions or interventions LO 05: apply (inter)national safety standards in flood risk management;
Education Method	The education will be a mix of: - Plenary Lectures - Online videos - Formative assignments and exercises - group discussion sessions.
Computer Use	Numerical exercises via python Notebooks will be a substantial part of the course.
Assessment	Modelling flow, Transport and Bed Protection: written exam (40%) Numerical assignments must be completed with a pass to be allowed to participate in the exam. Ocean waves: written exam (20%), project (13.3%) River Systems and Interventions: written exam in the regular period and oral exam in the resit period (26.7%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Contact	B.C. (Bram) van Prooijen B.C.vanProoijen@tudelft.nl
Expected prior Knowledge	The module expects prior knowledge of fluid mechanics, including Open Channel Flow. It furthermore builds on the Faculty Core and Program Core. We expect the students to work with Python.
Academic Skills	Critical thinking, (mono-disciplinary and partly multi-disciplinary) problem solving and the application of theory in new engineering situations are the main academic skills that will be trained.
Literature & Study Materials	Books: Unsteady flow in open channels by Jurjen Battjes and Robert Jan Labeur (Cambridge University Press), also as free ebook available via library.tudelft.nl Waves in Oceanic and Coastal Waters by Leo H. Holthuijsen (Cambridge University Press), also as free ebook available via library.tudelft.nl Introduction to bed, bank and shore protection by Schiereck. Additional material:

	Slides Syllabus Reading material
Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Materials that are allowed during the exam will be provided via Brightspace.
Collegerama	Yes

CIEM3110	Hydraulic Engineering Fundamentals and Environments Part II	9
Responsible Instructor	Dr.ir. B.C. van Prooijen	
Instructor	T.M. de Boer	
Instructor	A.J. van der Hout	
Instructor	Dr.ir. M. van Koningsveld	
Instructor	Dr.ir. J. Bosboom	
Instructor	P. Taneja	
Instructor	Drs. O.C. Koedijk	
Instructor	S.G. Pearson	
Instructor	Ir. F.R.S. Vinke	
Instructor	Drs. R. ter Hofstede	
Instructor	J.A. Álvarez Antolínez	
Instructor	M.C. Onderwater	
Instructor	Ir. L. de Boom	
Contact Hours / Week x/x/x/x	x/x/12-13/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The Hydraulic Engineering Extended Track Core of 24 EC consists of two parts. Part II (9 EC) contains two units: Coastal and Estuarine Systems and Introduction to Port and Waterway Systems. Together with the unit River Systems and Interventions of Part I, they cover the Hydraulic Engineering Environments or systems. 1. Coastal and Estuarine Systems (6EC). This unit covers: (1) the characteristics of a range of coastal systems (seas, open coasts, estuaries and tidal basins), (2) the morphodynamic behaviour of these systems (including the response to human interventions) based on the underlying hydrodynamic and sediment transport processes, and (3) typical coastal engineering problems and methods for coastal protection. 2. Introduction to Port and Waterway Systems (3EC) This unit explains the importance of port and waterway systems in supporting waterborne supply chains at different scales, and discusses performance optimization from an infrastructure perspective. The unit is organized around three themes: (1) Ports and terminals: challenges for port development, port planning, port layout, terminal development, (2) waterways: challenges for waterway development, waterway dimensions, waterway elements, ship and ship-waterway interactions, traffic management, (3) system performance: analyzing the performance of systems of ports/terminals and waterways (queueing theory).	
Course Contents Continuation	Various topics will be considered in this module. An overview per unit is provided below. Topics per week for Coastal and Estuarine Systems (and corresponding chapter in textbook): Week 1: Introduction (Ch1) and large-scale geographical variation of coastal systems (Ch2) Week 2: Wind waves and tides (Ch3), Week 3: Global wave and tidal environments (Ch4), Coastal hydrodynamics wind waves (Ch5) Week 4: Coastal hydrodynamics wind waves (Ch5), Coastal hydrodynamics tides (Ch5), Week 5: Sediment transport (Ch6), Cross-shore transport and profile development (Ch7) Week 6: Cross-shore transport and profile development (Ch7), Longshore transport and coastline changes (Ch8) Week 7: Exam questions Cross-shore and Longshore, practical Ch8, Coastal inlets and tidal basins (Ch9) Week 8: Coastal inlets and tidal basins (Ch9), exam questions Tidal, Coastal protection (Ch10) Topics per week for Introduction to Ports and Waterways: Week 1: General introduction ports and waterways; integral port planning Week 2: Planning & design of port water areas Week 3: Terminal planning & design + Container terminals (example) Week 4: Queueing theory part 1 & 2 + method Kooman Week 5: Port economics, port-city interaction, port environment Week 6: Inland navigation: history + corridors + cargo + fleet & Performance of inland ships (ship-ship interaction) Week 7: Dimensioning of waterways and waterway elements; dimensions of channels Week 8: Guest lecture; Q&A session	
Study Goals	On completion of the Hydraulic Engineering Extended Track Core (Part I and Part II combined), the student will be able to: 1. Explain the physical principles behind flow, transport, turbulence, mixing, sediment transport, surface waves and hydraulic loads. 2. Derive analytical/numerical models to assess typical hydraulic engineering problems based on physical principles combined with data. 3. Design solutions for typical hydraulic engineering problems regarding scour, siltation, contamination, sediment transport, bed stability. 4. Explain the characteristics of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems. 5. Predict the behaviour of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems in response to natural and anthropogenic change taking uncertainties into account.	
Study Goals continuation	The specific Learning Objectives for the unit Coastal and Estuarine Systems are: LO 01 - Discuss the nature and complexity of typical coastal engineering problems (Ch1, but also Ch7-10). LO 02 - Explain the imprint on present-day coastal system (uninterrupted coastlines as well as tidal inlets and basins) of: - Geological processes, sea level and climate (Ch2). - Fluvial, wave and tidal processes and their relative influence (Ch2). LO 03 - Understand the global variation of: - Coastal systems (Ch2, Ch4) - Wave and tidal climate (Ch4) LO 04 - Analyse the hydrodynamics of waves and tides: - in the oceans (Ch3); and - along open coasts and in tidal basins (Ch5).	

	LO 05 - Evaluate sediment transport processes under waves, tides and currents (Ch6), distinguishing between: - cross-shore transport (Ch7) - longshore transport (Ch8) - transport in tidal basins and inlets (Ch9). LO 06 - Apply system knowledge to determine the morphological response to environmental conditions as well as interventions (groins, breakwaters, nourishments, land reclamations, sand mining) of: - open coasts (Ch7-8); and - tidal inlets and basins (Ch9). LO 07 - Assess the merits and disadvantages of various coastal interventions for protection against flooding and erosion (Ch10) The specific Learning Objectives for the unit Introduction to Ports and Waterways Systems LO 01 - general: - Explain the functions of ports and waterways in the total transport chain to a level that is necessary for functional design and capacity analysis; - Discuss the importance of performance indicators such as capacity, efficiency, safety and sustainability, and - Describe the environmental issues relevant to ports and waterways and explain the importance of integrating environmental considerations in (stakeholder inclusive) port planning and design. LO 02 - ports/terminals: - Explain the principles of port master planning and develop basic port layouts based on principles of robustness and adaptability in an uncertain future; - Develop functional designs of port terminals (number of elements and their order-of-magnitude dimensions) using state-of-the-art design methods and rules (Module A focuses on container terminals, Module B04-01 will address other terminal types); - Explain the basic elements of investment analysis and can perform such an analysis for a simple case (analysing trade-offs). LO 03 - waterways and port water areas: - Discuss the importance of inland waterway transport corridor policies/plans and perform a basic capacity analysis taking into account forecasted traffic volumes and waterway infrastructure in place or planned. - Explain the basics of ship motions, ship-ship interaction and ship-fairway interaction - Develop functional designs of inland waterways and port waterway areas using state-of-the-art design methods and rules (canals, locks, bridges, approach channels, turning basins, berths). LO 04 - system behaviour: - Analyse relevant processes influencing capacity - Compare the performance of port and waterway systems using rules of thumb and queueing theory (waiting time as a fraction of service time) and the Kooman method for lock capacity assessment.
Education Method	Coastal and Estuarine Systems: 1. Mix between video lectures, lectures, practicals with exam questions, question hours. 2. Weekly formative digital assignments. Active participation and demonstration of a learning curve in the formative assignments is required to access the exam.
Assessment	Introduction to Port and Waterway Systems: Mix between lectures and flipped class room; work sessions Exam 1: Coastal and Estuarine Systems - Written Exam (66,7%) Exam 2: Introduction to Ports and Waterways Systems (33,3%): Written Exam and Design Exercise (the written exam counts for 70% of the grade and the design exercise counts for 30%).
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Academic Skills	We build on the Faculty Core, Program Core and Part 1 of the Hydraulic Engineering Extended Track Core. This module will strengthen various academic skills, ranging from critical thinking, conceptualization, dealing with data, to problem solving.
Literature & Study Materials	The Hydraulic Engineering Track Core will use open-source material. This includes the following books: Bosboom, J. and Stive, M. J. F. (2022 or 2023 - latest version). Coastal Dynamics. TU Delft Open. ISBN (softback/paperback) 978-94-6366-370-0, ISBN (e-book) 978-94-6366-371, DOI 10.5074/T.2021.001 Lecture Notes, Van Koningsveld, M. (Ed.), Verheij, H.J., Taneja, P. and De Vriend, H.J. (2021). Ports and Waterways - Navigating the changing world TU Delft, Hydraulic engineering, Ports and Waterways, Delft, The Netherlands. Ligteringen H. (2022), Ports and Terminals TU Delft, Hydraulic Engineering, Ports and Waterways, Delft, The Netherlands.
Judgement	In addition to these books, we will provide Lecture Slides and Open Source software via Brightspace. See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Permitted Material during the exam will be announced via Brightspace.
Collegerama	Yes

CIEM3210	Coastal Engineering	15
Responsible Instructor	Dr.ir. J. Bosboom	
Instructor	Dr.ir. B.C. van Prooijen	
Instructor	Dr.ir. B. Hofland	
Instructor	Dr.ir. S. de Vries	
Instructor	M.A. de Schipper	
Instructor	A.P. Luijendijk	
Instructor	Dr. A. Antonini	
Contact Hours / Week x/x/x/x	x/x/x/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Central to this B-module are coastal engineering problems and interventions and specifically the assessment of the impact of these interventions and their technical design. Impact assessment and design require advanced system knowledge and proper application of tools. These three elements in connection (interventions, knowledge and tools) are the basis of this B-module. Ethical issues and the role of the engineer will be addressed in the model assignments part of the second two units described below by addressing conflicting perspectives of various vocal and non-vocal stakeholders. The module consists of three units of equal weights, which run in parallel (with especially strong connections between the first and second units): 1. Advanced Coastal and Estuarine Systems 2. Coastal and Estuarine Modelling 3. Shore protection & Breakwaters Advanced Coastal and Estuarine Systems treats, in-depth, coastal and estuarine processes and system behaviour (a.o. waves, tides, currents, coarse and fine sediment transport, morphodynamic change) and response to interventions (a.o. storm surge barriers, groynes, dredging and nourishments). A wide range of coastal systems is considered: deltas, estuaries, beaches and dunes, tidal plains, barrier and lagoon coasts. Not only the physical principles but also their complex interplay in the built and natural environment is discussed. This process knowledge is linked to coastal engineering problems and interventions. Coastal and Estuarine Modelling combines knowledge on prediction, modelling and monitoring concepts with a hands-on approach. Students apply various state-of-the-art models in order to solve complex coastal engineering problems taking into account its uncertainties. Students (functionally) design and evaluate coastal engineering interventions related to amongst others coastal safety or sediment management. Students collaborate in the group assignment. Shore protection & Breakwaters treats the mechanisms of (in)stability of protection layers under wave loads. Aspects needed for practical implementation of these protection layers are introduced, like construction, environment, uncertainty, maintenance, boundary conditions, and modelling techniques. The technical design of all parts of a breakwater cross-section is treated in-depth.	
Study Goals	(B1-1) Analyse and assess in depth the complex behaviour of coastal and estuarine systems (deltas, estuaries, beaches and dunes, tidal plains, barrier and lagoon coasts) including the response of these systems to human interventions (B1-2) Apply models to predict the behaviour of coastal and estuarine systems and assess the impact of coastal engineering interventions (B1-3) Apply the theories of wave loads on different types of slope protection (B1-4) In collaboration propose and design coastal engineering interventions for purposes such as protection against flooding and erosion and construction in water (B1-5) Account for uncertainties in relation to monitoring, modelling and data analysis for research and design (B1-6) Clearly present and report design and modelling results	
Education Method	Advanced Coastal and Estuarine Systems: Lectures and/or knowledge clips, Practicals (inquiry-based learning), Coastal and/or estuarine field trip (with assignment) Coastal and Estuarine Modelling: Lectures and/or knowledge clips, Practicals (inquiry-based learning), Master classes (Guests are invited to discuss ongoing projects), Modelling assignments Shore protection & Breakwaters: Lectures and/or knowledge clips, Master classes by guests, Possibly participation in field trip of Unit 1	
Assessment	Exam 1: Advanced Coastal and Estuarine Systems - Oral Exam (33%) Exam 2: Coastal and Estuarine Modelling - Project (33%) Exam 3: Shore protection and Breakwaters - Project (33%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Academic Skills	Track Base (units Flow and Transport and Ocean Waves), module A (unit Coastal and Estuarine Systems)	
Literature & Study Materials	Critical thinking, problem-based learning and solving, synthesis of system knowledge and response, the application of theory in new engineering situations, making inferences of coastal systems by analysing model results and data, communicating results by oral presentation (live or videos) and to a lesser extent by writing are the main academic skills that will be trained Advanced Coastal and Estuarine Systems: Bosboom, J. and Stive, M. J. F. (2021). Coastal Dynamics. TU Delft, Open. ISBN (softback/paperback) 978-94-6366-370-0, ISBN (e-book) 978-94-6366-371, DOI 10.5074/T.2021.001, possibly revised to include and match the new courses, Annotated lecture slides (Selected sections of) Journal papers, Selected chapters from Winterwerp et al (2021), Fine Sediments in Open Water from fundamentals to modelling. Coastal and Estuarine Modelling: Annotated slides and tutorials, Selected journal papers, In-depth material for specific problems/systems, Selected chapters from: 1) Roelvink & Reniers, A guide to modelling, Coastal Morphology and 2) Winterwerp et al, Fine Sediments in Open Water from fundamentals to modelling. Shore protection and Breakwaters: Part A: Bed, Bank, and Shore Protection book by G.J Schiereck Part B: Breakwater design lecture notes Online lectures, videos and links to literature	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

CIEM3220	River Engineering	15
Responsible Instructor	C. Piccoli	
Instructor	Prof.dr.ir. W.S.J. Uijtewaald	
Instructor	Dr.ir. B. Hofland	
Instructor	Dr.ir. E. Mosselman	
Instructor	Dr.ir. C.J. Sloff	
Instructor	Dr.ir. M. Zijlema	
Instructor	Dr.ir. W.M. Kranenburg	
Instructor	mr.dr.ir. C. Mai Van	
Instructor	Dr.ir. A. Blom	
Instructor	Dr. T.A. Bogaard	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	E. Ragno	
Instructor	Dr. R.M.J. Schielen	
Instructor	C. Piccoli	
Instructor	Dr. D. Wüthrich	
Instructor	Dr. L.M. Stancanelli	
Instructor	J.A. Arriaga Garcia	
Contact Hours / Week x/x/x/x	x/x/x/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	HE Module B2 consists of three units: B2-1. Physics of River Systems (4EC), which focusses on the physics of river systems and their response to anthropogenic or natural change; B2-2. River Functions and Management (6EC), which provides students with a broad knowledge on the wide range of purposes or functions of a river system (flood safety, navigation, freshwater availability, nature, etc) and stakeholders, as well as river interventions; B2-3. Schematization and Modelling of River Problems (5EC), which teaches students how to schematize and simplify a river-related problem, formulate research questions and alternative solutions, assess the problem using appropriate tools (such as modelling and monitoring), propose a solution and mitigate unwanted impact.	
Study Goals	On completion of this module, the student will be able to 1. understand and apply scientific knowledge that contributes to solving river system-related problems in an ethically sound manner; 2. analyse and assess the behaviour of river systems and catchment, and assess and predict the system response to interventions, while taking into account neighbouring/adjacent systems; 3. (a) analyze the problem, (b) formulate functional requirements and constraints, (c) formulate alternative solutions (including autonomous development), (d) design human interventions in the river or catchment environment for purposes such as flood protection, nature development or restoration, navigation, freshwater supply, ecology, port operability, water quality, and construction in water; 4. account for societal, environmental, and economic aspects, and uncertainties in research and design, at various scales; 5. account for the complexities of multidisciplinary environments and the range of stakeholders: (inter)national river management authorities, nature organisations, NGOs, dredging companies, navigation stakeholders, other governments.	
Education Method	Lectures, discussion sessions, exercises, presentations, field trip, project groups, group sessions	
Assessment	Unit B2-1: Physics of River Systems - Oral Exam (26,7%) Unit B2-2: River Functions & Management - Portfolio (40%) Unit B2-3: Schematization and Modelling of River Problems - Portfolio (33,3%) A successful participation in the module, requires a minimum evaluation of 5.8 in each of the 3 units. There will be a resit opportunity for the oral exam. A minimum grade of 5.5 is required for all individual components of both portfolios. If the grade of a portfolio component is less than 5.5, you will get the opportunity to submit a revised version and defend it orally. The maximum grade of that component will then be a 6.0.	
Expected prior Knowledge	MUDE, and Track Base plus Module A of CE track Hydraulic Engineering	
Academic Skills	Critical thinking, problem solving, and application of theory in new engineering situations are the main academic skills that will be trained.	
Literature & Study Materials	B2-1. Physics of River Systems: Powerpoint slides, selected journal papers and other reading material (available through links on Brightspace), home exercises B2-2. River Functions & Management: Powerpoint slides, collected reading material and journal papers B2-3. Schematization and Modelling of River Problems: Collected reading material	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Materials that are allowed during the exam will be provided via Brightspace.	

Collegerama	Yes
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CIEM3230	Advanced Design of Ports and Waterways Systems and Interventions	15
Responsible Instructor	O. Kirichek	
Instructor	Dr.ir. M. van Koningsveld	
Instructor	F. Baart	
Contact Hours / Week x/x/x/x	x/x/x/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The world's economy relies heavily on waterborne supply chains. Approximately 80% of all global trade is at some point transported by shipping. Overall efficiency of global supply chains is to a great extent determined by the in-port and hinterland transport networks they are connected with. A major challenge for the field of ports and waterways engineering is timely adaptation of transport networks, and their associated infrastructure, to ever changing external circumstances (changing vessel sizes, shifting trade patterns, increased attention for sustainability, digitalization etc.). In this module, we teach students advanced design methods for ports, waterways, and systems of ports and waterways in a societal, environmental, and economic context. Students will learn to analyze performance of these systems and design interventions for improvement (e.g. port accessibility, locks, waterway dimensions, breakwaters, quays), accounting for the logistical, nautical and natural processes (e.g. waves, tides, siltation) at play. Data science and advanced simulation techniques are essential for the quantification of trade-offs and therefore an integral part of this B Module. As such students that follow this B-line take a first step towards becoming research software and data engineers. Students will also become familiar with ethical issues arising around large scale engineering projects through real life case studies that involve multiple stakeholders. Unit 1 will cover the full extent of Q4 and Unit 2 will start in week 5. The students are expected to spend about 20% of their time on lectures, 30% on self-study, 20% on formative exercises and 30% on data-driven simulations. The guest lectures will be given by the key industry experts, who will share the most recent insights on innovations and developments in the ports and waterways. The guest lectures will be combined with the field visits, where the students will explore monumental engineering facilities (e.g. container terminals, smart quay walls, locks, breakwaters) in ports. Topics to be addressed in Unit 1 are: 1) Waterways Advanced dimensioning of waterways and waterway elements (e.g. locks, crossings, bends) Sediment management (e.g. accessibility design, dredging and maintenance strategies, maneuverability) Design of hard infrastructure (e.g. locks, slope protections) the latter are part of HOS B1 New developments (e.g. climate change, nature-inclusive solutions) 2) Ports and Terminals Functional design of various terminal types: liquid bulk, dry bulk (including mooring arrangements), general cargo, multipurpose terminals, Ro-Ro, ferry terminals, fishery ports, inland ports, marinas and cruise terminals Port infrastructure and environmental conditions (quay walls, breakwaters, port operability, safe navigation) Trade-offs and financial feasibility New developments (e.g. energy transition, digitalization) 3) Ports and Waterways systems Simulation: Introduction into simulation for P&W (planning and trade-offs) Simulation: Introduction into simulation models (deterministic, stochastic, discrete time/event) Data science for P&W: key concepts Topics to be addressed in Unit 2 are: 1) Advanced data-driven simulations 2) Simulation for ports and waterways (planning and trade-offs) 3) Simulation models (e.g. for analyzing the effect of climate change, sea level rise, emission mitigation measures, etc.) 4) Data science for ports and waterways: key concepts	
Study Goals	On completion of the module B3 (B03-01 and B03-02), the student will be able to: LO1. Analyse logistic and operational processes in ports and waterways systems and discuss the capacity controlling parameters; LO2. Design port/terminal and waterway infrastructure to pre-defined objectives covering economic, societal and environmental aspects; LO3. Design interventions in ports and waterways systems, model interactions ranging from simple to complex, between physical processes and port and waterways systems, and apply simulations for the evaluation of performance; LO4. Design hard and soft infrastructure interventions to improve port, terminal, and waterway performance including quantifying trade-offs and discussing these trade-offs with peers; LO5. Demonstrate the learning over intervention design and performance evaluation in real-life case studies, and LO6. Apply data science techniques and concepts for a quantitative analysis of processes in port and waterway systems in various design exercises.	
Education Method	Interactive lectures, design exercises, Q&A sessions	
Assessment	Unit 1: Ports and Waterways - Intervention, Design and Performance evaluation - Written Exam (60%), formative exercise (in groups) Unit 2: Advanced simulation and quantification methods - Individual Assignment (40%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Module A, old curriculum: CIE4330, CIE4301, CIE5308	
Academic Skills	The student will acquire essential skills in simulation and data science and apply these to understand and quantify the interactions between the state of the water infrastructure and the (economic, environmental and societal) performance of port and waterway systems. The student will integrate multidisciplinary knowledge and apply it in design exercises addressing real life situations. The exercises will be designed to stimulate critical thinking and develop conceptual, analytical, scientific reporting and communication skills.	
Literature & Study Materials	Van Koningsveld, M. (Ed.), Verheij, H.J., Taneja, P. and De Vriend, H.J. (2021). Ports and Waterways - Navigating the changing world. Lecture notes. TU Delft, Hydraulic engineering, Ports and Waterways, Delft, The Netherlands. Ligteringen, H. (2022) Ports and Terminals. TU Delft, Hydraulic engineering, Ports and Waterways, Delft, The Netherlands Schiereck G.J., Verhagen, H.J. and Hofland B. (2019). Introduction to Bed, Bank and Shore Protection, Delft, The Netherlands.	

	Lecture Slides Open Source software Handouts
Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	none
Collegerama	Yes

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Hydraulic & Offshore Structures (HOS)
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CIEM4000	Base: Hydraulic and Offshore Structures	15
Responsible Instructor	Dr. E. Kementzetzidis	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Dr.ir. B. Hofland	
Instructor	Ir. J.W. Welleman	
Instructor	mr.dr.ir. C. Mai Van	
Instructor	Dr.ir. H. Hendrikse	
Instructor	A. Tsouvalas	
Instructor	Dr. M.F.S. Tissier	
Instructor	Dr. E. Kementzetzidis	
Instructor	Dr. A. Antonini	
Contact Hours / Week x/x/x/x	14 hrs. Q2 4 hrs. Q3	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The CE HOS track addresses the challenges associated with the design, construction and operation of hydraulic and offshore structures by unifying the knowledge from structural, hydraulic, and geotechnical engineering disciplines. The role of the Track Base Module is to equip students with the fundamental knowledge necessary to tackle the engineering challenges associated with the structural design, construction and operation of structures in river, coastal and marine environments.	
Study Goals	Upon completion of this module, the student will be able to: MLO-1. Analyse fluid- and soil-structure interaction processes, as well as structural mechanics principles, Relevant for the engineering of hydraulic and offshore structures. MLO-2. Analyse scientific knowledge on fluid- and soil-structure interaction, as well as structural mechanics, that contributes to solving complex hydraulic and offshore structures. MLO-3. Perform, statistically model selection and identification for fluid- and soil-structure interaction analyses. MLO-4. Quantify uncertainty in models and measurements for fluid and soil-structure interaction analyses. MLO-5. Evaluate uncertainty propagation for fluid-and soil-structure interaction analyses.	
Education Method	Class lectures and recordings, group assignments	
Assessment	The Track Base Module is assessed both in Q2 and Q3. The gravity associated with each unit module is as follows, Waves (HOS-TC-1), Q2 : Written exam [19%] Fluid-Structure Interaction (HOS TC-2), Q2-Q3 : Group assignment [27%] Structural Mechanics (HOS-TC-3), Q2 : Written exam [27%] Soil-Structure interaction (HOS-TC-4), Q3 : Written exam [27%] A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	BSc Civil Engineering or equivalent, CE Programme Base Module.	
Academic Skills	Critical and interdisciplinary thinking, identification of problem-oriented solution procedures, information collection and synthesis, application of theory to new situations using the relevant equations and correlating mathematical solution with physical insights and design procedures, use of data for critical assessment of modelling assumptions and design considerations.	
Literature & Study Materials	Information available in each module unit.	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Information available in each module unit.	
Collegerama	No	

CIEM4110	Hydraulic Structures	9
Responsible Instructor	Dr. D. Wüthrich	
Instructor	Ir. W.F. Molenaar	
Instructor	S.C. Barbosa Nunes	
Instructor	D.J. Peters	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. D. Wüthrich	
Contact Hours / Week x/x/x/x	x/x/10/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	HOS-A1 Hydraulic Structures: 9ECs - HOS-A-1 Ultimate Strength & Fatigue: 3ECs - HOS-A-2 Design of Hydraulic Structures: 6ECs The module enables students to produce a conceptual engineering design of common hydraulic structures, building on already acquired skills. Students learn to analyse a problem to formulate a design objective, which leads to a functional-spatial design. This is followed by a structural design where next to stability and strength checks, constructability plays a major role. The student further develops skills to motivate assumptions, methods and values of design parameters, using basic sketches. Hydraulic, geotechnical and structural aspects are addressed in coherence to understand the behaviour of these structures. The content of the module contributes to the MUDE and Ethics learning lines.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Generate sound preliminary designs for hydraulic structures based on current natural and societal needs, accounting for its interaction with the environment its life-cycle aspects, using available scientific knowledge. (HOS-1,2,3) MLO-2. Analyse principles of ultimate resistance and fatigue relevant for the design of hydraulic structures, using available scientific knowledge. (HOS-2,3) MLO-3. Perform statistically model selection and identification for the analysis and verification of the structural health of hydraulic structures. (HOS-3,4) MLO-4. Effectively collaborate and communicate with the aim to generate and optimize the design of hydraulic structures. (HOS-4) MLO-5. Critically reflect on the design and reliability of hydraulic structures and its structural components. (HOS-2,4) MLO-6. Translate current natural and societal needs into possible technical designs of offshore structures by accounting for relevant moral values, norms and principles. (HOS-4)	
Education Method	LECTURES & TUTORIALS During lectures, the main theories related to hydraulic structures are described and its applications discussed. Topics needed in support of the design exercise are also introduced and further elaborated. Tutorials apply theories or techniques previously explained during lectures. During the tutorials students work in class, while the lecturer is available for answering questions. Students will implement the tutorial work within the design exercise. DESIGN EXERCISE To stimulate the active use of the lectured theory and tutorials, a design exercise has to be completed, usually in teams of two students. Team formation and subject selection for the exercise will start during the 1st teaching week of the course. Progress of the exercise will be as follow (see Brightspace for detailed deadlines): - End of week 2 - 1st check : workplan and basic design - End of week 4 - 2nd check : progress & results - End of week 6 - 3rd check : progress & results - End of week 8 - Final report	
Assessment	The assessment for this module consists of: - Individual Exam: 50 % - Group Design Project: 50 % Both the individual exam and the final deadline for the group design project take place at the end of the education period (Q3). The content of the individual exam is evenly divided over the involved units, so that 50% of the exam problems are related to unit HOS A1-1 (Ultimate Strength and Fatigue) and 50% of the exam problems are related to the content of unit HOS A1-2 (Design of Hydraulic Structures). To successfully pass the module, all components of the assessment must be evaluated with a pass-grade. A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	BSc Civil Engineering or equivalent, MUDE Module, CE Programme Base Module.	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well.	
Literature & Study Materials	Lecture notes and handouts will be available digitally on Brightspace.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Permitted Materials during Exam will be announced on Brightspace	
Collegerama	No	

CIEM4120	Offshore Structures	9
Responsible Instructor	Ir. J.S. Hoving	
Instructor	Ir. J.S. Hoving	
Instructor	S.C. Barbosa Nunes	
Instructor	Dr.ir. P. van der Male	
Instructor	Prof. M. Veljkovic	
Contact Hours / Week x/x/x/x	x/x/12.x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The module Offshore Structures of the Hydraulic and Offshore Structures (HOS) track, is worth 9 ECTS and consists of the following units: - Ultimate Strength and Fatigue = 3 ECTS - Design of Offshore Structures = 6 ECTS	
Course Contents	Note here that the unit on Ultimate Strength & Fatigue is shared with the module CIEM4110 Hydraulic Structures. This module enables students to produce conceptual engineering designs for fixed offshore structures accounting for ultimate strength and fatigue criteria. Within this module, students gain a thorough theoretical understanding of the design process, its structuring, available design methodologies and the philosophy behind the design codes. Furthermore, this module addresses offshore, geotechnical, structural and reliability aspects in coherence to analyse the behaviour of offshore structures. The ultimate strength and fatigue part of the module specifically focuses on the ultimate strength and fatigue assessment of steel and concrete structures. The provided knowledge and skills are required to be able to perform the verification of offshore structures design and are therefore directly applied in the design component of this module. Throughout the module students work on a group design project, where each student team designs an offshore support structure from scratch up to a preliminary design by performing a full offshore support structure design-cycle. This design-cycle includes I) the assembly of the structural configuration, II) the determination and assessment of the permanent, variable and environmental loads on the structure, III) the design and analysis of its foundation and IV) the static and dynamic structural analysis of the structure based on ultimate strength and fatigue principles. By means of this group design project, students will learn to judge their design against ULS and FLS design criteria and be able to modify and improve their design based on static and dynamic structural analyses. Thereby, students will be able to analyse and evaluate the dimensioning and performance of structural components for strength and stability. Furthermore, students learn to predict how changing (parts of) their design will influence the corresponding loads and based on their insight in these loads, be able to defend their design choices. Moreover, students will learn what the practical implications of their design choices are for operations and other feasibility and life-cycle aspects. To reinforce the understanding of these aspects, the course program also includes a visit to an offshore structures fabrication yard. During this module, students will not only learn how to design an offshore structure, but they also learn how to collaborate effectively in a team and will further develop skills and competences to motivate their assumptions and provide argumentation of their design choices by accounting for relevant moral values, norms and principles.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Generate sound preliminary designs for offshore structures, accounting for its interaction with the environment and its life-cycle aspects, using available scientific knowledge. MLO-2. Analyse principles of ultimate resistance and fatigue relevant for the design of offshore structures, using available scientific knowledge. MLO-3. Perform, statistically sound and context-based, model selection and identification for the analysis and verification of the structural health of offshore structures. MLO-4. Effectively collaborate and communicate with the aim to generate and optimize offshore structures designs. MLO-5. Critically reflect on the design and reliability of offshore structures and its structural components. MLO-6. Translate current natural and societal needs into possible technical designs of offshore structures by accounting for relevant moral values, norms and principles.	
Education Method	The education in this module includes problem-based lectures, workshops, a group design project, home assignments and self-study. In total, this module has 12 contact hours per week, consisting of: - two 2-hour lectures (2x45 min) regarding methods to assess the ultimate strength and fatigue of steel and concrete structures, - two 2-hour lectures (2x45 min) focused on comprehension and application of theory and methodology of offshore structures design, and - a 4-hour workshop (4x45 mins) focused on problem analysis and evaluation, treating and discussing example problems, as well as to provide feedback and reflection on the group design project as well as allowing for discussion and Q&A regarding the group design project. The group design project is performed in groups and consists of 3 phases. Each phase comes with a formative assessment & feedback session per group in interaction with the lecturers and/or your peers, followed by a final deadline at the end of the teaching period at which the assignment is graded. The groups generally consist of 3 or 4 students, but the group size may be adapted depending on cohort size. Additionally, under reservation of unforeseen and/or external circumstances, the module includes an excursion to a fabrication yard where offshore support structures are constructed. Here note that the number of participants to this excursion is generally limited by the capacity at the yard and the excursion is therefore not a compulsory part of the module.	
Prerequisites	Relevant BSc in Engineering, CEGM1000 Modelling, Uncertainty and Data for Engineers, CIEM000 Programme Base: Mechanics and Interdisciplinary Perspectives.	
Assessment	Participants of this module are summatively assessed through a Group Design Project, worth 4 credits, and an individual written exam, worth 4 credits. In addition, participants are formatively assessed through home assignments, worth 1 credit. This means that the home assignments are checked to provide feedback but are not graded, this however does not mean that the home assignments are optional. The final grade for the module is the weighted average of the Group Design Project (50%) and the individual written exam (50%). To pass the course, participants must score a pass grade, i.e. at least a 6.0 (rounded to halves), on both the project and the exam. Group Design Project:	

	The final deadline for the Group Design Project is set at the end of the Q3 education period. If you fail to get a pass grade for the assignment, but the assignment grade is equal to or higher than a 5.0, you are allowed to improve your grade by submitting an addendum to your assignment report in the following education period, i.e. during Q4. Here, please note that the maximum grade that can be obtained after submitting an addendum to improve an insufficient grade is 6.0. Written exam: The regular individual written exam for this course is organised during the exam period of Q3, and a re-exam is organised in the exam period of Q4. The written exam is an open-book exam, meaning that any written or printed material is permitted during the exam sessions. The content of the individual exam includes content of all the involved units, so that approximately half of the exam problems are related to Ultimate Strength and Fatigue and approximately half of the exam problems are related to the Design of Offshore Structures. If due to any unforeseen circumstances the written exam for this course may not be given in a plenary on-campus session, alternatively the written exam for this course may be given as an online open-book exam through Brightspace. To participate in the online exam you must be enrolled for this course in Brightspace. For further practicalities regarding any online examinations, you will be informed through Brightspace. A successful participation in the module, requires a pass grade for all components.
Expected prior Knowledge	Relevant BSc in Engineering, MUDE Module, CE Programme Base Module.
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills addressed in this course. Furthermore, students will collaborate intensively with their colleagues in a module-wide group design project thereby developing organizational and interpersonal skills. In addition, students develop written communication by presenting and reflecting on their designs in a compact design report, requiring to focus on the essentials of their design and their work.
Literature & Study Materials	The lecture material for this course consists of the lecture slides, lecture notes, papers and chapters from relevant literature, a basis for design document for the group design project and the stick-model hand-out, which will all be available through Brightspace. Additional reading material that provides useful information in support of the course are: > The books Handbook of Bottom Founded Offshore Structures, parts 1 and 2, by prof.dr.ir. Jan H. Vugts, available online through the website of the Publisher Eburon. > The book Hollow Sections in Structural Applications, by prof.dr.ir. J. Wardenier, available at VSSD.
Judgement	See Assessment. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	The written exam is an open-book exam, meaning that any books, slide print-outs, personal notes, i.e. any written or printed material is permitted during the exam sessions. However, to limit the amount of paper waste, a maximum of 10 A4-pages (written or printed) are allowed during the exam. This rule does not include books. Additionally, it is allowed and advised to bring a scientific calculator to the exam. Nevertheless, any devices that allow for communication are not allowed, i.e. no laptops, pads or phones.

CIEM4210	Marine Renewables	15
Responsible Instructor	Dr. G. Lavidas	
Instructor	B.C. Ummels	
Instructor	Dr. G. Lavidas	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	Dr. A. Antonini	
Instructor	Dr.ir. D.J.N. Allaerts	
Instructor	J.O. Colomes Gene	
Contact Hours / Week x/x/x/x	16-18 hours a week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module concerns the design, monitoring and assessment of offshore wind and ocean energy farms, in general, and different technologies (both bottom-fixed and floating), specifically, in an integrated manner, including control, installation, maintenance and economics. Regarding the development of offshore energy farms, this means that all relevant aspects and stakeholders are addressed and their implications for design assessed. From a structural perspective, this implies that the different technologies should be understood as a system as a whole and the varying environmental interactions aerodynamic, hydrodynamic and soil be considered in conjunction. The content of the module contributes to the MUDE and Ethics learning lines.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Create offshore renewable solutions based on future natural and societal needs, using available scientific knowledge (HOS- 1, 3). MLO-2. Evaluate offshore renewable structures in interaction with the surrounding environment, using available scientific knowledge (HOS-2, 3). MLO-3. Convey quantified uncertain outcomes corresponding the design of hydraulic and offshore structures (HOS-4). MLO-4. Propose new ways for dealing with ethical issues within the field of hydraulic and offshore structures from the perspective of different relevant moral frameworks (HOS-4). MLO-5. Conduct a solution-oriented debate with stakeholders about ethical issues within the field of hydraulic and offshore structures, while acknowledging real-world constraints in terms of technological possibilities, governance and institutional options (HOS-4).	
Education Method	Blended	
Assessment	The module is assessed at the end of Q4: - Group assignment: 80% - Individual exam: 20% Relation between assessment module units (weighing) - HOS-B-1 Probabilistic Design: 20%, - HOS-B-2 Numerical Modelling: 25% - HOS-B-3 Offshore renewable Structures: 55% A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	BSc Civil Engineering or Applied Science (or equivalent). MUDE Module, CE Programme Base Module, HOS Track Base Module and HOS Modules A1 or HOS Module A2	
Academic Skills	Engineering, Group-work, project management, coding	
Literature & Study Materials	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well.	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Pen & calculator	
Collegerama	No	

CIEM4220	Dams, Dikes and Breakwaters	15
Responsible Instructor	Dr. A. Antonini	
Instructor	Prof.dr.ir. S.N. Jonkman	
Instructor	Dr.ir. B. Hofland	
Instructor	mr.dr.ir. C. Mai Van	
Instructor	Ir. R.E. Jorissen	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	J.O. Colomes Gene	
Contact Hours / Week x/x/x/x	0/0/0/12	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module aims at the engineering design of hydraulic structures that protect against the effects or impact of waves, currents and high-water levels, starting with understanding the underlying processes and related systems. It concerns the design of breakwaters, dikes and closure dams; flood risk reduction strategies, and structures to protect beds, banks and shorelines. The content of the module contributes to the MUDE and Ethics learning lines	
Study Goals	On completion of this module, the student will be able to: MLO-1. Create solutions for dams, dikes and breakwaters based on future natural and societal needs, using available scientific knowledge (HOS-1, 3). MLO-2. Evaluate dams, dikes and breakwaters in interaction with the surrounding environment, using available scientific knowledge (HOS-2, 3). MLO-3. Convey quantified uncertain outcomes corresponding to the design of dams, dikes and breakwaters (HOS-4). MLO-4. Consider ways of dealing with ethical issues within the field of dams, dikes and breakwaters from the perspective of different relevant moral frameworks (HOS-4). MLO-5. Conduct a solution-oriented debate with stakeholders about ethical issues within the field of dams, dikes and breakwaters, while acknowledging real-world constraints in terms of technological possibilities, governance and institutional options (HOS-4).	
Education Method	Frontal lectures	
Assessment	The module is assessed at the end of Q4: - Exercise portfolio: 60% - Individual exam: 40% Relation between assessment module units (weighing) - HOS-B1 Probabilistic Design: 20%, - HOS-B2 Numerical Modelling: 25% - HOS-B4 Dams, Dikes and Breakwaters: 55% A successful participation in the module, requires a minimum evaluation of 5.8 for all assessments	
Expected prior Knowledge	BSc Civil Engineering or Applied Science (or equivalent). MUDE Module, CE Programme Base Module, HOS Track Base Module and HOS Modules A1 or HOS Module A2	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well	
Literature & Study Materials	Lecture notes: Breakwater design, Van den Bos and Verhagen, 2018. Lecture notes: Closure works, Verhagen, 2017. Lecture notes: Introduction to Bed, bank and shore protection. 2nd edition, Schiereck and Verhagen 2019. Lecture notes: Flood defences, Jonkman, Jorissen, Schweckendiek and van den Bos, 2021 Manual: Hydraulic Structures, Voorendt and Molenaar, 2021 Lecture notes: Lanzafame (2023) Probabilistic Design Lecture Notes (online website)	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Lecture note and books	
Collegerama	Yes	

CIEM4230	Floating and Submerged Structures	15
Responsible Instructor	J.O. Colomes Gene	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	J.O. Colomes Gene	
Contact Hours / Week x/x/x/x	appr. 18 hours a week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module addresses the design, monitoring and assessment of floating and (semi)-submerged civil engineering solutions for infrastructural and urban development applications, for instance floating bridges and moored tunnels, including control, installation, maintenance and economics. This implies that the structure should be understood as a system as a whole and the varying environmental interactions aerodynamic, hydrodynamic and soil be considered in conjunction. The content of the module contributes to the MUDE and Ethics learning lines.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Create floating and submerged solutions based on future natural and societal needs, using available scientific knowledge (HOS-1, 3). MLO-2. Evaluate floating and submerged structures in interaction with the surrounding environment, using available scientific knowledge (HOS-2, 3). MLO-3. Convey quantified uncertain outcomes corresponding the design of hydraulic and offshore structures (HOS-4). MLO-4. Propose new ways for dealing with ethical issues within the field of hydraulic and offshore structures from the perspective of different relevant moral frameworks (HOS-4). MLO-5. Conduct a solution-oriented debate with stakeholders about ethical issues within the field of hydraulic and offshore structures, while acknowledging real-world constraints in terms of technological possibilities, governance and institutional options (HOS-4).	
Education Method	This module is based on project-based learning methodology. Students will mainly work on hands-on group activities, supported by online pre-recorded videos and lecture notes.	
Assessment	The module is assessed at the end of Q4: - Group assignment: 60% - Individual exam: 40% Relation between assessment module units (weighing) - HOS-B-1 Probabilistic Design: 20%, - HOS-B-2 Numerical Modelling: 25% - HOS-B-5 Floating and Submerged Structures: 55% A successful participation in the module, requires a minimum evaluation of 5.8 for all assessments.	
Expected prior Knowledge	BSc Civil Engineering or Applied Science (or equivalent). MUDE Module, CE Programme Base Module, HOS Track Base Module and HOS Modules A1 or HOS Module A2.	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well	
Literature & Study Materials	Probabilistic design unit: - Lecture notes: Lanzafame (2023) Probabilistic Design Lecture Notes (online website) - Recommended textbooks: Der Kiureghian, A. (2022). Structural and System Reliability. Cambridge University Press. Moss, Robb Eric S. (2020) Applied civil engineering risk analysis. Springer. Computational Modeling unit: - Lecture notes and online tutorials - Recommended textbooks: Numerical Methods for Engineers by Steven C. Chapra and Raymond P. Canale. Floating and Submerged Structures unit: - Lecture notes	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations No additional material is expected to be allowed in exams. If the instructors consider necessary the use of additional material, this will be communicated to students in sufficient time in advance.	
Colleggerama	Yes	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Structural Engineering (SE)
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CIEM5000	Base: Structural Engineering	15
Responsible Instructor	A. Tsouvalas	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Prof.dr. A. Metrikine	
Instructor	Ir. J.W. Welleman	
Instructor	S.C. Barbosa Nunes	
Instructor	Dr. H.M. Jonkers	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	Dr.ir. M. Ottele	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr.ir. H.R. Schipper	
Instructor	A. Tsouvalas	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. F. Kavoura	
Instructor	A. Jagadeesh	
Instructor	S. Hiller Bernal	
Contact Hours / Week x/x/x/x	xx/14/8/xx Q3 week 1-4 = 8 hours per week Q3 week 5-8 = 6 hours per week	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	In this module, the learning line on Modelling, data monitoring and sensing and Uncertainty and risk analysis is continued from the Faculty base module. Modelling of structures is central in the SE Track, since both mechanics and codes are in their essence models of real structural behaviour. Data and sensing is addressed sideways when it comes to the determination of mechanical properties and behaviour of materials and structures from laboratory experiments. Experimental results are discussed to illustrate e.g. failure phenomena during lectures and to explain the basis of codes and design models. Ethics is reflected through sustainability and circularity that are the major aspects that future structural engineers need to develop in order to preserve environment. Designing skills are important for all structural engineers and in this module the goal is that students embrace principles and good feeling of structural design with traditional construction systems and materials. The module is composed of 2 units: Unit 1 - Sustainable Construction Members and Systems (8EC) and Unit 2 - Mechanics of Slender Structures (7EC). Unit 1 focuses on loads and sustainability aspects of structural design and design and analysis of steel, timber and concrete members. More specifically, the following topics are covered: Identification and quantification of loads on structures. This includes short-term, long term, and moveable loads, and the combination of different types of loads. Design of steel cross sections and members. Influence of properties of base material, production routes and execution tolerances on verification of critical steel cross sections loaded in bending and shear, members loaded in compression and steel bolted and welded connections. Design of timber cross sections and members. Identification of material properties and their use in the design and verification for strength, stiffness and stability at member level. Design and verification of timber joints and connections. Analysis and verification of response of prestressed concrete members exposed to forces, i.e.: axial, bending and shear, in order to meet requirements for ultimate and serviceability limit state. Application of the principles of sustainability and circularity in civil engineering structures. The content on sustainability covers general LCA approaches and system boundaries supported by specific environmental product data, fabrication processes used in construction of structures, railways and roads. Main principles of circularity and reusability will be covered supported by examples of real structures. The content of Unit 1 is divided into 6 themes offered over 8 weeks in Q2 and 8 weeks in Q3 with a workload of 5EC and 3EC, respectively. In Q2, loads and sustainability aspects of structural design and design and analysis of steel and timber members are covered. In Q3, the design and analysis of prestressed concrete systems and connections in steel and timber structures are covered. Unit 2 treats the mechanics of slender structures. In Q2, simple cases are treated in which analytical solutions of single elements are possible. The goal is to explain physics and the basics of the methods. In Q3, the focus is shifted towards more design-type assignments in which the students apply the solution methods in more complex systems consisting of combined one-dimensional structural elements and plates together with complex boundary/interface conditions. The unit covers: Formulation of governing equations of one-dimensional and two-dimensional slender structures. Statics and dynamics of cables, rods, shear beams and beams in bending. Statics and dynamics of plates in bending. Statics and dynamics of combinations of one-dimensional slender structures. Statics and dynamics of plates undergoing in-plane deformations. Statics and dynamics of combinations of one-dimensional and two-dimensional slender structures.	
Study Goals	After completion of this module, students will be able to: 1. Identify and quantify the most important loads on structures, roads and railway and schematize and combine them as input for the determination of the force distributions and deformations. 2. Design critical steel cross sections, members and joints with optimized dimensions and reliable performance against structural and environmental requirements under the influence of single and combined bending, shear, torsion and axial loads. 3. Design and verify structural performance of timber members and connections subjected to single and combined bending, shear and axial loads through understanding of fundamental material properties and force flows in timber joints. 4. Design, analyse and optimize prestressed concrete structures, considering influence of internal stress distribution, time dependent effects, type of pre-tensioning, boundary conditions, prestress losses, cracks and reinforcement details. 5. Understand and apply the principles of sustainability to identify, analyse and evaluate negative effects of civil engineering structures at component and system level on natural environment and society and to use circularity strategies to reduce the environmental impact of structures, railways and roads. 6. Formulate the governing equations of motion of a class of one-dimensional slender structures (beams, cables, rods) and plates subjected to static and dynamic loads in- and out- of plane. 7. Analyse 1D slender structures subjected to static loads and dynamic loads and evaluate the structural response. 8. Analyse plates subjected to static and dynamic loads in- and out- of plane and evaluate the structural response. 9. Create mathematical models of complex structures consisting of one-dimensional elements and plates subjected to static and dynamic loads and evaluate their structural response.	

Education Method	Lectures. Practical sessions. For more information please check the Brightspace page of the module.
Computer Use	Maple, Python. Please check the Brightspace page of the module for more detailed information.
Assessment	Unit 1: There are two written exams, one at the end of each quarter, related to content of those quarters. Knowledge on theoretical background and problem-solving skills are tested by open-ended (non-repeating) questions. The assessment of design and analysis skills are tested through analysis and calculation exam questions that are related to home assignments, but not limited to. Unit 2: Material delivered in Q2 will be assessed on the basis of a written exam. In Q3, the assessment will be on the basis of a final Assignment Portfolio (Report). The weight for the assessments above are as follows: - Sustainable Construction Members and Systems - Written Exam (Q2) 33.3% - Sustainable Construction Members and Systems - Written Exam (Q3) 20% - Mechanics of Slender Structures - Written Exam (Q2) 27.7% - Mechanics of Slender Structures - Report (Q3) 20%
Expected prior Knowledge	A successful completion of the module, requires a minimum evaluation of 5.8 in each individual assessment listed above. The design of Sustainable Construction Members and Systems relies on pre-knowledge of students related to: Uncertainty quantification, probabilistic and stochastic processes. Basic principles of reliability of structure and partial safety factors. Basic knowledge of building principles and construction methods. Basic knowledge of mechanics of structures to determine sectional forces and deformation of schematized structural systems (simple beam, cantilever, frame, etc.) from engineering tables. Knowledge related to problems of stability of 1D members (basic flexural buckling Euler cases). Basic knowledge of mechanics of materials: elastic, plastic behaviour, stress-strain relations, relaxation. Basic design and verification of steel-cross sections subjected to single bending and shear loads. Knowledge of basic material properties of steel, concrete and timber. Basic design of reinforced concrete members (beams and slabs) and verification of ULS and SLS. The required pre-knowledge of students to follow the unit on Mechanics of Slender Structures is based on base faculty module, program base module and existing Bachelor level courses: - Differential Equations (CTC2100) - Dynamics & Modelling (CTC1210) - Dynamics of Systems (CTC2300) - Structural Mechanics 1-3 (CTC1110, CTC1310, CTC2210). More specifically for the structural mechanics part: - Knowledge on computing statically determinate structures, including support reactions and force/moment diagrams. - Basic knowledge on computing structures using the Principle of Virtual Work. - Knowledge on kinematic, constitutive and equilibrium relations for basic deformation cases (extension, bending, torsion). - Knowledge on stress distributions on basic prismatic members and on deformations of trusses and frames. - Knowledge on computing statically indeterminate structures, including the force and displacement methods.
Academic Skills	Analytical thinking; critical appraisal
Literature & Study Materials	Several textbooks and reading notes will be distributed by the lecturers. For more information, please check the Brightspace page of the module.
Judgement	See assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations For the open-book exams, any written text brought by the students is permitted. For other exams, please check instruction on the Brightspace page of the module.
Collegerama	Yes

CIEM5110	Structural Mechanics and Dynamics	9
Responsible Instructor	Dr. E. Lourens	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	F.P. van der Meer	
Instructor	Dr. E. Lourens	
Instructor	C. Shen	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	x/x/9-12/x Contact hours from 9-12 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The A1 module on Mechanics and Dynamics of structures and structural components provides the students with advanced knowledge on mechanics and dynamics of structures which are typically encountered in civil engineering. The structure of the module balances a strong theoretical basis with application cases relevant to the structural engineer. By following this module, students will learn how to analyse the stability of structures to both static and dynamics loads. They will get acquainted with the various computational techniques used in the static and dynamic analysis of complex structural systems and they will learn how to deal with measurements and analysis of vibrations of engineering structures. The learning line on Modelling, Data and Uncertainty is continued from the Faculty Base and Track Base module. In the third unit of this module (SE-A1-3), the student will extract modal parameters from spectral estimates obtained using measurement data from self-designed set-ups in the laboratory, quantify and interpret the uncertainty in these estimates and modal parameters, and statistically judge consistency between mode shapes. Also comparisons will be made between the estimates and their modelled equivalents as obtained in the second unit. The learning line is addressed explicitly in the LO's 2, 3, 7 and 8 of the third unit. In the first unit, the learning line on design will be continued from the preceding modules. In this unit students will learn to explain the criteria for designing stable structures. A range of instabilities that needs to be considered in a conceptual design phase is considered, including buckling, plastic collapse of frames, galloping, flutter, and vortex-induced vibrations.	
Study Goals	On completion of this module, the student will be able to: 1. Analyse the stability of structures subjected to both static and dynamics loads. 2. Explain the theoretical background of finite element formulations for structural analysis. 3. Perform finite element analysis to analyse the force distribution, stability and dynamic response of frame structures. 4. Analyse and interpret vibration measurements to identify modal parameters, including their associated uncertainty. 5. Design a measurement set-up for vibration analysis, with the purpose of quantifying modelling and measurement uncertainties.	
Education Method	Differs per unit. Please consult the Brightspace page.	
Assessment	Stability of structures - Written Exam (33%) Finite Elements for Structural Analysis - Written Exam (33%) Measurement and Analysis of Vibrations Report (34%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis. Coding skills: Python (MUDE level) Partial Differential Equations, Statics, Beam theory (BSc level) Numerical methods, Introduction to Finite Elements (MUDE level) Signal processing (MUDE level)	
Academic Skills	Skills that students will be working on in this course are: - Analytical and critical thinking - Interpretation of results - Performing laboratory experiments - Data analysis - Writing professional reports	
Literature & Study Materials	1. V.V. Bolotin. The dynamic stability of structures. San Francisco, Holden-Day (1964). 2. M. Paidoussis. Fluid-structure interactions: Slender Structures and Axial Flow. Academic press (1998). 3. Thomsen JMT. Vibrations and Stability, Advance Theory, Analysis and Tools. 2nd Edn. Verlag: Springer (2003). 4. Dewey H. Hodges, G. Alvin Pierce: Introduction to Structural Dynamics and Aeroelasticity. Cambridge University Press (2011) Noise and Vibration Analysis, Anders Brandt; Fundamentals of Signal Processing for Sound and Vibration Engineers, Kihong Shin & Joseph Hammond; Modal testing, Ewins	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Unit-specific.	
Colleggerama	No	

CIEM5120	Design of Structural Components	9
Responsible Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	S.C. Barbosa Nunes	
Instructor	Prof.dr.ir. P.C. Louter	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Prof. M. Veljkovic	
Instructor	A.T. Gebremariam	
Instructor	Dr. F. Kavoura	
Instructor	Dr. M.A. Popescu	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The A2 Module focuses on the Design of Structural Components. Potential for reducing environmental impact of structures and achieving an optimum structural design lies in integrating construction materials in structures where they best fit, given their mechanical and functional properties. By successfully completing this module, you will learn to understand the possibilities and limitations of both conventional and innovative construction materials, select and or adjust their properties according the functionality of the structures. You will learn to design structural members which combine the advantages of various materials such as steel, concrete, FRP or timber. Designing future structures, using innovative construction materials, optimised for circularity (prefabricated and demountable constructions) will be emphasised. The module consists of two units: (1) Prefabricated and composite structures and (2) Design with innovative materials. You will learn to design prefabricated and composite structures made of conventional construction materials, and will also explore improvements with novel construction materials, (composite) systems and fabrication methods. A2-1: Prefabricated and composite structures: The content is divided into 4 themes offered in Q3. More specifically the envisioned content is as follows: Theme 1: STEEL Design verification of steel members (beam-columns) exposed to interaction, axial force and transversal load Hollow section steel structures; Main topology of joints Steel-concrete composite structures; Principles of design; prerequisites and consequences of full and partial composite action; Theme 2: TIMBER Engineered wood products Massive timber buildings and timber frame housing Connections in prefabricated timber structures Timber trusses and portal frames Timber concrete composite structures Theme 3: CONCRETE External prestressing Strut-and-tie models and detailing Prefabricated elements for building and infrastructure Theme 4: RAILWAY Conventional track structures and innovations, such as prefab concrete slab tracks; prestressed concrete sleepers, timber sleepers, composite sleepers; composite ballast bed Modular in railway infrastructure: current experiences and perspectives A2-2: Design with innovative materials: The content deals with a wide overview of innovative construction materials, their applications, functional, sustainability and safety benefits and basic principles of design. The unit is offered in Q3. More specifically the envisioned content is as follows: Theme 1: HIGH PERFORMANCE CONCRETE Introduce new types of cementitious materials ((SHCC), UHPFRC, Geopolymer concrete, auxetic, phase change, self-sensing, self-healing, 3d printed, recycled) Theme 2: RECYCLED MATERIALS Introduction of recycled construction materials and their circularity Adaptive design for easy dismantling and reusing structural materials Challenges and opportunities. Theme 3: HIGH PERFORMANCE STEEL AND COMPOSITE STRUCTURES Weathering and stainless steel; High strength steel Composite materials and members Theme 4: STRUCTURAL GLASS basic information on how structural glass is used and what the material properties are, basic calculations Theme 5: DIGITAL MANUFACTURING 3D printed steel (material, members, connections) 3D printed concrete; Wood -plastic composites. Theme 6: INNOVATIONS IN PAVEMENT ENGINEERING Introduction of pavement structure with emerging materials Emerging materials to enhance performance and functionality of pavement Theme 7: INNOVATIONS IN RAILWAY ENGINEERING Conventional track structures and innovations, such as embedded rail (elastomeric materials), polymer sleepers, coated ballast.	
Study Goals	On successful completion of this module, you will be able to: LO1. Explain the principles to design prefabricated and structural (composite) components in a circular manner for buildings and infrastructure by using conventional and novel civil engineering materials. (SE1, SE3, SE5, SE8) LO2. Apply the advanced design aspects in structural engineering, focusing on stability problems, understanding and designing composite members, connections, detailing, external prestressing and prefabricated members with conventional construction materials (timber, steel, concrete). (SE1, SE3, SE6) LO3. Design, analyse and optimize composite structural members, e.g. steel-concrete, timber-concrete, timber-steel beams, decks and columns, with respect to short and long term loads and environmental conditions. (SE1, SE3, SE7) LO4. illustrate the influence of connections and detailing aspects in prefabricated and composite structures and apply this in the design and verification. (SE1, SE3, SE5) LO5. Identify the innovative, high performance and sustainable civil engineering materials and simulate their behaviour. (SE4, SE5, SE6, SE8, SE9) LO6. Discuss limitations and advantages of applying novel materials, advanced manufacturing and innovative systems in design of structural members and optimize design accordingly. (SE7, SE8, SE9) LO7. Discover and justify the possibilities of applying innovative materials in real engineering problems to improve the design and behaviour of the structure or the road and railway system. (SE1, SE3, SE5, SE6, SE7, SE8, SE9)	
Education Method	-Lectures in lecture halls, based on lecture notes, parts of lecture books, readers etc. - Quizzes and discussions in lectures -Self study of lecture material -Writing group reports, cooperating in groups. - Performing literature search and interpreting this literature for group reports	
Assessment	Summative assessments are all done on unit level. First unit has two summative assessments, one assignment (which accounts for 20% of the module grade) and a written exam (which accounts for 45% of the module grade). The assignment consists of parts on timber, steel and concrete and has to be submitted complete including all 3 parts. Assessment of the second unit is done on basis of a project (which accounts for 35% of the module grade). The total grade for the module is calculated based on total contribution of these 3 summative assessments. A successful participation in the module, requires a minimum evaluation of 5.8 for all partial assessments.	
Expected prior Knowledge	The module relies on pre-knowledge of Sustainable Construction Members and Systems (from CE-SE Track Base in Q2) A2-1: Prefabricated and composite structures: Assessment is based on Witten exam and final assignment reports. In the written exam, knowledge on theoretical background and problemsolving skills are tested by open-ended (non-repeating) questions. The assessment of design and analysis skills and collaborative work are tested through the assignments that can be done partially individually (calculation and analysis) and partially in the group (design aspects), depending on the number of students. The preferred option is that it is done individually.	

	The summative assessment will be on the basis of final report related to collection of three parts on timber, steel and concrete. All parts have to be submitted complete to obtain a grading result. The written exam can contain questions related to analysis and calculation exam questions that are related to home assignments, but not limited to. A2-2: A2-2: Design with innovative materials: The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam and the quality of the final assignment reports. Individual knowledge on theoretical background and problem-solving skills are tested by open-ended (non-repeating) questions in oral exam (group presentation). The assessment of design and analysis skills and collaborative work are tested through group project. The assessment will be on the basis of final report related to collective of (group) project.
Academic Skills	- To be able to discuss content with teachers and fellow students - To be able to search for literature on specific topics and be able to interpret these - To be able to write clear and consistent reports - To be able to prepare for exam questions that require insight on an academic level
Literature & Study Materials	- Lecture notes - (parts of) text books - papers
Judgement	See Assessment
Permitted Materials during Exam	- calculator - equation sheet

CIEM5130	Design of Civil Structures and Infrastructures	9
Responsible Instructor	Dr. K. Anupam	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr. Y. Yang	
Instructor	Dr.ir. H.R. Schipper	
Instructor	Dr. K. Anupam	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. M. Pavlovic	
Instructor	C. Noteboom	
Instructor	M.P. Felicita	
Instructor	Ir. J.P.G. Ramler	
Instructor	S. Hiller Bernal	
Contact Hours / Week x/x/x/x	0/0/14/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The SE_A3 module teaches you how to design engineering (infra)structures within a realistic context: from scratch, investigating multiple variants and materials to find the most suitable design. You will make a functional and technical design, with considerable level of detail. A3 will be organised as an engaging module: during the first weeks of the module keynote lectures from industry and expert staff members will be organised; broad and advanced content on design methodology, and the design of buildings, bridges, rail track and roads/pavements. in parallel you will be trained in various (soft/hard) skill-related lectures/workshops (teamwork, parametric design, presentation, setting up a clear calculation, visual communication) the central task is a group assignment around a realistic clients brief, containing buildings, bridges, rail and/or road, and including an individual contribution of each team member in the form of a preliminary design of one structural type. for deepening your knowledge on subjects, which is necessary for your individual contribution to the team design, various self-study content will be offered from a learning pool. In A3, attention will be given to ethical aspects of design of engineering structures by means of addressing a realistic design context and working in a realistic team atmosphere. Considerations related to ethics are: Which team member will take responsibility for which part of the design? Can we guarantee that the design is safe? Is this design addressing the societal need in a correct way? To what extent is our global environment loaded by the presented design? To what extent will people be affected by the construction process of this design? Can the design be constructed safely and how? The Core faculty module (CFM) will provide the fundamental basis for ethics, on which the course will build forth. Structural Design is also about optimisation and graphical representation CFM content that prepares for this will be built upon. This module also considers topics that will continue the line of both modelling and coding from the MUDE module. MUDE: Coding skills (Python, Matlab etc.) and MUDE on modelling, data and uncertainty analysis are pre-requisites. This module will follow the lines of MUDE by analysing advanced methods with application on bridges, building, road, railway structures and circularity. The MUDE course in this aspect prepare students to understand reliable design parameters, criteria and reliable design choices out of several options.	
Study Goals	On successful completion of this module, you will be able to 1. Integrate steps of conceptual and preliminary design phases of (infra)structures to design a (infra)structure from scratch taking into account the demand of the client and the society, the interaction between the structure and its environment and the need of lifetime maintenance, including design for circularity, while quantifying uncertainties in the brief, the boundary conditions, the loads, the design, and the construction phase (SE3, SE5). 2. Select optimal structural systems (traditional or innovative) for specific design task of buildings and bridges and combination of construction materials steel, timber and concrete, to be able to realize which options can be chosen for the structural system, taken into account the uncertainty in the end-use, lifespan and degradation mechanisms.(SE1, SE4, SE7) 3. Explain how to analyse the interfaces and interaction of track infrastructure and train and their effects on the railway system and the various influencing structural pavement design parameters on different supporting structures (SE1, SE4, SE7) 4. Explain principles of parametric geometry design and the uncertainty in tolerances that have to be taken into account, and apply these principles with appropriate software tools cooperatively in a design team.(SE1, SE7, SE9). 5. Examine transfer of forces in different structural systems and with different structural materials starting from the applied load down to the foundation. 6. Design the structural systems consisting of structural members and joints with appropriate dimensions and materials in a realistic design context (SE1, SE2, SE9) 7. Learn to work in a design team and to make and visualize and present a functional design of a site including buildings, bridges, rails, and roads (SE9) 8. Develop general design of a building, bridge, railways and road structural system to a detailed design level through cooperation within a design team. (SE1, SE2, SE3, SE4, SE5, SE9)	
Education Method	A3-1: Traditional lectures, possibly with use of blended-learning materials or flipped-classroom pedagogy A3-2: Self-study during week 3-10 Availability of lecturers during Q&A-sessions or office hours A3-3: Problem-based learning, discussion in teams, tutor sessions (individual or with full group)	
Assessment	The assessment is as follows: 1. General design principles of (infra)structural systems - Written Exam (35%) 2. Specific design aspects of buildings, bridges, rail and road structural systems - Report (25%) 3. Design challenge - Project (40%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	

	A3-1: Written exam (digital midterm exam via Möbius) A3-2: The level of the submitted work for the design assignment (SE-A3- Learning Unit 2) will be assessed against the frame of reference and learning objectives of the available self-study content. In this assessment, students will be assessed by means of some prescribed questions about their learnings of A3-2. The assessment of A3-2 will be done both in formative and in summative way. In the formative assessment, students will be asked to peer-review each others answer sheet and prepare a report on the assessment. In the second step, students will be asked to submit these questions (individual summative assessment and the peer review report). This means that the final report that the students will submit will contain two parts (a chapter containing A3-2 & peer report; and second report of A3-3). While carrying out the final assessment, attention will also be given on the improvements that the students made on the A3-2 questionnaire on the basis of peer review comments. The description of A3-3 assessment has been provided later in the document. A3-3: A3-3 module will be evaluated as a project on the basis of a final report and group presentations. The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam (presentation) and the quality of the final assignment reports. This submodule will be assessed against the frame of reference and learning 259 objectives of A3-3. Final report (individual and group part) and presentation will be rubrics in one summative assessment format. Final report (80 points): It is expected that each team will provide a report containing chapters both from their individual learning together with their teams final design. Please note that since one student per group will be a leader of a specific topic, this will provide the assessor a mean to judge the performance of the individual student. 25 points will be given on the basis of the overall report, whereas 55 points will be given on the basis of the individual chapter. Group presentation (20 points): A group presentation per team will be scheduled at the end of A3-3 sub module. This will be an overall evaluation of the teams final design choices. Students are expected to defend these choices in from of team of assessors.
Expected prior Knowledge	Basic and advanced structural mechanics (from BSc and SE Track Base Q2) - advanced knowledge on loads on structures (from BSc and SE Track Base Q2) - advanced knowledge of reinforced concrete members and basic knowledge of prestressed concrete and joints/connections (from BSc and SE Track Base Q2) - advanced knowledge of steel members and basic knowledge of joints/connections (from BSc and SE Track Base Q2) - advanced knowledge of timber members and basic knowledge of joints/connections (from BSc and SE Track Base Q2) - basic knowledge of circularity and sustainability (from BSc and SE Track Base Q2) - basic skills in CAD, teamwork, report writing, framework software, hand sketching, information literacy (from BSc ctc1000) - basic skills in structural design (BSc ctc1410 and ctc2320) - basic knowledge of road & railway design (BSc ctc3320) - knowledge of integral design and management of civil engineering projects (BSc ctc1220) - MUDE: Coding skills(Python, Matlab etc.) and MUDE on modelling, data and uncertainty analysis
Academic Skills	Team-based work, planning a design task, setting up a clear calculation, convincing visual presentation, oral presentation, report writing, sketching, self-study (problem-based-learning) that is functional to role and responsibility in team
Literature & Study Materials	Content on convincing visual presentation Content on setting up a clear calculation Content on team-based working Content on sketching Content on example details and drawings Quick Reference: Content on principles of design through phases Delft design Method"
Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Calculator and a Formula sheet.
Collegerama	No

CIEM5210	Applied Mechanics of Structures	15
Responsible Instructor	F.P. van der Meer	
Instructor	Ir. C. Kasbergen	
Instructor	Prof.dr.ir. L.J. Sluijs	
Instructor	Dr.ir. P.C.J. Hoogenboom	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	A. Varveri	
Instructor	R. Esposito	
Instructor	Dr. K. Anupam	
Instructor	Dr. A.A. Mehrotra	
Contact Hours / Week x/x/x/x	0/0/0/18	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module provides students with in-depth expertise on the analysis of the mechanical response of structures. Analytical and computational techniques are explained and applied. The module offers students the possibility to become applied mechanics specialists with the ability to use state-of-the-art computational tools and analytical tools across the structural engineering field. The module also prepares students to contribute to academic research that advances the limits of computational and analytical tools for analysis of the mechanical response of structures. Monitoring & data science learning line: This module builds on the modelling and coding line from the MUDE, with Unit 1 focused exclusively on modelling and Unit 2 addressing a specific aspect of modelling of structures including python coding workshops. The broader theme of uncertainty is addressed in discussions of the reliability of different models. Ethics learning line: Ethics is given attention in this module when the limits of different analysis approaches are discussed. From research perspective, being critical towards the accuracy and reliability of available methods and tools is a first step towards advancing their limits. In a pragmatic engineering environment, having the expertise to be critical of models may lead to moral dilemmas: What level of uncertainty about model reliability is acceptable? Having discussed the uncertainties of relatively high-fidelity models, what is a the good way to deal with even larger uncertainties of simpler modeling approaches? Moreover, ethical aspects of reporting in academia and engineering practice will be discussed in the module, such as acknowledgement of sources and fair representation of results. Design: This module equips students to analyse the behaviour of materials, structures and structural elements. Analysis is an important aspect of the design cycle for civil engineering structures. Units Unit 1 and Unit 2 focus primarily on methods and their underlying formulation. Unit 3 is more applied in the sense that it provides insight in the mechanical response of structures. It will be illustrated with practical examples how this insight can be used in the design of structures. Pitfalls and best practices in structural design will be discussed in the light of the theory provided in the unit. Unit 1 Computational modelling of structures: Introduction to the use of the finite element method; Nonlinear finite element analysis of steel structures; Nonlinear finite element analysis of structures made of a quasi-brittle material; Application of the finite element method to solve engineering problems. Unit 2 Advanced constitutive modelling: Mathematical and mechanical preliminaries; Computational plasticity; Fracture models and continuum damage mechanics; Large deformation mechanics; Computational modelling of visco-elastic materials; Large deformation elasto-plasticity; Introduction to thermodynamics. Unit 3 Advanced mechanics of structural elements: Theory of elasticity: energy principles, closed form solutions and limitations; Theory of plasticity: frame analysis and plate analysis	
Study Goals	On completion of this module, the student will be able to [1] Analyze the mechanical response of structures where that is complicated by the presence of geometric and material nonlinearity [2] Calculate the ultimate load of plates and frames accounting for plasticity [3] Explore the limits of analytical tools for evaluation of elastic stress and strain [4] Formulate, implement and test constitutive laws for detailed numerical analysis of the response of structures [5] Perform nonlinear finite element analysis for structural design verification Unit 1: 1. Select an appropriate mechanical model that idealizes the physical problem of analysis of a structure (boundary conditions, loading, dimension reduction, geometric order of approximation, material model) [1,4] 2. Build an appropriate finite element model that discretizes the selected mechanical model [1, 5] 3. Carry out nonlinear finite element analysis for a range of different applications (steel/concrete structures, seismic response of buildings) [1,5] 4. Interpret finite element solution results to solve engineering problems [1,5] 5. Create a technical report of a performed finite element analysis [5] Unit 2: 1. Select an appropriate constitutive model to account for physical phenomena of time-dependent behavior, direction dependent behavior, permanent deformations and failure in the analysis of material response [1,4] 2. Derive and apply the stiffness relations for orthotropic elasticity and classical laminate theory [1,4] 3. Understand and apply the concepts of large deformation mechanics [1] 4. Explain the concepts of thermodynamics and apply it to large deformation elasto-thermal material modelling [1,4] 5. Explain and apply concepts of elasto-plasticity, visco-elasticity, viscoplasticity in small and large deformation formulations [1,4] 6. Formulate constitutive models to describe the nonlinear response of materials 7. Implement constitutive models as user material in finite element code [1,4] 8. Verify consistency of a user material implementation with its intended behavior [1,4] Unit 3: 1. Reproduce the main steps in the derivation of analytical elasticity solutions, for example the point load on a half space, point	

	load on a half disk, stresses around a spherical cavity in a continuum, stresses around a circular opening in a plate, moments in circular plates, stresses in curved beams, the differential equations for torsion [3] 2. Demonstrate understanding of the physical meaning of these analytical solutions [3] 3. Argue about the usefulness and limitations of these solutions and of elasticity theory in general [3] 4. Explain the upperbound and lowerbound theorems for plastic collapse [1,2] 5. Calculate, by hand, the ultimate load of elastic/plastic frames with or without accounting for moment-normal force interaction [1,2] 6. Calculate, by hand, the ultimate load of elastic/plastic plates [1,2]
Education Method	Unit 1: Lectures, computer lab sessions Unit 2: Lectures, computer lab sessions Unit 3: Lectures
Assessment	Unit 1: The unit will be assessed in an oral exam covering the content of the lectures as well as the assignments (27% of the module grade). The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam and the quality of the final assignment reports. Reporting skills are also taught and assessed. Unit 2: The unit will be assessed in an oral exam covering the content of the lectures as well as the assignments (33% of the module grade). The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam and the quality of the final assignment reports. Unit 3: Students receive a grade based on peer feedback from presentation sessions (4% of the module grade). Final assessment is done with a written exam (36% of the module grade). A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Numerical Analysis. Coding skills: Python (MUDE level) Mechanics of slender structures (SE track base) Numerical methods, Introduction to Finite Elements (MUDE level)
Academic Skills	Analytical thinking, coding, reporting, presenting.
Literature & Study Materials	Unit 1: Lecture slides, DIANA user manual Unit 2: - R. de Borst and L.J. Sluys, (2001) Computational methods in non-linear solid mechanics - G.A. Holzapfel (2000), Nonlinear Solid Mechanics: A Continuum Approach for Engineering, John Wiley & Sons Ltd, ISBN 978-0-471- 82319-3 - Brannon, R.M. (2003), Functional and Structured Tensor Analysis for Engineering, UNM Book Draft, (https://my.mech.utah.edu/~brannon/public/Tensors.pdf) Unit 3: - Vrouwenvelder, A.C.W.M. and Witteveen, J. (2003) "Plastic Analysis of Structures, The plastic behaviour and the calculation of beams and frames subjected to bending", Lecture notes Delft University of Technology. - Vrouwenvelder, A.C.W.M. and Witteveen, J. (2003) "Plastic Analysis of Structures, The plastic behaviour and the calculation of plates subjected to bending", Lecture notes Delft University of Technology. - Blaauwendraad, J. (2001) Theory of Elasticity Direct methods. Lecture notes Delft University of Technology - Blaauwendraad, J. (2001) Theory of Elasticity Energy principles and variational methods. Lecture notes Delft University of Technology - Handouts on selected topics
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	None
Collegerama	No

CIEM5220	Applied Dynamics of Structures	15
Responsible Instructor	Prof.dr. A. Metrikine	
Instructor	Ir. C. Kasbergen	
Instructor	Dr.ir. M.J.M.M. Steenbergen	
Instructor	Dr.ir. K.N. van Dalen	
Instructor	R. Esposito	
Instructor	A. Tsouvalas	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	Dr. G. Lavidas	
Contact Hours / Week x/x/x/x	0/0/0/18	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Study Goals	Overall Module ILOs: 1. Describe conceptually the earthquakes, wind, wave and vehicle loads on structures (SE1, SE3). 2. Formulate and interpret the physical foundations of mathematical models for the dynamics of structures under earthquakes, wind, wave and vehicle loads (SE1, SE3). 3. Describe, model and analyse the earthquakes, waves, wind, and rail and road irregularities as random signals (SE1, SE3). 4. Understand and apply analytical methods of mathematical analysis of the dynamic response of structures to earthquakes, wind, waves and moving loads (SE1, SE3). 5. Describe and explain the main mechanisms of vibration excitation of structures by moving loads (SE1, SE3). 6. Analyse the dynamic response of structures in earthquake-prone regions and use it in design considerations (SE2, SE5, SE6). 7. Design structures for renewable energy generation based on their response to wind and waves (SE2, SE5, SE6). 8. Compute and evaluate the dynamic response of bridges to moving loads and use it in design considerations (SE2, SE5, SE6). 9. Compute and evaluate the dynamic response of rigid and flexible pavements at the tirepavement interface and use it in design considerations (SE2, SE4, SE5, SE6). 10. Compute and evaluate the dynamic behaviour of a railway track and its environment under train loading and use it in design considerations (SE2, SE4, SE5, SE6) B2-1: 1. Explain basic seismological concept (SE_B2_1,2). 2. Analyse recorded ground motions and distinguish local site effects (SE_B2_3). 3. Formulate the equations of motion of NDOF systems subjected to ground excitation and critically assess their correctness (SE_B2_2). 4. Apply different methods of seismic analysis in structures (linear and non-linear response) and analyse the results (SE_B2_4,6). 5. Consider soil-structure interaction and fluid-structure interaction in seismic analysis of structures (SE_B2_4,6). 6. Analyse and design structures in earthquake-prone regions using code provisions (SE_B2_4,6). B2-2: 1. Describe wind and waves stochastically (SE_B2_1,2,3). 2. Formulate and interpret the physical foundations of mathematical models for the structural wind and wave loads on structures (SE_B2_2). 3. Formulate and evaluate the equations of motion of NDOF systems subjected to wind and wave excitations (SE_B2_3,4). 4. Define wind and wave interactions with basic structural concepts (SE_B2_3,4). 5. Analyse the dynamic response of simple structures to wind and wave excitations (SE_B2_3,4). 6. Describe the phenomena of galloping, flutter and vortex-induced vibrations (SE_B2_1,4,7). 7. Design and analyse the dynamic response of structures for renewable energy generation from wind and waves (SE_B2_7). B2-3: 1. Describe and formulate mathematical models for the structural elements and structures in interaction with moving loads (SE_B2_1,2,3,4). 2. Describe and explain the major types of structural and environmental responses to moving loads (SE_B2_1,5). 3. Derive and analyse the response of 1D structures and 3D layered continua to moving loads (SE_B2_7). 4. Design a concept bridge model under moving loads and evaluate its dynamic response (SE_B2_8). 5. Design rigid and flexible pavements under moving loads and evaluate their dynamic responses and of their environment, taking into account the stochastic nature of the loading (SE_B2_9). 6. Compute and evaluate the dynamic behaviour of a railway track under train loading and design environmentally friendly tracks, taking into account the stochastic nature of the loading (SE_B2_10). 7. Describe the degradation mechanisms of road pavements and railway tracks (SE_B2_10) based on targeted monitoring.	
Education Method	B2-1: Lectures, discussion of assignments, lecturer-assisted computational exercises. B2-2: Lectures, discussion of assignments, lecturer-assisted computational exercises B2-3: Lectures, discussion of assignments, lecturer-assisted computational exercises.	
Assessment	One summative assessment on module level (all units) is based on an assignment that encompasses the basics of the random vibrations and numerical modelling (both subject matters are taught to all students of the module). In addition, at the end of the quarter, one written exam per unit is held. The total number of summative assessments in this module is four. The assessment of the module is as follows: 1. Structural Response to Earthquakes - Written Exam (25%) 2. Dynamics of structures subjected to wind & waves - Written Exam (25%) 3. Dynamics of structures under moving loads - Written Exam (25%) 4. Applied Dynamics of Structures - Group Assignment (25%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	- structural mechanics at the BSc level; - formulation of equations of motion for NDOF systems and onedimensional continuous structural systems (cables, beams, rods); - solution of ordinary differential equations at the BSc level; - modal and Fourier analysis of undammed and damped discrete and continuous systems.	
Academic Skills	The following skills will be further developed in this module: critical and interdisciplinary thinking, identification of problem-oriented solution procedures, information collection and synthesis, application of theory to new situations using the relevant equations and correlating mathematical solution with physical insights and design procedures, information collection and synthesis, use of data for critical assessment of modelling assumptions and design considerations.	
Literature & Study Materials	B2-1: Material distributed by the lecturer: readers of CIE5260 and CIE5340 (adjusted to the new learning unit); Dynamics of Structures: Theory and Application to Earthquake Engineering by Anil K. Chopra; Maple files and Matlab (Python) scripts for seismic analysis (prepared by the teaching team). B2-2: Lecture slides and notes, relevant book chapters B2-3:	

Judgement

Lecture slides and notes, book chapters (Fryba, Vibration of solids and structures under moving loads; Yang, Yau and Wu, Vehicle-bridge interaction dynamics).

See Assessment

For more information on grading, see article 14 in the Rules and Guidelines (RGBE):

<https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations>

CIEM5230	Concrete Structures	15
Responsible Instructor	Dr. Y. Yang	
Instructor	G. Ye	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	Dr. Y. Yang	
Instructor	Ir. E.O.L. Lantsoght	
Instructor	R. Esposito	
Instructor	M. Lukovi	
Instructor	M. Poliotti	
Contact Hours / Week x/x/x/x	0/0/0/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Concrete is the worlds mostly used construction material. The module Concrete Structures offers a comprehensive cluster of courses that, through the application of existing knowledge and insights in new scientific developments, prepares students for the current and future challenges of society relating concrete structures. Various sectors of Civil Engineering and the entire life-span of concrete structures are covered: from the understanding and design of concrete material, the advanced modelling of concrete members taking into account the unique behaviour of concrete, to the design, maintenance of complex concrete structures, such and concrete bridges. The course will include 5 units, namely: B3-1: Advanced modelling of concrete structures. B3-2: Concrete science and technology. B3-3: Numerical modelling of concrete structures. B3-4: Concrete structures under special loadings. B3-5: Concrete bridges. On completion of this module, the student will be able to: 1. Design concrete structures and members using plasticity theory and fracture mechanics for concrete. 2. Explain and design cement-based building materials and the advanced concrete mixture design methodology, further apply to design, control, and execute durable and sustainable concrete in practice. 3. Create, carry out, interpret, and document nonlinear finite element analysis of reinforced concrete structures or elements. 4. Model the behaviour of concrete members and structures subject to various environmental and anthropogenic loads, and to create appropriate design accordingly. 5. Design and assess precast and cast-in-situ concrete bridges and bridge elements considering the construction methods and the loading conditions.	
Study Goals	A breakdown of the learning objectives in each unit given in the course is explained as following: B3-1: 1. Interpreted simplified theoretical models with limit state theorems of the theory of plasticity 2. Design and assess various structural types using limit analyses, the types of structures include beam, wall, slabs, and joints. 3. Point out the fracture process of concrete and its difference between linear elastic fracture mechanics. 4. Outline the influence of the local fracture process of concrete on the global structural behaviour in terms of fatigue behaviour, shear and punching shear behaviour, etc. B3-2: 1. Outline the fundamentals of concrete materials 2. Explain the mechanisms and the chemical process relate to the strength development and the resultant mechanical properties 3. Control volume change during the hardening process 4. Design concrete mixture and advise corresponding execution methodology. B3-3: 1. Create appropriate finite element models for reinforced concrete structures or elements 2. Interpret nonlinear finite element analysis results 3. Explain how to account for uncertainties when using finite element analysis 4. Compile adequate finite element analysis reports. B3-4: 1. Explain the behaviour of concrete structures under temperature effects in the early stage of hardening and hardened concrete, including (technological and structural) measures to influence this behaviour. 2. Discuss the difference between structural response under external loads and imposed deformations, and design concrete structures subjected to imposed deformations. 3. Design and execution of storage systems and reservoirs in reinforced and prestressed concrete, focusing on tightness criteria. 4. Design of concrete protective structures under extreme loads, e.g. impact, blast, fire, cryogenic loads. 5. Design concrete structures for cyclic loading with various amplitude and duration such as seismic loading and fatigue loading. 6. Outline the main environmental loads and possible deteriorations of concrete structures. B3-5: 1. Outline the considerations of designing a concrete bridge in the Netherlands. 2. Evaluate the advantages and disadvantages between different construction methods (cast in situ) and able to explain the relationship between the construction method and the alignment of the reinforcement and/or prestressing cables. 3. Apply the different traffic load models specified in the Eurocode. 4. Apply the methods of load distribution. 5. Evaluate the different alternatives and analyse and calculate roughly the main dimensions of the bridge. 6. Choose the different types of bearings and expansion joints properly.	
Education Method	The course will be given though various education methods, including conventionally in person lectures, self-study with prerecorded video and reading materials, and workshops in the lecture room.	
Assessment	B3-1: 100% from the combined final written exam with unit SE-B3-2 B3-2: 100% from the combined final written exam with unit SE-B3-1 B3-3: 100% from oral exam B3-4: 100% from oral exam B3-5: 100% from oral exam	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Knowledge of material properties of concrete, concrete mix design, mechanics of materials. Design principles and calculation methodes of reinforced and prestressed concrete structural members in both ULS and SLS. Structural analysis and matrix analysis of structures. Understand the safety filosofies and theoretical background of Eurocode.	
Academic Skills	Evaluate and interpretation of results and development of judgemental skills, writing scientific reports, learn how to solve problems, work and discussing in a group, present your results orally, and simplify practical problems into physical models.	
Literature & Study Materials	B3-1: Learning materials, slides and prerecorded videos publishehd on Brightspace. B3-2: Suggested text books and Lecture notes published on Brightspace. B3-3: Lecture slides and selected papers published on Brightspace. B3-4: Textbook and slides published on Brightspace. B3-5: Learning materials, slides and videos published on Brightspace.	

Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	None
Collegerama	Yes

CIEM5240	Steel and Composite Structures	15
Responsible Instructor	Prof. M. Veljkovic	
Instructor	Ir. P.A. de Vries	
Instructor	F.P. van der Meer	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. M. Pavlovic	
Instructor	Dr. F. Kavoura	
Instructor	O. Karpenko	
Instructor	Dr. T. Tankova	
Responsible for assignments	Dr. T. Tankova	
Contact Hours / Week x/x/x/x	0/0/0/24	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module is composed of 3 parallel units. Unit 1 on Strength and Durability of Steel and Composite Structures and Joints provides in-depth knowledge on how to design, model, analyse and critically assess results of prediction models of structural members and joints. The principles and theory of using experiments and reliability-based assisted design and verification of Steel and Composite Structures (SCS) and joints are explained and practised for monotonic and cyclic loading. Unit 2 on Heavy-duty and Unit 3 on Lightweight steel and composite structures will allow students to apply and extend their knowledge from Unit 1 in real case structures, such as bridges, support structures for renewable energy, lightweight FRP structures, cold-formed steel members, etc. The learning line on Modelling, Data management and sensing, risk analysis is continued from the Faculty Base and Track base module. Students will have the opportunity to participate in monitoring and sensing lab (or virtual) experiments during the module and evaluate data. Ethics related to the design and reporting, construction, ownership and decommission of steel and composite structures are discussed throughout the module. Incidences and accidents which lead to a collapse of structures will be discussed from (un)ethical behaviour of partners in a project. The balance between economy-safety-sustainability of construction work will be addressed, showing future engineers' responsibility. B4-1: The content of the units is defined within 3 themes: Theme 1 (30%): Modelling of Steel and Composite structures and design by experiments Elastic, plastic and damage material modelling of steel and FRP; Modelling of local and global elastic buckling; Anisotropic and layered modelling; Evaluation of design values for resistance of various failure modes of steel members. Theme 2 (30%): Design verification of steel connections and joints- Connection component resistance: resistance of bolts, welds, net cross-section in tension and shear; Shear and axially loaded joints Simple connections; Beam to column moment resistant joints; Steel column base joints. Theme 3 (40%): Fatigue and fracture Design of steel structures against fatigue failure: Welded details; Bolted details; Detail classes; SN curves; HSS curves; Post-welded treatment for welded joints; Local prediction models (Hot spot stress approach, SCF, effective notch stress) by FE analysis on examples of hollow section joints, OSD. Fracture mechanics - theoretical background; Material properties for fracture mechanics; Crack initiation and propagation by analytical models and FEA B4-2: The content of the units is defined within 2 themes: Theme 1: Steel and hybrid bridges - Structural design of steel bridges - Bridge loads, traffic accounting and analysis - Design verifications of steel bridge elements - Advanced bridge analysis - Execution and details of steel bridges - Renovation and maintenance of steel bridges Theme 2: Supporting structures for renewable energy - Hollow section steel joints in jacket structures- Monopile supporting structures for wind turbines and connections. B4-3: The content of the units is defined within 2 themes: Theme 1 (40%): Lightweight and slender steel structures, Stability and resistance of complex columns and frames. Stability of welded and rolled steel members. Cold-formed thin-walled steel members. Theme 2 (60%): FRP Composite Structures Design of FRP structures FRP material behaviour. Analysis and verification of FRP structural members Analysis and verification of FRP joints and details.	
Study Goals	B4-1: Students will be able to: 1. Create, carry out, interpret and document linear and nonlinear finite element analysis of steel members and connections/joints. (B4-LO1) 2. Improve design models for resistance of structural members and joints based on quantification and understanding of scattering of material and structural properties (e.g. partial factors). (B4-LO3, LO4, LO5) 3. Explain the concept of the Monte Carlo method and perform Monte Carlo simulations using Excel and Python programming. (B4-LO3, LO5) 4. Explain the fatigue initiation and crack propagation process and mechanism and determine the main factors influencing the fatigue strength of materials, connections, and structures. (B4-LO4, LO5, LO6) 5. Explain the basics of the fatigue behaviour of critical details (welded connections, bolted connections, tubular joints) and the theoretical background of prediction methods. (B4-LO3, LO4, LO5) 6. Design steel structures exposed to cyclic loading (fatigue strength) and explain design concepts in the presence of structural damage based on the lifetime prediction methodologies. (B4-LO3, LO4, LO5, LO6) 7. Design ductile, damage tolerant and reusable moment resisting and simple joints in steel structures for their beam-to-column and column to base-plate connections. (B4-LO3, LO4, LO5, LO6) 8. Apply the theoretical background to create engineering models for the design verifications of the structural connections/joints	

	exposed to monotonic, cyclic and impact loading. (B4-LO3) B4-2: Students will be able to: 1. Understand the design requirements and principles of designing diverse fixed or movable highway, railway and pedestrian bridges. (B4- LO2,LO3,LO4) 2. Design durable, sustainable and cost-efficient bridges by integrating global structural, cross-section and deck systems. (B4-LO3,LO6) 3. Analyse structural behaviour of steel and steel-concrete bridges in response to construction phases, traffic and environmental loads. (B4- LO4,LO6) 4. Analyse ultimate, serviceability and fatigue limit states of new and existing bridges. (B4-LO3,LO5,LO6) 5. Optimise construction time and costs by combining diverse production and execution methods. (B4-LO3,LO4,LO6) 6. Create strengthening solutions for existing bridges. (B4-LO2,LO3,LO4) 7. Calculate the loads applied in the supporting structure of wind turbine towers and evaluate their structural performance. (B4-LO3,LO6) 8. Validate the design of the tubular and multi-membered towers/jackets and their connections by using calculations according to structural design codes, guidelines and linear and non-linear FE models, including imperfections. (B4-LO1,LO5,LO6) B4-3: Students will be able to: 1. Design lightweight rolled, welded, cold-formed, stiffened and unstiffened cross-sectional steel members of plated and cylindrical steel structures. (B4-LO2,LO3,LO6) 2. analyse the effects of local buckling, imperfections and post-buckling behaviour in thin-walled plated and cylindrical structural members and apply engineering methods to perform the design verification. (B4- LO2,LO3,LO6) 3. Design and analyse steel frames by considering members imperfections and limit state provisions of design codes. (B4-LO5,LO6) 4. Choose optimal combinations of FRP materials and production processes for a specific application in infrastructure and set of loading and environment conditions. (B4-LO4,LO5,LO6) 5. Design FRP structures by considering the structure's specific material behaviour and mechanical and functional requirements. (B4-LO4,LO6) 6. Analyse the mechanical performance and failure modes of FRP structures and joints on the level of plies, laminates and structural components using Classical Laminate Theory, progressive failure analysis and Finite Element models. (B4-LO1,LO4,LO6) 7. Verify designs of FRP structural members and joints by using calculations according to structural design codes. (B4-LO3,LO4,LO6)
Education Method	B4-1: Pre-recorded short video lectures will be embedded into weekly instructions for self-studying aligned with blended learning, live/online lectures during contact hours, problem-based learning through exercises, theoretical problems and home assignments, numerical examples, discussion in teams during contact hours, tutor sessions (FEM software), guest lectures from industry. B4-2: Pre-recorded short video lectures will be embedded into weekly instructions for self-studying aligned with blended learning, live/online lectures during contact hours, problem-based learning through exercises, theoretical problems and home assignments, numerical examples, discussion in teams during contact hours, tutor sessions (individual or with a complete group), role-plays, guest lectures from industry, site visits. B4-3: Pre-recorded short video lectures will be embedded into weekly instructions for self-studying aligned with blended learning, live/online lectures during contact hours, problem-based learning through exercises, theoretical problems and home assignments, numerical examples, discussion in teams during contact hours, tutor sessions (individual or with a complete group), role-plays, guest lectures from industry, site visits.
Assessment	B4-1: LO2,3,4,5,6,7: Application of theoretical background, design and problem-solving skills are tested in open-book written exam with open-ended questions related to steel joints, fatigue resistance prediction by local approach (FEA), composite structures and connection details. LO1,2,3,8: The assessment of design and analysis skills, and collaborative work are tested through an integral report on analysis and calculations connecting content of all 3 modules. Part of report related to unit 1 will focus on analysis and verification specific fatigue detail, local buckling or joint resistance of e.g. a bridge, structure, wind tower or FRP structure from unit 2 and 3. Students will apply engineering tools and verification standards (Eurocodes), create and analyse own finite element models for specific structures and details. B4-2: LO1,2,5,6,7: Application of theoretical background, design and problem-solving skills are tested in open-book written exam with open-ended questions related to steel bridges and wind turbine supporting structures. LO3,4,7,8: The assessment of design and analysis skills and collaborative work are tested through an integral report on analysis and calculations that connect content of all 3 modules. Part of the report related to unit 2 will focus on analysis and verification of limit states (ultimate, serviceability, fatigue) of critical cross section or component of a steel bridge and wind turbine tower. The FE analysis from unit 1 will be implement in the report and compare the engineering tools and design verification standard. B4-3: LO1,3,4,5,6: Application of theoretical background, design and problem-solving skills are tested in open-book written exam with open-ended questions related to FRP composite material, structures and details, and buckling prone, thin-walled steel structure components. LO2,3,5,6,7: The assessment of design and analysis skills and collaborative work are tested through an integral report on analysis and calculations that connect content of all 3 modules. Part of report related to unit 3 will focus on analysis and verification of limit states (ultimate, serviceability, fatigue) of critical cross-section or component of FRP composite structures and slender steel structures. The FE analysis from unit 3 will be implemented in the report, and compared to engineering tools and design verification standards.
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. To be able to follow and efficiently follow this module, pre-knowledge from BSc studies, and preceding modules of MSc study (faculty base module, track base module, A module) are necessary: Basics of structural analysis (incl. principles of second order theory). Basics of continuum mechanics and strength of materials. Design and verification skills for steel cross sections and members class 1, 2 and 3. Basics of quantification and analysis of loads in civil engineering structures. Concepts of limit state analysis and design. Principles of design of structural joints and connections in structures. Basics of the design of steel-concrete composite beams. Basics of stability of 1D and 2D structures. general understanding of numerical methods) Reliability and statistics Basic and advanced structural mechanics Basic knowledge of circularity and sustainability Basic skills in CAD, teamwork, report writing, framework software, hand sketching, information literacy.
Academic Skills	Team-based work, planning a design task, setting up a clear calculation, convincing visual presentation, oral presentation, report writing, sketching, drafting, self-study.
Literature & Study Materials	B4-1: Text books; Reading materials; Readers; Journal/Conference Papers; Pre-recorded Video Lectures; R&D Reports

	B4-2: Text books; Reading materials; Readers; Journal/Conference Papers; Pre-recorded Video Lectures; R&D Reports B4-3: Text books; Reading materials; Readers; Journal/Conference Papers; Pre-recorded Video Lectures; R&D Reports See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Judgement	
Permitted Materials during Exam	Open-book written exam means that all theoretical teaching material is allowed. Numerical examples are not allowed.
Collegerama	No

CIEM5250	Building Engineering	15
Responsible Instructor	Ir. A. Borgart	
Responsible Instructor	A.C.B. Schuurman	
Responsible Instructor	Ir. S. Pasterkamp	
Responsible Instructor	Dr.ir. H.R. Schipper	
Module Manager	Dr.ir. H.R. Schipper	
Instructor	Ir. A. Borgart	
Instructor	Dr. I. Nevzgodin	
Instructor	A.C.B. Schuurman	
Instructor	Prof.dr.ir. P.C. Louter	
Instructor	Ir. E.R. van den Ham	
Instructor	Ir. S. Pasterkamp	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr.ir. H.R. Schipper	
Instructor	H. Alkisaai	
Instructor	P. Eigenraam	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	Ir. C.J. Janssen	
Instructor	Dr. E. Brembilla	
Instructor	Dr. F. Kavoura	
Instructor	Dr.ir. A.C. Boerstra	
Instructor	M.P. Felicita	
Instructor	Ir. J.P.G. Ramler	
Contact Hours / Week x/x/x/x	0/0/0/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module will focus on the engineering of buildings as complex systems in which many facets need to be aligned in an integrated interdisciplinary design process. From an economy point of view, a share of about 70% of construction industry is involved with buildings, which makes this module very relevant for future civil engineers. The module content is grouped in three learning units: LEARNING UNIT 1 (LU1, Architecture, Building Physics and Façades, 4 EC) will provide you with the specific building-related background knowledge that is not directly structural engineering: contemporary and 19th century architectural styles façade design/detailing building physics (heat transport, acoustics, wind flow) building services (heating, ventilation, air-conditioning, vertical transport, water). LEARNING UNIT 2 (LU2, Building Structures, Construction Methods and Foundations, 6 EC) is about the flow of forces in regular building structures and stability in various typologies of buildings (office, commercial, industrial, residential, etc.). Unit 2 is grouped around the following three themes: Theme 1 (Building Structures): advanced knowledge on the flow of forces in building structures stability and lateral resistance and known solutions for various typologies of buildings according to their function (office, commercial, industrial, residential) use and reuse of various building materials, to give you a clear understanding on which options are available and which solutions are appropriate for every part of the new or existing building structure complexity of the design process and responsibilities of different partners in the design and construction chain Theme 2 (Foundations): interaction with the subsoil via the foundation is an important part of the building design, and you will learn how to develop a good foundation in cooperation with the geotechnical advisor. attention will also be paid to deep excavations, that are often required for foundation and parking garages or other underground spaces under a building. Theme 3 (Construction methods): The design of the building is depending on available construction methods, and by studying these, you will get a clear understanding of the limitations and options each of these methods offer. Impact on surroundings and urban context will be addressed, as well as safety of construction, economy and material efficiency (zero-waste) on the construction site. LEARNING UNIT 3 (LU3, Design of Spatial Structures, 5 EC) is an advanced unit in the field of special and spatial building structures, their design process, structural mechanics, structural behaviour and performance, computational modelling, combined with practical use of design- and analysis tools for structural engineering. Topics include structures in nature biomimetics plates, shells and vaults membrane and pneumatic structures cable-nets tensegrity structures (space) frames grid(shell) structures and domes adaptive and deployable structures. Whereas Learning Unit 2 will focus on more regular building types, Learning Unit 3 broadens the scope with an extensive introduction into complex geometries and typologies. Apart from design skills, also academic skills will be trained during this	

	unit since the unit includes a mini conference, including the writing and presentation of scientific papers. Working on an integrating design exercise of a realistic building that connects all three learning units, you will experience the consequences (limitations, options and uncertainties) of available design and construction methods. During the exercise, you will receive feedback from professional building engineers.
Study Goals	On completion of this module, you will be able to: 1. recognise and discuss architectural styles and construction methods of the 19th, 20th and 21st century; 2. distinguish façade typologies and use those in design, while correctly applying the laws of building physics in the design evaluation to meet relevant criteria for climate, energy-efficiency and durability, taking into account stochastic material properties, climate data and user behaviour; 3. discuss the relationship and integration between design of load bearing structure, façade structure and other relevant parts/aspects of the building, such as services, vertical transport, plumbing, climate, ducts; 4. analyse whether reuse of existing structures or structural components in a project is economically and technically feasible and, where it is, reuse such existing building structures in the design of the newly developed building, correctly assessing the reliability of the existing structure or components based on measurements or probabilistic reasoning; 5. correctly identify functional, structural and construction requirements and loads for low-rise and multi-storey buildings with different targeted functions; 6. explain and apply, in the context of a design, the most important typologies and stability systems of low-rise and multi-storey buildings, using different construction materials, and being able to model and simulate the behaviour of such systems both with self-written code and existing FEM software; 7. develop, analyse and evaluate various reliable solutions for building foundations and deep basements in close cooperation with a geoengineer, considering the close interaction between structure and foundation, recognising and limiting risks of uncertain underground soil conditions, surveying and foundation methods; 8. explain how a building design can be constructed in an economic, safe and practical manner, acknowledging what the consequences and risks of construction methods are for the design and vice versa, and acting as a structural engineer with a professional, responsible and ethical attitude; 9. Design spatial structures (such as domes, (grid)shells, inflatables and membrane structures, space frames, cable systems and tensegrity structures), analyse and explain their 3D structural action; 10. Use advanced design tools, numerical analysis tools, and physical models or measurements to analyse and visualise 3D structural behaviour, and develop such models in self-written code.
Education Method	LU1: Lectures, weekly short exercises, discussion in class, design studio LU2: Lectures, individual feedback sessions centred around design exercise LU3: Lectures, hands-on physical modelling workshops, discussion in teams, presentations, tutor sessions (individual or with full group) Approximately half of the weeks will be used for a series of 50+ lectures, while the other half of the week is used for self-study in studio format: an instruction room will be reserved for the participating students to allow for collaboration in groups on the design assignment.
Assessment	Each of the three learning units will have a separate written exam, focused on the theory (lecture notes, textbooks) and skills that can best be assessed in a written exam. Since building engineering is very much design-oriented and aiming for a cognitive level that is hard to assess in the limited time of a written exam, formative assessment moments will be included, for example by tutor feedback, discussion sessions, presentations, and peer review of designs. On overall-module level, for each academic year, a different building for a specific clients brief will have to be designed by the students. This design assignment will function as the integrator of the module. The final grade will be determined in a single rubric based on the level of understanding displayed at the presentation (oral exam) and the quality of the final assignment reports. LU1: individual written exam (2 EC, 13%), design assignment on module level, see below LU2: individual written exam (3.5 EC, 23%), design assignment on module level, see below LU3: individual written exam (2.5 EC, 17%), design assignment on module level, see below The design assignment on module level will comprise 7.0 EC (47%), which ECs are spread over the above 3 learning units (approx. in a proportion 13%, 17%, 17% for LU1/2/3). The assignment will include presentations, poster and paper. The design assignment is likely to be a team exercise, depending on number of participants. A successful participation in the module requires a minimum grade of 5.8 for all partial grades, so for each of the three written exams and for the design assignment.
Tags	Building & Spatial Development Building Engineering Design Ethics Group work Integrated Practicals Soil Mechanics Structural Mechanics Sustainability
Expected prior Knowledge	The module can only successfully be completed with sufficient prior knowledge in the following subjects: - general understanding of optimisation methods and numerical methods (from CFM/MUDE - necessary for LU3) - reliability and statistics (from CFM/MUDE and BSc - necessary for understanding safety of building structures/materials, stochastic loads and soil/foundations) - energy flow through solids (from Programme Base, necessary for building physics) - basic and advanced structural mechanics (from BSc and SE Track Base) - advanced knowledge on loads on structures (from BSc and SE Track Base) - advanced knowledge of reinforced concrete members and basic knowledge of prestressed concrete and joints/connections (from BSc and SE Track Base) - advanced knowledge of steel members and basic knowledge of joints/connections (from BSc and SE Track Base) - Basic knowledge of timber members and basic knowledge of joints/connections (from BSc and SE Track Base) - basic knowledge of circularity and sustainability (from BSc and SE Track Base) - basic skills in CAD, teamwork, report writing, framework software, hand sketching, information literacy (from BSC) - theory of ethics (from BSc and Programme CE Base)
Academic Skills	Team-based work and collaboration, planning a design task, setting up a clear calculation, convincing visual presentation, oral presentation, academic literature review, academic writing and presenting, sketching, drafting, physical modelling, self-study, effective communication
Literature & Study Materials	LU1: The Great Builders, Kenneth Powell (Ed.), London: Thames & Hudson, 2021 (selected chapters) Basics of Building Physics, A. Zeegers and A.C. van der Linden, I.M. Kuijpers, Van Gaalen, 2022 (selected chapters) Exterior Building Enclosures : Design Process and Composition for Innovative Facades, Keith Boswell and C Keith Boswell, 2006 (selected chapters) Facades: Principles of Construction, Ulrich Knaack, Tillman Klein, Marcel Bilow, and Thomas Auer, 2007 (as reference work) Modern Construction Envelopes, Andrew Watts, 2014 (as reference work) LU2: Reader Building Structures 1, 2018 ed. K.C. Terwel

	Reader Building Structures 2 - R. Abspoel 2013 Concrete Building Structures 2016 Structural calculations of High Rise Structures, 2017 Reader Renovation and Reuse, 2022, S. Pasterkamp LU3: To be decided, will be announced at start module
Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Will be announced on Brightspace at the start of the module.
Collegerama	Yes

CIEM5260	Transportation Infrastructures	15
Responsible Instructor	Dr. X. Liu	
Responsible Instructor	Dr. V.L. Markine	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Prof.dr. Z. Li	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Dr. K. Anupam	
Instructor	Prof.dr.ir. R.P.B.J. Dollevoet	
Instructor	Dr. Z. Yang	
Instructor	Dr. H. Wang	
Contact Hours / Week x/x/x/x	0/0/0/18	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module provides knowledge on modern tools for design, assessment, performance prediction and optimisation of engineering structures in application to transportation infrastructures, such as roads, railways, bridges, and tunnels. In this process, the entire design-construction-service cycle will be considered. The analysis of structural behaviour commences with defining the loading conditions. Therefore, the dynamic interactions of road/track/bridge/tunnel and moving vehicles will be introduced first. The students will learn how to numerically analyse such interactions and their consequences like rutting, vibrations and noise. The methods for analysis of the long-term behaviour of engineering structures, including wear, (rolling contact) fatigue damage, etc., will be considered as well. Additionally, the students will learn how to apply the structural reliability methods to account for the uncertainties of real life, even including the effects of the weather. The module also focuses on critical issues related to the aging of transportation infrastructures. Topics will cover aspects of material and structural degradation due to increased dynamic loading, environmental factors, and climate change effects, and familiarize with tools to assess and predict structural durability. The use of innovative materials to enhance the performance and durability of transportation infrastructures will be also discussed. The students will get acquainted with modern general-purpose finite element software: ANSYS, ABAQUS, PLAXIS, Discrete element methods, etc., as well as with some dedicated design software, and will use these tools to assess structural behaviour at multi-scale level. To assure the long-term safe performance, the methods of condition monitoring of engineering structures in application to transportation infrastructures will be studied. Several non-destructive testing (NDT) technologies like Digital Image Correlation (DIC), laser scanning, and satellite data used to assess the performance and condition of infrastructure will be covered. Examples of monitoring solutions using static, moving, and crowd sensing technologies will be analysed. Application of the NDT methods for parameter identification of the numerical models will be considered as well. Combined with state-of-the-art data-analysis methods and machine learning, the monitoring information will turn into maintenance strategies. In this module we organise excursions to companies to discuss their engineering approaches for integral design for transportation infrastructures. All knowledge will be applied to real-life case studies to be performed in close collaboration with manufacturers and engineering companies. All these will serve as preparation for MSc theses.	
Study Goals	On successful completion of this module, the student will be able to B6-LO1: Analyse the design-construction-service cycle of transportation infrastructures (SE2, SE3, SE5). B6-LO2: Evaluate state-of-the-art construction processes and techniques for transportation infrastructures (SE2, SE7). B6-LO3: Evaluate transportation infrastructures modelling and design solutions, considering the effects of the inherent dynamics, operational conditions, and relevant sources of uncertainties (SE1, SE4). B6-LO4: Propose physics-based and data-based monitoring and maintenance solutions for transportation infrastructures (SE6, SE8). B6-LO5: Create innovative and ethical solutions and compare results with the state-of-the-art knowledge in real-life challenges from selected case studies from the industry (SE7, SE8). B6-LO6: Practice and further develop effective communication skills (both written and oral) to ethically share results and opinions with peers, engineers, academics, and stakeholders from the transportation infrastructure industry (SE9). B6-LO7: Provide constructive feedback, resolve possible conflicts in diverse teams, and ethically tackle critical transportation infrastructure challenges (SE9). B6-1: After completing this unit, the students will be able to: B6.1-LO1 Analyse and evaluate methods for the construction process of transportation infrastructures. B6.1-LO2 Evaluate design aspects and procedures for transportation infrastructures. B6.1-LO3 Explain the capabilities of different design methods and solutions under different parameters, variations, and uncertainties in properties and loading patterns. B6.1-LO4 Select and apply design methods, and propose improvements in the construction and design of transportation infrastructures by tackling real-life case studies from the transportation industry. B6.1-LO5 Evaluate current practices of maintenance in the transportation infrastructure industry. B6.1-LO6 Ethically and effectively discuss and analyze results with main stakeholders. B6-2: After completing this unit, the students will be able to: B6.2-LO1 Evaluate the most important dynamics and operational conditions of existing transportation infrastructures. B6.2-LO2 Explain degradation mechanisms that can affect transportation infrastructures during their whole life cycle. B6.2-LO3 Assess the performance over time of transportation infrastructures, considering both dynamics and the influence of degradation mechanism. B6.2-LO4 Select and apply methodologies to estimate the remaining lifetime of existing transportation infrastructures. B6.2-LO5 Discuss and present results to different stakeholders effectively and ethically. B6-3: After completing this unit, the students will be able to: B6.3-LO1 Implement state-of-the-art monitoring solutions for transportation infrastructures. B6.3-LO2 Evaluate current practices of maintenance in the transportation infrastructure industry. B6.3-LO3 Design intelligent condition-based maintenance strategies including both physics and data-based characteristics of transportation infrastructures. B6.3-LO4 Assess the performance of transportation infrastructures over their whole life-cycle,	

Education Method Assessment	including sustainability aspects. B6.3-LO5 Effectively and ethically present results and discuss their advantages and disadvantages to different stakeholders. Lectures / Blended Learning / Laboratory and Simulation Experiments / Excursions The module is assessed through four summative assessments: three on the unit level and one on the module level. 25%: B6-1 written exam. A set of questions from this unit will be included in the Module assessment (practical questions on engineering/analytical methods, software for a real-life design problem). 20%: B6-2 written exam. Knowledge of the theoretical background and problem-solving skills are tested in a written exam with open questions. 15%: B6-3 report. The assessment of this unit consists of a final individual report that covers the following topics: state-of-the art monitoring solutions, practical monitoring tools, hands-on experiments, maintenance strategies, life-cycle management, sustainability aspects, and challenges in AI, data analysis, and digitalization. Students will utilize commonly available pavement and railway monitoring tools for infrastructure condition evaluation. Selected questions of this unit will be evaluated as part of the module assessment. Students will select a real-life case study involving transportation infrastructures, summarize current practices and future challenges, present the case to infrastructure stakeholders, and submit a technical report. A presentation for the selected case study will be presented in a colloquium. 40%: Module assessment which consists of both presentation and final report. Designing, analysing, and communication skills are evaluated through this module assessment. A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Expected prior Knowledge	Students from all preceding SE-A Modules can follow SE-B6. Students from the different SE-A Modules will have different pre-knowledge: - Knowledge about modelling approaches, such as FEM (SE-A1). - Knowledge about material aspects (SE-A2). - Knowledge about aspects of integral design of civil structures including road and rail (SE-A3). From Track-Base, the following are topics we expect students to know about: 305 - Loads on infrastructure engineering, short-term and long-term loads. - Material properties and the corresponding mechanical behaviour. For example, steel, concrete, timber structures. - Sustainability, circularity and LCA approaches used in the construction of infrastructures. - One-dimensional slender structures and plates formulations of the governing equations and fundamental solutions (in both statics and dynamics). From Program-Base and Faculty-Base, the following are topics we expect students to know about: - Coding (Python, Matlab, etc.) tools from MUDE (Faculty-Base) and Program-Base. - Topics from modules in the Program-Base such as deriving, analysing and solving fundamental problems via simpler engineering forms. - Basics of statics, kinetics, mechanics of materials, dynamics, elasticity, etc. - Fundamentals of the Ethics line. We will use specialized software for different examples, including ANSYS. For a further understanding, knowledge about FE is desirable but not mandatory.
Academic Skills	Working with transportation infrastructures is inherently multidisciplinary, making interdisciplinary thinking very relevant in this module. We address challenges from both theory and practice; thus, problem learning and solving, and applying theory to new situations are academic skills the students will use and develop in this module. The assessment considers reports and presentations to stakeholders, so scientific writing, presentation, and time management are essential personal skills.
Literature & Study Materials	B6-1: AASHTO (1986). AASHTO guide for design of pavement structures. Washington D.C., USA. Ansys, VI-Rail, 3D Move manuals. CROW (2002). Design and overlay design for pavements with thin asphalt surfacings (in Dutch). Publication 157. Ede, The Netherlands. Handbook of Railway Vehicle Dynamics. NCHRP 1-37A (2004). Mechanistic-empirical design of new and rehabilitated pavement structures, Transportation Research Board, National Research Council, Washington, D.C., USA. NEN-EN 1991-2:2003 - Eurocode 1 - Actions on structures - Part 2: Traffic loads on bridges, 2003. NEN-EN 1990:2002/a1:2006 - Eurocode: Basis of structural design - Annex 2: Application for bridges, 2002. Online learning material based on Railway Engineering (TUD) ProfEd online courses. Papagiannakis A.T and Masad E.A. (2008). Pavement Design and Materials, 1st edition, Wiley. Shell International Petroleum Company Ltd., Shell pavement design manual (1978), London. UIC. USP project. Modelling method for static loads. UIC. USP project. Modelling method for dynamic loads. Yang H. Huang (2008) Pavement Analysis and Design, 2nd Edition, Pearson Education B6-2: Dawson, A. (2014). Anticipating and Responding to Pavement Performance as Climate Changes. In Climate Change, Energy, Sustainability and Pavements (Eds.: Gopalakrishnan, Steyn and Harvey). Springer. Esveld, C. (2001). Modern railway track, second edition. In Modern railway track, second edition (pp. 1-653). MRT productions. Huang and Di Benedetto (Eds) (2015). Advances in Asphalt Materials: Road and Pavement Construction. Woodhead Publishing Limited. Iwnicki, S., Spiryagin, M., Cole, C., & McSweeney, T. (Eds.). (2019). Handbook of Railway Vehicle Dynamics second edition. CRC Press. https://doi.org/10.1201/9780429469398 Jing, R. (2019). Ageing of bituminous materials : Experimental and numerical characterization, Dissertation TU Delft, 232 p. Lewis, R. Olofsson, U. (2009). WheelRail Interface Handbook. Woodhead Publishing. https://doi.org/10.1016/B978-1-84569-412-8.50031-8. Online learning material based on Railway Engineering (TUD) ProfEd online courses. 311 Srirangam, S.K. (2015). Numerical simulation of tire-pavement interaction, Dissertation TU Delft, 204 p. Varveri, A. (2017). Moisture damage susceptibility of asphalt mixtures: Experimental characterization and modelling, Dissertation TU Delft, 180 p. B6-3: Barriera, M., et al., In Situ Pavement Monitoring: A Review. Farrar, C.R. and Worden K. (2013), Structural Health Monitoring: A Machine Learning Perspective, Wiley, Chichester. Fraden, J. (2010), Handbook of Modern Sensors. Springer, fourth edition. Fwa, T.F., The Handbook of Highway Engineering. Kobbacy, K.A.H. and Murthy, D.P. eds., 2008. Complex system maintenance handbook. Springer. Lajnef, N., Smart Pavement Monitoring System - Federal Highway. Márquez, A.C., (2007). The Maintenance Management Framework: Models and Methods for Complex Systems Maintenance, Springer, London, pp. 1-341. Miradi, M., Knowledge Discovery and Pavement Performance Intelligent Data Mining. Online learning material based on Railway Engineering (TUD) ProfEd online courses. Parmenter, D., 2015. Key performance indicators: developing, implementing, and using winning KPIs. John Wiley & Sons. Sharma, S., A data-driven approach for pavement surface distress classification.
Judgement	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Calculator
Collegegrama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Traffic & Transport Engineering (TTE)
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CIEM6000	Base: Traffic and Transport Engineering	15
Responsible Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Prof.dr.ir. S.P. Hoogendoorn	
Instructor	Prof.dr. Z. Li	
Instructor	Dr. V.L. Knoop	
Instructor	Ir. A.J. Pel	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	E. Quaglietta	
Instructor	Dr. O. Cats	
Instructor	Dr.ir. H. Farah	
Instructor	Dr. S. Sharif Azadeh	
Contact Hours / Week x/x/x/x	x/12/6/x	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The track base provides students with a solid basis of the fundamentals of traffic and transport engineering including planning and design of traffic and transport networks and infrastructure, transport modelling and analysis, traffic modelling and management, and identifying and assessing the ethical and societal implications of traffic and transport engineering solutions. Ethical aspects are important for engineers, practitioners, and researchers in the sector of traffic and transport engineering. In the track base, relevant ethical aspects and societal impacts of traffic and transport engineering solutions (e.g., safety, environment, health, equity, privacy) and the responsibilities when creating a design or developing models will be addressed across the three different units. In addition, the first unit dedicates two lectures to discuss these aspects in more depth.	
Study Goals	On completion of this module, the student will be able to: TTE-TB-LO1: Design the traffic and transport infrastructure and networks. TTE-TB-LO2: Develop a traffic and transport engineering solution and analyse its performance. TTE-TB-LO3: Discuss, analyse and apply traffic and transport models. TTE-TB-LO4: Set-up, perform and report a transport engineering solution/ modelling study. TTE-TB-LO5: Assess the ethical and societal implications of traffic and transport engineering solutions.	
Education Method	Lectures, in class small groups practice problems, in-class discussions, self-assessment quizzes, Group work on the project, feedback on group work. Collegerama is only planned for some (parts of) units; check Brightspace.	
Assessment	Exam 1: Planning and design of Traffic and Transportation Networks and Infrastructure - Written Exam (13,3%) Exam 2: Planning and design of Traffic and Transportation Networks and Infrastructure - Group Assignment (13,3%) Exam 3: Transport Modelling and analysis - Written Exam (33,3%) Exam 4: Traffic Modelling and Management -Written Exam (40%)	
Tags	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Analysis Modelling Project Transport & Logistics Transport phenomena	
Expected prior Knowledge	BSc courses (or equivalent): CTB1002 (Linear Algebra), CTB2000 (Programming), CTB2200 (Probability and Statistics), CCTB2400 (Numerical Math)	
Academic Skills	Critical thinking, problem-based learning and solving, scientific writing and presentation, time management, communication, application of theory to new situations	
Literature & Study Materials	Lecture notes and scientific articles provided by the teachers and are made available through Brightspace.	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Varies per unit -- Check Brightspace	
Collegerama	Yes	

CIEM6110	Methods in Traffic and Transport Engineering	9
Responsible Instructor	Dr.ir. S.C. Calvert	
Instructor	Dr.ir. W. Daamen	
Instructor	Dr. M. Snelder	
Instructor	Dr.ir. S.C. Calvert	
Instructor	Dr.ir. W.J. Schakel	
Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr.ir. A. Gavrilidou	
Instructor	Dr. S. Sharif Azadeh	
Contact Hours / Week x/x/x/x	x/x/12/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This module provides the students with further deepening of the three learning lines (data, modelling, and risk) offered in the faculty base with a focus on applications from the Traffic and Transport Engineering track including empirical analysis for transport and traffic engineering, optimisation in transport engineering, simulation and uncertainty in transport engineering.	
Study Goals	On completion of this module, the student will be able to: TTE-A-LO1. Formulate research questions and hypotheses in relation to transport infrastructure, network, operations and behaviour TTE-A-LO2. Design and conduct experiments in the realm of transport infrastructure, network, operations and behaviour TTE-A-LO3. Analyze and interpret empirical and simulation data about transport infrastructure, network, operations and behaviour TTE-A-LO4. Formulate a mathematical optimisation model addressing problems related to transport infrastructure, network, operations and behaviour TTE-A-LO5. Solve optimisation problems using solvers and tailored designed efficient algorithms also considering stochasticity in the inputs. TTE-A-LO6. Develop and implement simulation model functionalities for representing transport infrastructure, network, operations and behaviour TTE-A-LO7. Develop suitable data-driven modelling approaches for verification, calibration and validation of models, input and parameters in the context of transport infrastructure, network, operations and behaviour TTE-A-LO8. Compare models properties and performance for analyzing transport infrastructure, network, operations and behaviour TTE-A-LO9. Apply techniques for risk and uncertainty analysis for transport data and models	
Education Method	Lectures, lab classes with formative assessments where students formulate problems and solve them with a computer.	
Assessment	Exam 1: Empirical Analysis for Transport and Traffic Engineering - Assignment (33,3%) Exam 2: Optimisation in Transport Engineering - Written Exam (33,3%) Exam 3: Simulation and Uncertainty in Transport Engineering - Assignment (33,3%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. 1. BSc courses (or equivalent): CTB1002 (Linear Algebra), CTB2000 (Programming), CTB2200 (Probability and Statistics), CCTB2400 (Numerical Math) 2. Design, organise and implement data collection (MUDE: data) 3. Analyse the collected data using statistics (MUDE: data and uncertainty) 4. Fundamentals of Discrete Choice Models/ Traffic Flow Theory (TTE Base track) 5. Formulate a mathematical problem (MUDE: modelling) 6. Solve continuous and integer optimisation problems (MUDE: modelling) 7. Monte Carlo simulation, random number generators (MUDE: uncertainty)	
Academic Skills	Completion of the MUD and track core	
Literature & Study Materials	Parts of the current reader for CIE4831 (Empirical Analyses for Transport & Planning) Introduction to Operations Research by Hillier and Lieberman. Discrete optimization: Introduction to logistics systems management by Gilbert Laporte. NLP: Numerical optimization by Wright and Nocedal. Lecture slides and a selection of key and relevant papers	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Will be communicated during lectures	
Collegerama	No	

CIEM6210	Transport Networks and Systems	15
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Dr. C. Maat	
Instructor	N. van Oort	
Instructor	Dr. M. Snelder	
Instructor	Prof.dr. J.H.G. Correia	
Instructor	Ir. A.J. Pel	
Instructor	Prof.dr.ir. L.A. Tavasszy	
Instructor	G.K. de Clercq	
Instructor	Dr. S. Fazi	
Instructor	M. Rinaldi	
Contact Hours / Week x/x/x/x	x/x/x/17	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In-depth analysis of the functioning of transport networks for passenger and freight transport, the relevant performance characteristics for the users (private persons, firms) including the resilience, the interaction with economical and spatial development, the broader sustainability effects (societal, economic, environmental) of networks. In-depth analysis of route choice and assignment in road networks, and mode choice and route assignment for passenger and freight transport in intermodal networks, the impact from disruptions and the impact on performance indicators for users and the society. Optimisation and assessment of the design of logistics services, road networks, and intermodal networks, in response and in interaction with socio-economic and spatial planning developments and by balancing the interests of users and society, so by taking into account network performance, resilience and sustainability. The module deepens understanding of the societal impacts of transport networks, and applying methods and approaches to evaluating and design these networks in an ethical and societal responsible way.	
Study Goals	On completion of this module, the student will be able to: B1-LO1. To create integrated solutions for transport networks for various modes, services and user classes to resolve performance challenges. B1-LO2. To evaluate the performance impact of current and (potential) future networks for users and society in different dimensions. B1-LO3. To assess the suitability of standard methods for real-world transport network design problems. B1-LO4. To analyse the current and the expected future performance of transport systems, services and networks and their tuning with land use. B1-LO5. To apply different standard methods for the design of transport networks. B1-LO6. To operationalise performance indicators for transport networks including links with norms and methods of measurement. B1-LO7. To explain which theoretical approaches, optimisation and calculation methods and models are needed for performance estimation, assessment and design steps. B1-LO8. To explain dynamic complexity between transport networks, and land use and other systems (system of systems). B1-LO9. To define road and freight transport networks in the societal, sustainability, perspective and define the main related design problems. B1-LO10. To define road and freight transport networks in their technical and user characteristics, and define the main related design problems.	
Education Method	Lectures, formative assignment (group project) with instruction, practical lecture/workshop, guest lectures	
Assessment	Exam Unit 1: Transport Modelling: Traffic Assignment - Written Exam (20%) Exam Unit 2: Regions, Transport and Networks - Written Exam (26,6%) Exam Unit 3: Freight transport networks and systems - Written Exam (26,6%) Exam Unit 4: Sustainable and Resilient Transport Networks and Systems - Presentation (26,6%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. 1. Basic insights performance, impacts, ethical and social implications and their indicators - of traffic and transport networks (TTE-TB-1, TTE-TB-1) 2. Basic insights in transport economics (individual choice preferences, business perspectives (TTE-TB-1) 3. Basic insights in Transport Modelling and Analysis, including aggregated, disaggregated models, gravity models and discrete choice models for both mobility and freight networks. (TTE-TB-2) 4. Basic knowledge on distribution (OD-estimation), assignment and equilibrium models (TTE-TB-2) Basic knowledge on mixed-integer linear optimisation (TTE-TB-2) 5. Insight in characteristics of unimodal and intermodal networks, physical and service networks for individual passenger transport, collective public transport and for freight transport (TTE-TB-3) 6. Insight in data-driven modelling approaches for verification, calibration and validation of models, input and parameters in the context of transport infrastructure, network, operations and behaviour (TTE-A1)	
Academic Skills	Critical, interdisciplinary and ethical thinking, social-impact aware and solution-oriented learning, information collection and synthesis, communication and responsiveness, scientific writing and presentation, application of theory to new situations.	
Literature & Study Materials	Transport Modelling: Traffic Assignment: - Lecture slides; - Chapter 2 of book: M.S. Fiorenzo-Catalano (2007) 'Choice set generation in multi-modal transportation networks', TRAIL; - Chapters 1 & 2 of paper: Luuk Brederode, Adam Pel, Luc Wismans, Erik de Romph, Serge Hoogendoorn (2019) 'Static Traffic Assignment with Queuing: model properties and applications', Transportmetrica A: Transport Science, Volume 15, Issue 2; - Chapters 1.5 & 3.1 of report: Joe Castiglione, Mark Bradley, John Gliebe (2015) 'Activity-Based Travel Demand Models: A Primer', TRB; - Paper: Menno Yap, Oded Cats, Bart van Arem (2020) 'Crowding valuation in urban tram and bus transportation based on smart card data', Transportmetrica A: Transport Science, Volume 16, 2020 Issue 1; - Paper: Carlo Giacomo Prato (2009) 'Route choice modeling: past, present and future research directions', Journal of Choice Modelling, Volume 2, Issue 1, 2009; pp. 65-100; - Paper: Chris M.J. Tampère, Ruben Corthout, Dirk Cattrysse, Lambertus H. Immers (2011) 'A generic class of first order node models for dynamic macroscopic simulation of traffic flows', Transportation Research Part B: Methodological, Volume 45, Issue 1; - Paper: Bliemer, Michiel C.J., Raadsen, Mark P.H., Brederode, Luuk, Bell, Michael G.H., Wismans, Luc J.J. (2014) 'A unified framework for traffic assignment: deriving static and quasi-dynamic models consistent with general first order dynamic traffic assignment models', working paper University of Sydney; http://hdl.handle.net/2123/19191;	

- Paper: Niels van Oort, Ties Brands, Erik de Romph, Jessica Aceves Flores (2014) 'Incorporating unreliability of transit in transport demand models: Theoretical and practical approach', Prepared for the 93rd Annual Meeting of the Transportation Research Board 2014

Regions, Transport and Networks:

- Lecture slides;
- Book: Jean-Paul Rodrigue (2020) 'The geography of Transport Systems'. Fifth Edition; New York: Routledge, 456 pages. ISBN 978-0-367-36463-2

Freight transport networks and systems:

- Lecture slides;
- Book: Tavasszy, L., G. de Jong (eds) (2014), Modelling Freight Transport, Elsevier (digital edition)
- Report: CE Delft (2020) 'STREAM Freight Transport' (digital edition; annual update)
- Paper: Jin Y. Yen (2015) Finding the K shortest Loopless Paths in a Network, Management Science 17(11)

Sustainable and Resilient Transport Networks and Systems:

- Lecture slides;
- Assignment book and tutorial (dedicated for the course): Sustainable and Resilient Transport Systems: Analysis and Interventions;
- Report: US Department of Transportation/FHWA (2017) 'Travel time reliability: making it there on time, all the time' (digital edition)
- Report: United Nations (2017) 'Transforming our World: the 2030 Agenda for Sustainable Development' (digital edition)
- Report: Department for Communities and Local Government (2009) 'Multi-criteria analysis: a manual'
- Book Chapter: Pascual, U., R. Muradian (eds) (2010) 'The Economics of Valuing Ecosystem Services and Biodiversity'
- Paper: Henry J. Liu, Peter E.D. Love, Michael C.P. Sing, Baozhuang Niu and Jianfeng Zhao (2019) 'Conceptual Framework of Life-Cycle Performance Measurement: Ensuring the Resilience of Transport Infrastructure Assets', Transportation Research D;
- Paper: Karst T. Geurs, W. Boon and B. van Wee (2009) 'Social Impacts of Transport: Literature Review and the State of Practice of Transport Appraisal in The Netherlands and the United Kingdom';
- Handbook of Emission Factors for Road Transport (HBEFA 4.1, 2019 - digital edition);

Judgement

See Assessment

For more information on grading, see article 14 in the Rules and Guidelines (RGBE):

<https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations>

CIEM6220	Road Traffic Systems	15
Responsible Instructor	Dr.ir. H. Farah	
Instructor	Dr.ir. W. Daamen	
Instructor	Prof.dr.ir. S.P. Hoogendoorn	
Instructor	Dr.ir. A. Hegyi	
Instructor	Dr.ir. A.M. Salomons	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. Y. Yuan	
Instructor	Dr. M.P. Hagenzieker	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	Dr.ir. W.J. Schakel	
Instructor	Dr.ir. D.C. Duives	
Instructor	Dr.ir. H. Farah	
Instructor	Dr.ir. A. Gavriilidou	
Instructor	M. Rinaldi	
Instructor	Dr.ir. I. Martínez Josemaría	
Contact Hours / Week x/x/x/x	x/x/x/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module teaches about (road) traffic systems. We consider car traffic, but also active mode traffic (cycling, walking). Note that the module considers the efficiency and safety of the traffic systems, and covers a qualitative and quantitative parts. The module also includes the ethical parts of transport research; these are explicitly included across all four units of the module. The module consists of the following units: - Active modes: Traffic and transport - Traffic safety - Intelligent vehicles for safe and efficient traffic: Design and assessment - Urban and motorway traffic flow modelling and control Active Modes: Traffic and Transport: Contact hours/week: 6h/week - Classroom lectures (4h) + Practicals (2h) List of instructors: Dorine Duives (responsible), Serge Hoogendoorn, Winnie Daamen, Alexandra Gavriilidou Traffic safety: Contact hours/week: 4h/week - Classroom lectures (4h) List of instructors: Haneen Farah (responsible), Marjan Hagenzieker Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment: Contact hours/week: 4h/week - Classroom lectures (4h) List of instructors: Marco Rinaldi (responsible), Irene Martínez Josemaría, Bart van Arem, Marjan Hagenzieker, Victor Knoop, Wouter Schakel Urban and Motorway Traffic Flow Modelling and Control: Contact hours/week: 2x2h classroom lectures + 1x4h pc-practical, i.e. 8 h/w List of instructors: Andreas Hegyi (responsible), Victor Knoop, Maria Salomons, Yufei Yuan	
Study Goals	On completion of this module, the student will be able to: TTE-B2-LO1. Analyse, assess, model, predict, and manage road infrastructure systems TTE-B2-LO2. Design and evaluate innovative solutions to increase traffic safety, efficiency and sustainability TTE-B2-LO3. Participate, contribute and effectively communicate in debates and professional projects involving international, multicultural and interdisciplinary teams. TTE-B2-LO4. Acquire new knowledge and skills to continue operating effectively.	
Education Method	Lectures, formative group assignments + feedback, in-class small groups practice problems, in-class discussions, self-assessment quizzes, guest speakers	
Assessment	Exam 1: Active Modes: Traffic and Transport - Written Exam (20%) Exam 2: Traffic Safety - Assignment (20%) Exam 3: Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment - Assignment (27%) Exam 4: Urban and Motorway Traffic Flow Modelling and Control - Written Exam (33%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. 1. Basic calculus (e.g. vectoren, numeriek, statistiek) 2. Basic modelling techniques (Python/ Biogeme script kunnen schrijven) 3. Basic statistics and regression analysis, basic probability theory, basic notion of optimization 4. Data collection methods 5. Basic knowledge on PDEs (both differential/difference eqs) 6. Traffic variables & fundamental diagrams 7. Shock wave theory 8. Basic Traffic control: static Traffic lights, ramp metering, basics of systems and control 9. SPECIALIST theory (An algorithm to resolve traffic jams) Note: prerequisites 6-9 are covered in the course Traffic modelling and management	
Academic Skills	critical and interdisciplinary thinking, problem-based learning and solving, continuous investigation, scientific writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations.	
Literature & Study Materials	Presentation slides and reading material will be provided after each lecture	

Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Exam 1 (Active Modes: Traffic and Transport): Simple calculator. Exam 2 (Traffic safety): assignment. Exam 3 (Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment): assignment. Exam 4 (Urban and Motorway Traffic Flow Modelling and Control): simple calculator; one two-sided A4 paper with own notes, handwritten in own handwriting, with normal font size.
Collegerama	No

CIEM6230	Public Transport and Railway Systems	15
Responsible Instructor	Dr. O. Cats	
Instructor	Prof.dr. R.M.P. Goverde	
Instructor	N. van Oort	
Instructor	Dr. D.M. van de Velde	
Instructor	Dr. O. Cats	
Contact Hours / Week x/x/x/x	x/x/x/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module Public Transport (PT) and railway systems deals with modelling, design and analysis of public transport and railway operations including the interactions with the customer demand and system governance. Three main units compose this module, namely: i) Public transport planning and operations, which focuses on PT service design, access and egress, PT control and emerging modes. ii) Public transport networks and modelling, which deals with behavioural and network modelling of PT as well as design and analysis of PT systems. iii) Railway Operations and Control, which tackles railway safety and signalling, railway automation and coordination in the railway value chain. The module will add to the learning line on Ethics related to planning, management and control of road- and rail-bound public transport operations. Each unit tackles possible impacts that design/operational decisions of public transport stakeholders and engineers can have on sustainability, public health, user privacy, social inclusion and fairness, presenting methods and solutions to meet international requirements, standards and policies. In addition, lecture material together with the group projects and formative assessment provided in the module, will support shaping the personal and professional responsibility of the students in line with the ethical code of conduct required as a future public transport engineer.	
Study Goals	On completion of this module, the student will be able to: TTE-B3-LO1. Analyse key dependencies among design variables, parameters, and criteria concerning organisation, service, infrastructure, operations and demand of road- and rail-bound public transport. TTE-B3-LO2. Apply and evaluate strategic, tactical, and operational design methods to align public transport/railway services with emerging societal and environmental needs. TTE-B3-LO3. Analyse processes and develop models for capturing interactions among system components including functional relations with other transport systems, society, and environment. TTE-B3-LO4. Design and evaluate intervention plans to build/upgrade, manage and operate public transport and railways in line with customer and stakeholders requirements. TTE-B3-LO5. Define and apply methods to predict and assess service performances of public transport/railway systems and the impacts on other systems, society, and the environment. TTE-B3-LO6. Demonstrate team cooperation, communication, and work management capabilities in three group projects (in unit B3-2) and formative assignments (in B3-1 and B3-3), to support the development of lifelong skills required in the professional world.	
Education Method	On-campus or on-line lectures from instructors, Guest lectures, Active learning activities such as in-class quizzes, games and group exercises, in-class discussions, feedback sessions, pre- and post-lecture material such as reading scientific articles and watching videos, formative assignments.	
Assessment	Exam 1: Public Transport System and Supply Planning and Operations - Written Exam (33,3%) Exam 2: Public Transport Demand and Network Planning and Operations - Group Assignment (33,3%) Exam 3: Railway Operations and Control - Written Exam (33,3%)	
Tags	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Modelling Policy Analysis Rail & Road Engineering / Planning Sustainability Transport & Logistics Transport phenomena	
Expected prior Knowledge	As a pre-requisite to taking this module the student needs to have previous knowledge about: 1) Transport fundamentals including characteristics of the transport supply and the transport demand, traffic flow properties and attributes, network design variables and key performance indicators for the assessment of transport service quality and reliability. 2) Modelling, simulation and optimisation methods and possible applications to transport systems. 3) Data collection, management and analysis techniques and applications to the transport sector. 4) Statistical and uncertainty analysis with a general understanding of relevant risk mitigation related concepts including robustness, resilience and reliability.	
Academic Skills	The main academic transferable skills that will be developed in this module will be critical thinking, analytical thinking and problem solving, specifically applied to the planning and operations of public transport and railways. The main personal skills that the student will build are organizational, team-building, and interpersonal and presentation abilities by means of concrete cooperation with colleagues in assignments having the scope of emulating projects and interactions in a professional environment.	
Literature & Study Materials	Public Transport System and Supply Planning and Operations: Lecture slides and two papers per week provided during the course Public Transport Demand and Network Planning and Operations: Lecture slides and pre- and post-lecture material including 3-5 reference papers are made available through Brightspace. Railway Operations and Control: A set of papers or book chapters made available on BrightSpace. Each lecture will require reading one paper/book chapter.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Calculator	
Collegerama	No	

CIEM6240	Road and Railway Engineering	15
Responsible Instructor	Dr. K. Anupam	
Responsible Instructor	Dr. H. Wang	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Prof.dr. Z. Li	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Prof.dr.ir. R.P.B.J. Dollevoet	
Instructor	Dr.ir. H. Farah	
Instructor	Dr.ir. M. Saeednia	
Contact Hours / Week x/x/x/x	x/x/x/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module addresses the design of road and railway infrastructures and their management, where the interface between traffic and transport as a service and infrastructure requirements for road and rail are fundamental. In the module, students will learn about classical design criteria for road and railway infrastructures. Moreover, the module covers current developments on topics such as a) robustness and resilience under climate change conditions, b) the use of new and more sustainable materials and design techniques, c) advanced IoT, d) AI-based monitoring and maintenance methods for road and railway infrastructure, and e) the effect of new technologies on design and maintenance of road/railway infrastructure, i.e. automated/smart vehicles and truck platooning. The module also covers aspects of management of road/railway infrastructure networks, focusing on the inter-relation between road/railway traffic management, network performance, and network quality. Particularly, methods to evaluate geometry, surface characteristics, reliability and user safety will be analysed by defining performance criteria, continuous performance monitoring, and different maintenance techniques. The module will enable students to ethically assess the road/railway infrastructure condition and its effect on traffic, capacity, and performance over time.	
Study Goals	On completion of this module, the student will be able to TTE-B4-ILO1: Identify key components for road and railway infrastructure design. TTE-B4-ILO2: Explain fundamental -mechanics and -dynamics of road and railway infrastructures tailored for traffic operation. TTE-B4-ILO3: Assess the road and railway infrastructure condition and predict their performances in terms of reliability, availability, maintainability, safety, sustainability, resilience, serviceability, and robustness for smooth traffic management. TTE-B4-ILO4: Design road and railway infrastructure based on traffic operation demands. TTE-B4-ILO5: Improve the sustainability and reliability of the road and railway infrastructure during the design-construction-operation cycle. TTE-B4-ILO6: Assess the effectiveness of state-of-the-art monitoring solutions and the relation between usage, condition and maintenance strategies for road and railway infrastructure. TTE-B4-ILO7: Evaluate innovative and ethical solutions to current and future challenges of road and railway infrastructure asset management. TTE-B4-ILO8: Propose innovative and ethical solutions to actual challenges of the road and railway infrastructure industries, based on operational traffic demands. TTE-B4-ILO9: Effectively and ethically report and communicate results with stakeholders from the industry.	
Education Method	Lectures / Blended Learning / Laboratory and Simulation Experiments / Excursion	
Assessment	Exam 1: Road Engineering - Written Exam (40%) Exam 2: Railway Engineering - Written Exam (40%) Exam 3: Integrated Project - Report - Group and Individual (20%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. The following are pre-knowledge assumed that most students will have: - Geometric design principles of roads, railways and their implications for safety, efficiency, equity and sustainability. - Road and railway components and understanding of its design elements, including key components and infrastructure from subsurface to surface. - Innovations in the field of road and railway infrastructure design, including societal challenges, current trends and future needs. - Transport network design and service planning. - KPIs design for evaluation of impacts of the transportation networks. From Program-Base and Faculty-Base, the following are topics we expect students to know about: - Coding (Python, Matlab, etc.) tools from MUDE (Faculty-Base) and Program-Base. Particularly, probabilistic models, sensing data measurement, collection and processing. - Topics from modules in the Program-Base such as deriving, analysing and solving fundamental problems via simpler engineering forms. - Basics of statics, kinetics, mechanics of materials, dynamics, elasticity, etc., covered in the program-base. - Fundamentals of the Ethics line. We will use specialized software for different examples.	
Academic Skills	Developments in the field of road and railway infrastructures are inherently multi-disciplinary. Students will learn topics including the reliable design of road/railway infrastructures, operations, sensing technologies, IoT, management, etc. As such, interdisciplinary and ethical thinking is a skill required to be further developed in the students. This module considers real case studies coming from challenges currently being faced in the industry. Thus, students will have to apply theory to practice and to develop problem learning and solving skills, while working efficiently and ethically correct. Finally, scientific writing, presentation, and time management skills are critical in this module. Particularly, together with addressing challenges from the industry, the assessment in this module considers reports and presentations to stakeholders.	
Literature & Study Materials	Road Engineering: Barriera, M., et al., In Situ Pavement Monitoring: A Review. Farrar, C.R. and Worden K. (2013), Structural Health Monitoring: A Machine Learning Perspective, Wiley, Chichester. Fraden, J. (2010), Handbook of Modern Sensors. Springer, fourth edition. Fwa, T.F., The Handbook of Highway Engineering. Kobbacy, K.A.H. and Murthy, D.P. eds., 2008. Complex system	

	maintenance handbook. Springer. Lajnef, N., Smart Pavement Monitoring System - Federal Highway. Márquez, A.C., (2007). The Maintenance Management Framework: Models and Methods for Complex Systems Maintenance, Springer, London, pp. 1-341. Miradi, M., Knowledge Discovery and Pavement Performance Intelligent Data Mining. Parmenter, D., 2015. Key performance indicators: developing, implementing, and using winning KPIs. John Wiley & Sons. Sharma, S., A data-driven approach for pavement surface distress classification. Railway Engineering: Esveld, C. (2001). Modern railway track, second edition. In Modern railway track, second edition (pp. 1-653). MRT productions. Farrar, C.R. and Worden K. (2013), Structural Health Monitoring: A Machine Learning Perspective, Wiley, Chichester. Fraden, J. (2010), Handbook of Modern Sensors. Springer, fourth edition. Márquez, A.C., (2007). The Maintenance Management Framework: Models and Methods for Complex Systems Maintenance, Springer, London, pp. 1-341. Online learning material based on Railway Engineering (TUD) ProfEd online courses. Integrated Project: A selection of papers and reports available in Brightspace. Lecture slides and pre- and post-lecture materials.
Judgement	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Calculator
Collegerama	No

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Year 2 2023

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Electives Construction Materials (CM)
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CIEM1301	Advanced Computation Mechanics	5
Responsible Instructor	Dr. M. Lesueur	
Module Manager	Dr. M. Lesueur	
Instructor	J.O. Colomes Gene	
Contact Hours / Week x/x/x/x	4hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course is composed of three thematics relating to advanced computational mechanics concepts: multiphysics, multiscale and optimisation. The first three weeks will consist of lectures to give an overview about each of those concepts which includes: _ Multiphysics: physical equations coupling, interface coupling (such as fluid-structure interaction) _ Multiscale: homogenisation methods, mixed-dimensionality (such as Discrete Fracture Network) _ Optimisation: softening regularisation, machine learning During those weeks, you will decide on one of those topics for your project. In the following weeks, you will gather a literature review on the method selected, and run simulations implementing the method. The project is kept on track with a weekly follow-up by the project supervisor. Eventually, a report in the format of an academic paper needs to be submitted for the assessment. The last weeks of the course consist of presentations of the projects to the whole class.	
Study Goals	In this course, you will study advanced numerical Finite Element Method that is used to simulate complex mechanical phenomena. On completion of this module, you should be able to: LO1. Identify which numerical method to apply in case of complex physical behaviour. (ILO 2, 4) LO2. Explain the algorithms needed to implement one selected method of your choosing in a Finite Element simulation. (ILO 2) LO3. Run Finite Element simulations implementing the selected method. (ILO 7)	
Education Method	Problem-based learning. The students are expected to study background material as supplied to them on Brightspace. They will select a topic of interest to which they will apply the theory and methods covered. Students will independently run a research-oriented project and deliver a report in the form of research article. Online and on-campus lectures and on campus workshops.	
Assessment	Peer feedback, lecturer feedback and self-evaluation. The assessment will be based on an individual or group project. The group project will be assessed in the form of both a presentation and a report in the format of an academic paper. The pass grade should be a minimum of 5.8	
Expected prior Knowledge	Fundamentals of Finite Element Method from MUDE	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a project so organizational and interpersonal skills will be developed as well.	
Literature & Study Materials	Lecture slides, lecture notes, papers and chapters from relevant literature (to be provided via Brightspace)	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CIEM1302	Forensic Construction Materials Engineering	5
Responsible Instructor	O. Copuroglu	
Module Manager	O. Copuroglu	
Instructor	A. Varveri	
Contact Hours / Week x/x/x/x	6hrs lectures 4hrs workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is offered for the students who are interested in forensic construction materials engineering, which comprises techniques and protocols to associate construction damage and failure to the materials characteristics and design. The course is especially suited for those students who would like to work in the field of consultancy. The content covers the following topics. Each week contains a workshop devoted to the applications of the forensic techniques on the construction materials. Week 1. Introduction and damage types Week 2. Sampling and specimen preparation Week 3. Optical microscopy Week 4. Corrosion techniques Week 5. Electron microscopy and microanalysis Week 6. X-ray techniques Week 7. Selected topics Week 8. Practical aspects, reporting and communication	
Study Goals	On completion of this module, the student will be able to: 1. Define activities in a forensic engineering research. Recognize the need for a forensic research in civil engineering cases. Classify damage types in structures and compare damage and failure examples. Evaluate and select in-situ testing methods, laboratory tests and numerical methods to identify the causes of damage and failure. [ILO 7, 8, 10] 2. Associate damage features with durability problems. Compose a protocol by choosing the suitable forensic tools for a case. Evaluate and criticise forensic reports from other parties. Develop a systematic approach (preliminary investigation-hypothesis-data collection-evidence analysis) to analyse the causes of damage and (premature) failure. Communicate forensic investigation details with the (multidisciplinary) stakeholders. [ILO 2, 3, 6, 9] 3. Illustrate working principles of different analytical tools for forensic engineering research. Classify analytical techniques based on their use i.e., chemical composition, phase analysis, pore analysis etc. Compare and explain these techniques to show they are not alternatives but complementary to each other. [ILO 8, 10] 4. Summarize forensic research tools and protocols for commonly used construction materials e.g.; reinforced concrete, asphalt concrete, masonry, glass etc. Distinguish advantages and limitations of in-situ testing over lab testing. [3, 6, 8]	
Education Method	Lectures and workshops	
Assessment	Formative feedback will be provided on the weekly assignments. Summative assessment: portfolio (100%) In this course, a portfolio is defined as: participation in the workshops, writing a technical report based on the workshop activities and participating in an oral discussion with the lecturer(s) concerning the workshop activities. The summative assessment will cover the following workshop activities + discussion. The order and subject of the activities may change depending on the availability of the facilities. This will be communicated with the students timely. Workshop 1: Sampling /specimen prep. / NDT (10%) Workshop 2: Optical microscopy characterization (10%) Workshop 3: Forensic techniques for corrosion (10%) Workshop 4: Electron microscopy and microanalysis (10%) Workshop 5: X-ray techniques (10%) Workshop 6: Selected topics (10%) Discussion: Discussion of the final consultancy report (40%) The portfolio discussion will take place in an individual conversation with the student where questions about the content of the module may also be expected. A successful participation in the elective requires a minimum grade of 5.8	
Expected prior Knowledge	The course is specifically designed for the students who follow the Construction Materials MSc. Track of the CE program. The students who would like to follow this course are expected to have already completed an advanced CE materials course e.g. science of construction materials and construction materials technology. Students of the SE-track can follow the course provided they have successfully completed the concrete technology unit in module B3. All other students are equally encouraged to follow the course, provided they have sufficient background knowledge on construction materials. Before enrollment, these students must seek approval of the course manager for admission.	
Academic Skills	1. Knowledge (Identify the damage and failure of construction materials in a structure) 2. Comprehension & Application (Describe and interpret the characteristics of damage and failure and hypothesize the causation) 3. Analysing (Analyze and classify the damage. Connect findings of the analysis to the causation) 4. Synthesis (Formulate a scientific forensic explanation of the observations and prepare a consultancy report) 5. Evaluation (draw conclusions based on scientific evidence. Explain and summarize the casuation of the observed damage and failure in a convincing discussion)	
Literature & Study Materials	Main literature (freely and digitally available via TU Delft library): - Scrivener, Karen, Ruben Snellings, and Barbara Lothenbach, eds. A practical guide to microstructural analysis of cementitious materials. Vol. 540. Boca Raton, FL, USA:: Crc Press, 2016. - Additional readers, print-outs and papers will be provided by the lecturers	
Judgement	See Assessment	

	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

CIEM1303	Upscaling Techniques in Construction Materials Design and Engineering	5
Responsible Instructor	Ir. C. Kasbergen	
Module Manager	Ir. C. Kasbergen	
Instructor	G. Ye	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. B. Šavija	
Instructor	R. Esposito	
Instructor	Dr. K. Anupam	
Contact Hours / Week x/x/x/x	6hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course on upscaling techniques in construction materials, design and engineering will consist of the following topics: - Introduction to computational upscaling techniques. - Introduction to homogenization and viscoelasticity. - Micromechanical aspects of response focusing on the interplay between adhesion and cohesion mechanisms, the effects of moisture, aging etc. - Numerical simulation based on homogenized properties. - Back-calculation of stress/strain response at individual components based on numerical simulation results. - Linear poroelasticity for the determination of elastic properties of composite-like materials. - Microporoelast fracture mechanics (linear poroelasticity and linear fracture mechanics). - Multiscale modelling of the transport properties of concrete, the deterioration process under coupled service conditions (multi-action). - Obtaining the microstructure of cement paste numerically as well as experimentally. - Upscaling of mechanical properties of cement paste. - Modelling of rheology, time-dependent response on creep, shrinkage, and fatigue of concrete. These topics will be supported with several workshops and guest lectures. The course may suit students with different specializations having the below-mentioned prerequisites. Within the Civil Engineering MSc programme, the course is recommended for students from the tracks Construction Materials, Structural Engineering and Geo-Engineering.	
Study Goals	On completion of this module, the student will be able to: 1. Explain microstructural modelling and upscaling approaches of multi-component materials. [ILO 2, 10] 2. Analyse the influence of microstructural features on the (mechanical, transport) properties of building materials. [ILO 2, 3] 3. Select an appropriate upscaling technique (computational, numerical or analytical homogenisation) for a given multiscale problem. [ILO 2, 5] 4. Apply microstructural modelling and upscaling approaches to predict engineering properties for multi-component materials (e.g., asphalt, concrete, metal composite). [ILO 2, 10]	
Education Method	Lectures, assignments, workshops	
Assessment	Students will receive formative feedback on assignments during workshops.	
Expected prior Knowledge	Portfolio with minimum passing grade of 5.8	
Academic Skills	FEM knowledge (finite element method) - at the level of CIEM1110, CIEM2110, CIEM5110, Strength of Materials CTB2210	
Literature & Study Materials	Critical analysis of model results and measurements, model selection, reporting	
Judgement	Lecture slides, reading material provided before/during lectures and workshops	
	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CIEM1304	Glass Science and Engineering	5
Responsible Instructor	Prof.dr.ir. P.C. Louter	
Module Manager	Prof.dr.ir. P.C. Louter	
Instructor	C. Noteboom	
Contact Hours / Week x/x/x/x	8hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Glass plays an ever more prominent role in structures through its application in facades, roofs, floors, beams, columns, walls and complete glass constructions. Consequently, there is an increasing emphasis on glass technology both in engineering practice and in research, resulting in a strong need for engineers in this field. This course therefore focuses on the application of glass in structures and covers the state-of-the-art in glass research and engineering. Through lectures, workshops, analyses and a hands-on design of a glass structure, students learn about the engineering and research aspects of glass, such as its mechanical behaviour, manufacturing processes, testing procedures, design standards, connection typologies, safety aspects, sustainability considerations and innovative glass structures.	
Study Goals	On completion of this module, the student will be able to: 1. Design a glass structure, including the verification of structural glass components, stability measures and the detailing of glass connections. 2. Implement safety and sustainability considerations in the design of a glass structure. 3. Analyse the latest developments in glass research and engineering. 4. Apply the latest glass developments in the design of a glass structure. 5. Analyse the material behaviour of glass in structures. 6. Assess the influence of production and processing techniques on the performance of glass in structures. 7. Evaluate the effects of loading and environmental conditions on the performance of glass and address these in the design of a glass structure.	
Education Method	Lectures, workshops and (site) visits.	
Assessment	Formative feedback on the assignments is given in weekly workshops and via interim presentations. Written report, consisting of 2 parts: 1. Structural design of a glass structure. 2. Analysis of a glass engineering and research project. Background information: During the course the students work on two assignments, namely: 1. Design a glass structure. 2. Analyse a glass engineering and research project. The design project and project analyses are supported through the weekly lectures. In addition, during weekly workshops, the students get advice and (formative) feedback from the tutors on their assignments. At given moments, the students shortly present their results in an oral presentation and get (formative) feedback from the tutors and their fellow students. Finally, the students document their design project and analyses in a written report, which is graded (summative assessment). A successful participation in the elective, requires a minimum grade of 5.8.	
Elective	Yes	
Expected prior Knowledge	No prior knowledge on glass is required. Besides civil engineering students from the MSc Track Construction Materials and the MSc Track Structural Engineering, also students from other tracks and other faculties are encouraged to participate. In case you need prior advice about your participation, please feel free to contact the responsible course instructor at christian.louter@tudelft.nl.	
Academic Skills	1. Designing (problem-based learning and solving; application of theory to new situations using the relevant equations; critical and interdisciplinary thinking) 2. Analysing (continuous investigation; information collection and synthesis) 3. Reporting (scientific writing and presentation) 4. Presenting (communication and responsiveness)	
Literature & Study Materials	Main literature (freely and digitally available via TU Delft library): 1. Chapter Architectural Glass Belis, Louter, Nielsen, Schneider 2. Springer Handbook of Glass J. David Musgraves, Juejun Hu, Laurent Calvez 3. Glass design and product standards Additional literature (optional): 1. Glass Structures - Jan Wurm 2. Glass Construction Manual - Schittich, Staib, Balkow, Schuler, Sobek 3. Glass in Structures: Elements, Concepts, Designs Rob Nijse 4. Structural use of glass in buildings, 2nd edition The Institution of Structural Engineers 5. Structural use of glass, SED 10 IABSE Haldimann, Luible, Overend	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	Yes	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Electives Geotechnical Engineering (GE)
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CIEM2301	Offshore Geotechnical Engineering	5
Responsible Instructor	Dr. K.G. Gavin	
Module Manager	Dr. K.G. Gavin	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	A number of countries have ambitious targets to develop offshore renewables projects. There are many technical challenges to be solved. From a geotechnical engineering perspective these include: (i) characterising unusual soils, (ii) considering the impact of irregular loading, including earthquakes, typhoons, impacts etc. (iii) developing frameworks to understand the response of soils to cyclic loading, (iv) updating design models developed from the oil and gas sector to the foundation systems deployed in renewables projects and (v) the consideration of installation risks. A series of short lectures will be by researchers and practitioners involved in developing and implementing state of the art design procedures (see content below). The lectures will be supplemented by small-group workshops on real-world problems. Marine Geology and Site Characterisation Offshore Soil Behaviour including cyclic response modelling. Design of Shallow Foundations (incl. mudmats, suction caissons, GBS). Development of the new ISO Pile Design Method for Axially Loaded Foundations The PISA model and alternative approaches for monopile design Anchor Systems for floating wind Installation of foundation systems	
Study Goals	On completion of this module, the student will be able to: - Describe the response of marine soils subjected to monotonic and cyclic loading [ILO 1,4] - Plan and implement an offshore site survey for soil characterisation purposes [ILO 1,4] - Analyse/design foundation systems for both bottom-fixed and floating offshore structures [ILO 1,4] - Explain the application and limitations of design codes [ILO 4,9]	
Education Method	Lectures and workshops.	
Assessment	Feedback and review sessions on workshops. Oral Exam A successful participation in the elective, requires a minimum grade of 5.8.	
Elective	Yes	
Expected prior Knowledge	BSc soil mechanics MUDE and Program Base	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a project so organizational and interpersonal skills will be developed as well.	
Literature & Study Materials	Provided in Brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	No	

CIEM2302	Trenchless Technologies	5
Responsible Instructor	Dr.ir. W. Broere	
Module Manager	Dr.ir. W. Broere	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course covers the use of trenchless technologies, which is a versatile installation method for small infrastructure (gas, water, sewers, etc.). It is meant as an addition to other specialistic courses and the topics studied here can also be applied in other courses. Next to the installation process and the design of the linings, the organisation of a TT project will be discussed. The course deals with basic aspects of: - Cables and ducts - Geology and geotechnics in relation to boring techniques and bore fluids - The technique of Horizontal Directional Drilling (HDD) - The technique of Micro-tunnelling - Boring equipment - Measuring equipment - Steering equipment - Technical calculations for HDD and Micro-tunnelling - Technical calculations for stresses in pipelines - Renovation of existing pipelines - Research on trenchless technology and innovative applications - Influence of contract types on project execution - Role of the contractor and engineering office - Risks and risk management - Case discussions	
Study Goals	On completion of this module, the student will be able to: identify and describe the methods available for trenchless installation and rehabilitation of cables and ducts [ILO 1,4,9] make a preliminary design for new pipe line installations [ILO 1,4]	
Education Method	lectures, design assignment	
Assessment	As part of the course, the students (in groups of max. 2 students) have to make a preliminary design and risk assessment for a river crossing of a large diameter pipeline, using HDD or micro-tunnelling. This design is based on the lecture notes as well as the relevant national codes and standards. The resulting design is discussed and commented during an (online) oral examination, during which also other topics from the course can be discussed. The assessment is based on the design assignment (100%). Peer feedback is provided between groups during the assignment. Feedback from the lecturers and TA is provided in a review session at the end of the lecture series. On demand feedback is also an option. A successful participation in the elective requires a minimum grade of 5.8.	
Expected prior Knowledge	Basic understanding of soil mechanics and structural mechanics is advisable but not mandatory.	
Academic Skills	Knowledge, application, analysis, design, evaluation, writing, reporting	
Literature & Study Materials	Obligatory lecture note(s)/textbook(s): Reader "Reader Trenchless Technology CIE5741" by W. Broere, S. van der Woude, Available via Microweb or as pdf on BS	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations The report for the design assignment can be used during the exam.	
Collegerama	No	

CIEM2303	Rock Mechanics Applications	5
Responsible Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Module Manager	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Ir. J.S. van der Schrier	
Contact Hours / Week x/x/x/x	5hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course covers the fundamentals of rock mechanics and rock engineering. It deals with the following: - characterisation and laboratory and in situ testing for the determination of properties of intact rock, discontinuities in rock, and discontinuous rock masses - mechanical behaviour of rock masses, including dynamic, time-dependent, and post peak behaviour. - basic principles of flow through discontinuities and discontinuous rock masses. - weathering and susceptibility to weathering of rock masses. - principle of excavation methods. Excavatability, wear and performance of cutting tools. Influence of blasting and other vibrations. Influence of stress and stress changes. - principle of support methods. - rock mass classification systems. Possibilities for analytical and numerical modelling of discontinuous rock masses. - application of principles of rock mass excavation and support to slopes, tunnel, dam, foundation, rock dredging or sea water breaker design. - monitoring, analysis, prediction and mitigation of subsidence due to solid removal. - application of engineering tools such as Total Geological Approach and function (system) analysis to projects case histories.	
Study Goals	On completion of this module, the student will be able to: - explain how to characterize, test, and monitor rock material, rock discontinuities and rock masses for a wide range of geo-engineering- and environmental applications [ILO 2, 3, 4] - select models that capture the essentials of the rock behaviour observed in the lab or in situ and highlight limitations of these models. [ILO 2] - appraise ground conditions and evaluate ground parameters, even from sparse and incomplete data sets, and formulate ground risks associated to projects. [ILO 3, 4, 5] - evaluate analytically the stability of various structures founded in or on rocks. [ILO 3, 5] - design basic stabilization measures and recommend more advanced methods for critical situations. [ILO 1, 9] - apply to construction projects lessons learned from case histories and engineering tools such as the Total Geological History and Function analysis. [ILO 1, 5] - Work on rock mechanics and rock engineering problems independently, compile results in written reports, and defend them orally. [ILO 2, 7, 9]	
Education Method	Lectures, guest lectures, laboratory, fieldwork, project work.	
Assessment	Formative exercises are embedded into the lectures, a laboratory test session and a 2 day rock mass classification fieldwork. A compulsory multi-stage project is integrated into the course and is used to assess the students. The students (in groups of max. 4 students) have to analyse critically data and make a preliminary site investigation, design and risk assessment for a fictive but very realistic construction project in rocks proposed by the co-lecturer. Students present their work at several stages of the project in an oral or written form and get feedback from the lecturers or peers. Groups upgrade their work according to the formative feedback they received and submit a final project report that is graded. A peer evaluation tool (Buddycheck) is used to evaluate the performance of each team member. The instructors grade the project report and incorporate the results of the peer evaluations in the final grade. For the sake of transparency, the grading rubric will be published in the BS page of the course before the start of the course. In A successful participation in the elective, requires a minimum grade of 5.8 for the multi-stage project report.	
Expected prior Knowledge	Basic understanding of (engineering) geology and geomechanics, advisable but not mandatory.	
Academic Skills	Several aspects of an academic attitude (knowledge, application, analysis, design, evaluation, writing, reporting) are integrated into the course. Aspects such as Ethics, integrity and citizenship are part of the case study provided by the industry.	
Literature & Study Materials	BrightSpace ppts and associated course material. Reference literature: - Practical Rock Engineering, Hoek (http://www.rocsience.com/hoek/PracticalRockEngineering.asp) - Engineering Rock Mechanics, John Harrison and John Hudson - An introduction to Principles - Illustrative worked Examples	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CIEM2304	Environmental and Energy Geotechnics	5
Responsible Instructor	Dr.ir. A.A.M. Dieudonné	
Module Manager	Dr.ir. A.A.M. Dieudonné	
Instructor	Prof.dr. P.J. Vardon	
Contact Hours / Week x/x/x/x	5hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. Multi-physics (thermal, hydraulic, mechanical) processes in environmental and energy geotechnics 2. Conceptual and mathematical modelling 3. Design of environmental and/or energy geotechnics systems 4. Evaluation of designs	
Study Goals	On completion of this course, the student will be able to: [LO1]: Identify and describe processes in environmental and energy geotechnical systems [LO2]: Develop a conceptual model of coupled processes occurring in environmental and energy geotechnical systems [LO3]: Derive and solve a mathematical model to quantify the impact on geotechnical systems of environmental or energetic loads [LO4]: Design a geotechnical project optimising the environmental and/or energetic behaviour	
Education Method	Balance of lectures, supervised tutorials and self-study.	
Assessment	Written formative assessment (based on check-list / rubric). Written summative assessment based on 2 design assignments. The pass grade is a 5.8/10.	
Expected prior Knowledge	MUDE	
Academic Skills	Report writing, critical assessment, problem solving.	
Literature & Study Materials	Will be provided via lecture material on Brightspace. Fredlund and Rahardjo (1993) Soil Mechanics for Unsaturated Soils. Wiley.	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Electives Hydraulic Engineering (HE)

CIEM3301	Building with Nature	5
Responsible Instructor	Dr. B.K. van Wesenbeeck	
Module Manager	Dr. B.K. van Wesenbeeck	
Instructor	Prof.dr.ir. S.G.J. Aarninkhof	
Instructor	Ir. S.A. Kalloe	
Instructor	Ir. G.G. Hendrickx	
Contact Hours / Week x/x/x/x	6-8 hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is on integration of the natural system in engineering thinking and design, called Building with Nature. Lectures are used to explain how Building with Nature contributes to climate change adaptation, the interaction between abiotic and biotic processes, Building with Nature principles and wave-vegetation modelling. The central element in this course is a case study of an estuary or bay. The students work in groups to disentangle abiotic and biotic processes in this area and to describe functioning of the system. This constitutes the basis to evaluate the problem at hand from perspectives of different stakeholders and to formulate objectives to serve these different stakeholders. Interventions to meet the objectives can be listed and evaluated using a multicriteria analysis. For a single intervention a more detailed design will be made. For this design wave-vegetation models will be used. Building with nature concepts will be assessed in a discussion session. Guest lectures will inform students on implemented Building with Nature practices and bring students in contact with companies that work with Building with Nature. Problem analysis, objective setting and intervention evaluation will be done in groups. The modelling exercise will be done in pairs. During tutorials the students can work on the assignments. A field visit of a day to the Oosterschelde area is part of the course and during the field visit a stakeholder game will be played. Reports will be delivered and graded individually.	
Study Goals	On completion of this module, the student will be able to: Explain how hydrodynamic and morphological processes in coastal ecosystems determine ecosystem occurrence. [ILO 3, 4] Give a clear problem description including climate change, flood risk and ecological considerations. [ILO 4, 5, 7] Translate the problem to clear project objectives. [ILO 7, 8] Consider multi-actor perspectives in evaluating the problem and in setting the objectives. [ILO 7, 9] Apply Building with Nature principles and natural system understanding in engineering designs. [ILO 1, 3] Assess intervention options to mitigate the problem, following a structured approach [ILO 2, 4] Run a basic wave-vegetation model for multiple conditions and scenarios [ILO 7] Make a qualitatively underbuilt Building with Nature design. [ILO 1] Evaluate long-term and large-scale effects of their proposed design considering climate change effects and ecological objectives. [ILO 1, 3, 4, 5]	
Education Method	Lectures, discussion sessions, case study, field visit, modelling exercise, report.	
Assessment	During the workshops the students can work on the assignments and will have the opportunity to ask questions and receive feedback from lecturers and peers. For this course a report will be delivered (100%), containing the following elements: Natural system description Problem analysis Stakeholder list SMART objectives 3-5 intervention options Multi-criteria analysis Field work monitoring plan Results of wave-vegetation modelling Design of a single option A successful participation in the elective, requires a minimum grade of 5.8.	
Elective	Yes	
Tags	Design Group Dynamics/Project Organisation Modelling Project	
Expected prior Knowledge	Bachelor in Civil Engineering MUDE and Program Base Preferred: CIEM3000 Hydraulic Engineering base module and CIEM3110 Hydraulic Engineering Fundamentals and Environments	
Academic Skills	This course will support the development of the following skills: critical and interdisciplinary thinking, problem-based learning and solving, information collection and synthesis, communication and responsiveness.	
Literature & Study Materials	Knowledge clips, reader material, and recommended papers from literature. Assignment material. EcoShape guidelines for building with nature design	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CIEM3302	Dredging for Sustainable Infrastructure	5
Responsible Instructor	J.A. Álvarez Antolínez	
Responsible Instructor	Prof.dr.ir. S.G.J. Aarninkhof	
Module Manager	Prof.dr.ir. S.G.J. Aarninkhof	
Instructor	Prof.dr.ir. C. van Rhee	
Instructor	M. van Koningsveld	
Contact Hours / Week x/x/x/x	2x2 hours Lectures x 7 weeks = 28 hours 1x5 hours Dredging Excursion = 5 hours 5x2 hours Training Session & question hrs = 10 hours total 43 contact hours	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The elective Dredging for Sustainable Infrastructure consists of 11 themes: - History of dredging technology supporting hydraulic infrastructure. - Workability of dredging and offshore operations. - Dredging equipment. - Dredging processes: breaching, cutting theory, hydraulic transport, and hopper sedimentation. - Source to impact: processes and source terms. - Environmental impacts: turbidity, sound, and emissions. - Assessment of dredging-induced turbidity. - Assessment of energy consumption during dredging activities and CO2 emissions. - Land Reclamation by hydraulic filling. - Survey, environmental monitoring, adaptive management. - Dealing with uncertainties. - Data science for dredging practices.	
Study Goals	On completion of this module, the student will be able to: LO1. Explain the physical processes of dredging of soils supported by soil mechanics and hydraulics theory. [ILO 2, 3] LO2. Analyse functional requirements of dredging operations regarding specifications, tolerances, quality control, and physical constraints in the fill or dumping areas and bunds. [ILO 1, 2] LO3. Recommend between dredging equipment based on functional requirements. [ILO 1, 2, 4] LO4. Assess the dredging supply chain regarding alternatives, realisation strategy, environmental conditions, workability, energy consumption and CO2 emissions, and their impact in relation to project milestones. [ILO 3, 4, 5, 8] LO5. Incorporate environmental aspects in dredging operations following adaptive management. [ILO 8, 10]	
Education Method	Lectures, several Lectures by guests from dredging industry, a PhD research lecture, tutorials, question hours.	
Assessment	Preparatory materials prior to each lecture (recommended reading). Lecture questions/polls/discussion are addressed with oral feedback. Several group assignments. Training sessions provide feedback on how to address assignments. The student will be evaluated on the basis of one summative assessment consisting of a case study to be carried out as group project. A successful participation in the elective requires a minimum grade of 5.8	
Expected prior Knowledge	In the elective Dredging for Sustainable Infrastructure [CIEM3302] it is expected knowledge and skills on Soil Mechanics, Fluid Mechanics, and Coastal Dynamics.	
Academic Skills	Information, collection, and synthesis; design of problem-oriented solution procedures; critical and interdisciplinary thinking; application of theory to new situations; and contextualize solutions with social, economic, environmental, physical, and procedure restrictions. In addition, students will need to collaborate intensively with their colleagues so organizational and interpersonal skills will be developed as well.	
Literature & Study Materials	Laboyrie, H. P., Van Koningsveld, M., Aarninkhof, S. G. J., Van Parys, M., Lee, M., Jensen, A., Csiti, A. and Kolman, R. (2018). Dredging for Sustainable Infrastructure. CEDA / IADC, The Hague, the Netherlands. Lecture notes Dredging Technology by van der Schrieck. Hydraulic fill manual for dredging and reclamation works, J. Hoff, A. N. Kolff, 2012	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	Yes	

CIEM3303	Advanced Modelling of Turbulent Flows and Transport	5
Responsible Instructor	Prof.dr.ir. W.S.J. Uijtewaal	
Module Manager	Prof.dr.ir. W.S.J. Uijtewaal	
Instructor	Dr.ir. M. Zijlema	
Contact Hours / Week x/x/x/x	6hrs interactive learning and 3hrs exercises	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. introductory lecture on the course 2. statistical description of turbulence (mean flow, higher statistical moments, (auto)correlations, Taylors hypothesis) 3. experimental techniques (single point vs. field measurements, flow visualisation, measurement techniques) 4. balance equations I (continuity equation, Navier-Stokes equations, turbulent kinetic energy) 5. Reynolds decomposition (ensemble averaging, Reynolds equations, closure problem, energy transfer, Kolmogorogov scales, energy spectrum, Kolmogorovs 5/3 law) 6. balance equations II (2DH/3D shallow water equations for free surface flows, scaling, hydrostatic pressure, wind stress, bottom friction, horizontal mixing, vertical mixing, constituent transport) 7. numerical techniques for 2DH shallow water equations (recap computational modelling, staggered schemes, time marching, momentum conservation, rapidly varied flows, flooding and drying) 8. turbulence modelling (mixing layer hypothesis, eddy viscosity concept, k-epsilon, LES) 9. numerical aspects of 3D turbulence modelling (regular vs. flexible meshes, sigma layers vs. z-layers for free surface flows, artificial vertical mixing, finite volume method, higher order upwind schemes+flux limiting) 10. turbulent flows in practice (boundary layers, law-of-the-wall, wall roughness, free surface flows in complex geometries, separation and recirculation) 11. turbulence diffusion (Reynolds analogy, dispersion, spreading) 12. non-hydrostatic flow modelling (vertical acceleration, subgrid, lock exchange, high performance computing)	
Study Goals	On completion of this elective, the student will be able to: 1. Identify different concepts to describe turbulence and characterize various turbulence processes as well as discuss the statistical description of turbulence, the various measurement techniques, and computational modelling concepts. (ILO 2, 4, 10) 2. Describe the balance equations for turbulent flows and discuss the necessity of the Reynolds decomposition and appreciate the closure problem. (ILO 2, 4, 10) 3. Identify the various numerical modelling concepts and discuss the abilities and restrictions of these concepts as well as categorise typical numerical issues, e.g. numerical properties and possible artefacts. (ILO 2, 4, 10) 4. To analyse observed problems and suggest solutions supported by his/her findings with evidences or examples for a given case. (ILO 2, 4, 7) 5. To execute a CFD model for computations, invocation of special functions, and performance of some other technical work. (ILO 2, 10) 6. To identify the various turbulence models and explain their abilities and limitations. (ILO 2, 4, 7, 9) 7. To make estimates of length and velocity scales characterizing the turbulent motions and judge the impact of turbulence on bed material, structures and the transport of matter (ILO 2, 4, 10) 8. To develop an engineering problem and select software packages, and the turbulence models in particular, in the context of applications with open water bodies as seas, estuaries, rivers, lakes and channels. (ILO 7) 9. To compose a view on new developments in the field of turbulence research and in the field of CFD. (ILO 2, 9)	
Education Method	Lectures, exercises, hands-on practical work, modelling sessions	
Assessment	The module includes a computer practical that consists of a number of assignments and an experimental practical. The student is strongly recommended to do the computational assignments in his/her own time or he/she can spend his/her time in our computer lab where assistance is available. For student convenience knowledge and experience with Python (or Matlab) is recommended. The computational practical is intended to train the student in identifying and categorising different turbulence and computational modelling issues, evaluating the results based on analysis and calculations, and getting used to work with software packages effectively. They are consistent with the learning objectives (see Table). By working through the assignments the student will gain a deeper insight into various turbulence phenomena and the abilities and limitations of the used computer model. Besides the computer practical the student is offered to do an experimental assignment in the Waterlab. The obtained data can be used to validate the used computer model. Students who hand in a report on their findings regarding their computational and experimental work will receive feedback, which will aid them in the summative assessment. The summative assessment of this course is an oral exam, which is organized after the course. The final grade is determined by the oral exam. The grading is related to the learning objectives. The pass grade must be a minimum of 5.8.	
Elective	Yes	
Tags	Computer Engineering	
Expected prior Knowledge	MUDE and Program base Preferred knowledge: Experience and knowledge of Python and Matlab CIEM3000 Modelling of flow and transport and bed protection is also recommended	
Academic Skills	- Understanding the physics of turbulent flows - Setting up a physics-based model - Programming in Python - Critical thinking - Reflecting on computational results and reporting on it	
Literature & Study Materials	Lecture notes, Turbulence: Introduction to Theory and Applications of Turbulent Flows, 2016 Frans T.M. Nieuwstadt , Jerry Westerweel , Bendiks J. Boersma, Springer Nature, Switzerland Numerical methods for shallow-water flow, 1994 C.B. Vreugdenhil, Kluwer, Dordrecht	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE):	

	https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Elaborated computer exercise
Collegerama	Yes

CIEM3304	Physical Oceanography	5
Responsible Instructor	Prof.dr. C.A. Katsman	
Module Manager	Prof.dr. C.A. Katsman	
Instructor	Prof.dr. J.D. Pietrzak	
Contact Hours / Week x/x/x/x	2x2 hours lectures x 7 weeks = 28 hours 1x4 hours Waterlab practical = 4 hours 4x3 hours practical / question hrs = 12 hours total 44 contact hours	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is an introduction to physical oceanography for civil and offshore engineering students, introducing them to the basic physical concepts behind oceanography. The aim of the course is to provide an understanding of the basic physics governing our oceans, from our estuaries and coasts to the deep ocean. Therefore, in this course emphasis is placed on the fundamental physical processes occurring in the ocean. This provides a basis for understanding geostrophic flows, tides, internal waves, estuary and river plume dynamics, the wind-driven ocean circulation and the thermohaline (density-driven) deep ocean circulation and its role for climate.	
Study Goals	On completion of this elective, the student will be able to: 1. explain the influence of Coriolis and stratification on a number of coastal and ocean flows. - ILO2,3 2. explain the basics of tides, wind-driven circulations, thermohaline circulation. - ILO2,3 3. derive the key equations governing the ocean circulation on coastal and basin scales and apply them to gain insight into the fundamental concepts and balances. - ILO2 4. explain the concepts of buoyancy and mixing and their role in coastal oceanography. - ILO2,3,4,5,8 5. evaluate the properties of internal waves, instabilities and stationary flows. - ILO2,3 6. analyse the role of the Coriolis force for the wind-driven circulation and in particular for western boundary currents. - ILO2,3 7. analyse the thermohaline circulation and its role for climate - ILO2,3,5,8	
Education Method	Lectures, laboratory group practical, practical / question hours	
Assessment	- Written Group Practical Density-Driven Flows (participation mandatory): Formative feedback on plan and end product (report or short movie) - Practical Sessions: Formative feedback on assignments / practice exam questions Written exam. The pass grade must be a minimum of 5.8	
Elective	Yes	
Tags	Fluid Mechanics Group work Lab Research Physics Practicals	
Expected prior Knowledge	Basic knowledge of continuum / fluid dynamics, f.ex. from Program Base CIEM0000	
Academic Skills	Critical thinking	
Literature & Study Materials	Lecture notes (available as hardcopy only) + study materials provided via Brightspace	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	none	
Collegerama	Yes	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Electives Hydraulic & Offshore Structures (HOS)
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CIEM4301	Onshore Hydropower	5
Responsible Instructor	J.P. Aguilar Lopez	
Module Manager	J.P. Aguilar Lopez	
Instructor	J.P. Aguilar Lopez	
Instructor	Dr. M. Marence	
Contact Hours / Week x/x/x/x	4hrs lectures/practicals per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The principles and practice of hydropower engineering is introduced in this course. Focus is on design of dam, diversion, and run of river powerplants, including the civil (structural, geotech, hydraulic), mechanical (turbines), and electrical (generation) components of these facilities. Newer technologies to be introduced in this course include low-head hydro, small hydro, in-conduit pressure recovery hydro and pumped hydro storage. Practical in class exercises are included which range to finding optimal hydropower sites based on GIS-tools to Assessing the stability of different types of dams. A major component of the course evaluation will be a group design of a hydropower project. The main topics of the course are: 1. Hydropower fundamentals 2. Small hydropower 3. Low head hydropower 4. High head hydropower 5. Dams 6. Dam Safety 7. Extreme Value Theory for Hydropower 8. Spillways 9. Foundations 10. Power waterways 11. Powerhouses and electromechanical equipment 12. Turbines 13. Project planning and costs 14. Pumped storage 15. Tidal Barrages and Hydro-kinetic Hydropower 16. Reservoir Operation	
Study Goals	On completion of this module, the student will be able to: LO1: Review the principles of hydro power-based energy generation and storage [ILO 1,4]. LO2: Identify key components of a hydropower development project [ILO 1,4] LO3: Evaluate the technical and economic feasibility, viability, and desirability of old and more recent hydro power projects [ILO 1,4]. LO4: Design a hydro power development project considering local and global environmental conditions [ILO 1,4].	
Education Method	Lectures and recordings, Practical exercises and design project.	
Assessment	Practicals and in class exercises Design Project (100% Mark) A successful participation in the elective, requires a minimum grade of 5.8	
Expected prior Knowledge	Fluid mechanics, Open channel flow, Hydraulic structures, Geotechnical engineering (BSc level)	
Academic Skills	Basic numerical analysis (excel, python, Matlab, etc).	
Literature & Study Materials	Handouts, e.g. ppt's and pdf's of lecture material or interesting background or specialist articles, will be put on BrightSpace. Other optional lecturenote(s)/textbook(s): 1. Hydropower Engineering Handbook Gulliver, John S 2. The engineering of Large Dams, H.H. Thomas (2 volumes) 3. Low Head and High Head Power Plants, E. Mosony (3 volumes). 4. Te Chow, Open Channel Hydraulics Optional lecture notes 'Water Power Engineering, Principles and Characteristics'. Hardcopy can be purchased at the VSSD-office, but PDF will be made available via BrightSpace.	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CIEM4302	Cold Regions Engineering	5
Responsible Instructor	Ir. J.S. Hoving	
Responsible Instructor	Dr.ir. H. Hendrikse	
Instructor	Ir. J.S. Hoving	
Instructor	Dr.ir. H. Hendrikse	
Contact Hours / Week x/x/x/x	4 (lectures) + 2 (workshop / question hour related to assignment)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Cold regions engineering relates to all engineering activities related to coping with low temperatures, specifically freezing temperatures, in the cold regions of our planet. The formal education of most civil engineers does not include an adequate understanding of the unique characteristics that work in these cold regions brings. Challenges include engineering problems related to soil mechanics, hydraulics, heat flow, materials and construction, among others, in cold environments. Therefore, this module addresses a broad range of cold regions engineering design and construction activities. Topics covered in this module are: - Ice features and ice regimes; - Ice morphology, ice physics and ice mechanics; - Thermodynamics of ice growth and icing; - Cold regions structures and development; - Ice loads on port- and offshore structures; - Dynamic ice-structure interaction; - Material behavior at low temperatures*; - Construction on ice and permafrost: foundations, roads and airfields*; - Hydraulics in cold regions*; *) may not be covered depending on external circumstances	
Study Goals	On completion of this module, the student will be able to: LO1. Classify cold region structures and developments. [ILO 5, 10] LO2. Predict expected ice features and ice regimes in different cold regions. [ILO 2, 5] LO3. Account for material behaviour in cold environments. [ILO 1, 4, 5] LO4. Analyse the physics, mechanics and thermodynamics of ice. [ILO 4, 10] LO5. Assess static ice-induced loads on port, coastal, marine and offshore structures. [ILO 2, 3, 4] LO6. Analyse and select optimal structure types and structural designs for performance in cold regions depending on local conditions. [ILO 1, 2, 3, 4, 7 8] LO7. Characterize and model dynamic ice-structure interaction. [ILO 2, 4]	
Education Method	Lectures, discussion of examples during practicals and exercises, as well as formative assignments.	
Assessment	Several formative group assignments on: 1. Ice material: Growth & Thermodynamics 2. Ice-structure interaction 3. Ice dynamics These assignments do not count towards the final grade, but serve as a means to test your knowledge and receive feedback on your understanding. Note furthermore that the size of the groups depends on the number of students in the course and submission deadlines for the group assignments are communicated through Brightspace. The course is concluded with an individual written open-book exam, where the pass grade is a minimum of 5,8.	
Special Information	This course is a new and updated version of the former course OE44115 Arctic Engineering.	
Expected prior Knowledge	Structural Mechanics / Dynamics (BSc)	
Academic Skills	Critical thinking, knowledge, design of experiments, design, writing, reporting	
Literature & Study Materials	Chapters from the ASCE Handbook for Cold Regions Engineering, Lecture notes on offshore engineering in Cold Regions, lecture slides, selection of scientific publications.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Open book exam	
Colleggerama	Yes	

CIEM4303	Flood Risk	5
Responsible Instructor	mr.dr.ir. C. Mai Van	
Module Manager	mr.dr.ir. C. Mai Van	
Instructor	Prof.dr.ir. S.N. Jonkman	
Instructor	Dr.ir. M. Kok	
Instructor	Dr.ir. A.M.R. Bakker	
Instructor	mr.dr.ir. C. Mai Van	
Instructor	J.P. Aguilar Lopez	
Instructor	Ir. R.E. Jorissen	
Instructor	M. Pregnolato	
Co-responsible for assignments	M. Pregnolato	
Contact Hours / Week x/x/x/x	4hrs lectures per week plus 2 x 4 hours workshops during the course.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The Flood Risk Elective Module aims at the engineering performance of flood risk analysis, and evaluate flood risk reduction strategies and measures for flooding prone areas as well as flood defence systems that protect against the effects or impact of waves, currents and high-water levels, starting with understanding the underlying processes and related systems. It concerns the Quantitative flooding risk analyses of flood prone regions (polders) with and/or without the flood protection system which serve for addressing and quantify risks, anticipate changes and to continuously optimise risk mitigation measures and flood risk reduction strategies for the regions.	
Study Goals	On completion of this module, the student will be able to: - Identify water related risks for flood prone regions with and without flood defence systems in different geographical features and threats i.e. riverine, coastal, mountainous and urban areas [ILO 3,5]; - Perform a functional analysis of a flood protection system given its protected region [ILO 3,5]. - Quantify key elements of flood risk analysis i.e. flooding probabilities and consequences [ILO 3]; - Perform additional essential elements of flood risk analysis for flood prone regions: setting-up hydraulic boundary conditions, estimation of failure probability of defence systems, analysis of breach growth, (simple) overland flow modelling, damage assessment and estimation [ILO 3,10]. - Formulate flood risk reduction strategies and mitigation measures based on current and future needs [ILO 1,8,9]	
Education Method	Lectures, workshops, case study: how to determine safety standards for a certain dike ring(s) in the Netherlands / question hours	
Assessment	The CE HOS Flood Risk Elective Module is assessed by assignment portfolio (report) and oral exam (Q1): - Assignment portfolio (report): 70% - Oral exam by means of a final presentation and Q&A: 30% A successful participation in the module, requires a minimum evaluation of 5.8.	
Expected prior Knowledge	MUDE: Statistics, Probabilities, Extreme value Analysis, Uncertainties Analysis, Probabilistic design approach; and Modelling Techniques; Program Base: System perspective and Integrated system design	
Academic Skills	Critical and interdisciplinary thinking, identification of problem-oriented solution procedures, information collection and synthesis, application of theory to new situations using the relevant approaches and correlating solution with social, economic, physical insights and design procedures. Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well.	
Literature & Study Materials	General lecture notes and books used in the master programme + specific information on Brightspace.	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	yes	
Collegerama	Yes	

CIEM4304	Hydraulics Fieldwork	5
Responsible Instructor	Dr.ing. M.Z. Voorendt	
Responsible Instructor	M.A. de Schipper	
Module Manager	Dr.ing. M.Z. Voorendt	
Instructor	Dr. S.E. Vos	
Instructor	Dr. A. Amiri Simkooei	
Instructor	Dr. R.C. Lindenberg	
Contact Hours / Week x/x/x/x	4 hrs/week lectures + tutorials	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course is given as a block course. The duration of the fieldwork itself is a full-time week and usually takes place at the end of September / beginning of October. The elective is offered to a limited (25 to 30) number of students. The application procedure is announced on Brightspace in Q4 preceding the fieldwork. There are two potential options for the execution of the fieldwork. It is announced in Q4 preceding the fieldwork which option(s) will be organised. Option 1: A hydraulic structures oriented fieldwork. This type of fieldwork acquires data for the evaluation of an existing configuration, or for the design of a new structure like a breakwater, marina or artificial island. Typical components of such a fieldwork are beach profile measurements, echo sounding for bathymetry measurement, sand sampling, observation of instantaneous wave conditions, inspection of breakwaters. Option 2: A morphologically oriented fieldwork. Measurements are done on the beach, like profile measurements, sand sampling, sediment transport measurements, etc. The collected data will have to be checked to the deep-water boundary conditions (waves and currents) which are available via the routine measuring devices. The introductory lectures are given in collaboration with the Coastal Remote Sensing elective within the MSc program Applied Earth Sciences, organised by the Department of Geoscience and Remote Sensing.	
Study Goals	On completion of this module, the student will be able to: - plan and execute a measuring plan needed to solve problems in the field of hydraulic engineering [ILO 2,3,7]. - decide what types of data have to be collected, according to what method, with a useful number of samples (with respect to resolution and frequency) and of sufficient quality [ILO 4,7]. - process raw data and organize them into a measurement record, including metadata [ILO 2]. - analyse these data and draw conclusions relevant to hydraulic engineering. Uncertainties like weather changes, unexpected circumstances, unknown environment and lacking local background are dealt with [ILO 1,2,3,8,10]. - organize and operate in a group performing a measurement campaign [ILO 9,10]. - Present the findings in a report and in an oral presentation [ILO 9].	
Education Method	Fieldwork with introductory lectures. The three weeks preceding the fieldwork are used for preparation: 4 hrs/week lectures + tutorials. The weeks after the fieldwork are used for processing and analysing the data, writing a report and giving a final presentation. Assignment (100%): participation, report, presentation and reflection. A successful participation in the elective, requires a minimum grade of 5.8.	
Assessment	Hints and feedback on the activities and analyses are given throughout the module.	
Expected prior Knowledge	Knowledge and skills regarding: - Assessment of the impact of coastal engineering interventions or structures and their technical design, or: - Earth observation technologies. Preferred prerequisites: Dams, Dikes and Breakwaters (CIEM4220) for option 1; Coastal Engineering (CIEM3210) for option 2. or AES MSC discipline core Earth Observation Technologies	
Academic Skills	Critical and analytical thinking is needed to determine what data are required to solve specific engineering problems, and to determine the required accuracy and amount of data (sample frequency and distance between measuring points). Raw data should be processed and interpreted. Problems during measuring, processing and interpretation should be solved. Methods and planning will have to be adapted to changing circumstances. The findings have to be reported.	
Literature & Study Materials	General lecture notes and books used in the master programme + specific information on Brightspace.	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Electives Structural Engineering (SE)
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CIEM5301	Shell Structures	5
Responsible Instructor	Dr.ir. P.C.J. Hoogenboom	
Module Manager	Dr.ir. P.C.J. Hoogenboom	
Contact Hours / Week x/x/x/x	5hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course covers analytical and numerical methods for analysing shell structures. The following is an overview of all the key terms that will be further explored within this course: Sagitta, a/t Ratio, Corbel dome, Arches, Catenary, Pressure lines, Middle third rule, Optimal arch shape, Barlow's formula, Thin or thick shells, Section forces and moments, Force flow in a dome, Edge disturbance, Comparison of moments in arches and domes, Optimal dome shape, Curvature, Gaussian curvature and mean curvature, Orthogonal parameterisation, Curvilinear coordinate system, Lamé parameters, Equations of Gauss and Codazzi, In plane curvature, Shell membrane equations, Membrane forces in a spherical dome, Derivation of the membrane equations, Stresses in a simply supported tube, Comparison to a beam model, Bernoulli's hypothesis, Deformation of a simply supported tube, Misfit at the boundaries, Tube edge disturbance, Sanders-Koiter equations, Compatibility equation, Shell differential equations, Plate boundary conditions, Edge shear stresses, In plane shear forces, Derivation of Sander-Koiter equation 6, Shell boundary conditions, Dependent quantities, Generalised edge disturbance, Beam supported by springs, Influence length, Shell finite element types, Nodes and integration points, Mesh locking and hourglass modes, Benchmarks, Model size restrictions, Finite element analysis of a 4-hypar shell, Interpreting shell finite element results, Hypar curvature, Reinforcement, Particle-spring method, Timber grid shells, In-extensional deformation, Theorema egregium, Gauß-Bonnet theorem, Formulas for point loads, Trajectories, Umbilical patterns, Finite element analysis of a conoid, Natural frequencies, Linear buckling analysis, Non-linear analysis, Cylinder buckling shapes, Differential equation for shell buckling, Buckling load factor, Design check of buckling, Imperfection sensitivity, Exceptions to imperfection sensitivity, Koiter's law, Knock down factor, Linear buckling analysis, Nonlinear finite element analysis, Model for imperfection sensitivity, Measuring shape imperfections, Stiffeners, Modal analysis, Rigid body modes, Equation of motion, Wave numbers, Vibration of hemispheres, spheres and cylinders, Natural frequency of a square shell, Measuring vibrations, Transient analysis, Damping ratio, Partial safety factors, Computation of failure probability, Acceptable failure probability, Many examples of remarkable shell structures.	
Study Goals	On completion of this module, - you can explain the force flow in shell structures [Civil Engineering MSc ILO 1]. - you can manually calculate stresses, deformations, buckling loads and natural frequencies of elementary shell shapes [2]; - you can explain the scientific approach to deriving and solving the governing differential equations and boundary conditions [2, 8]; - you can perform finite element analyses of shell structures. This includes linear analysis, vibration analysis, linear buckling analysis and nonlinear buckling analysis [2]. - you can interpret and verify the finite element results [4]. - you can answer the following questions about your shell designs: Will it deflect too much? Will it yield? Will it crack or break? Will it vibrate annoyingly? Will it buckle? Will it be unsafe? What causes this and how can I improve it? [1, 2, 4, 10]	
Education Method	Lectures and two workshops	
Assessment	Questions will be put forward in the lectures. Students are encouraged to submit answers. The teacher will give feedback on each answer. Written exam, 3 hours, exam grade is elective grade. A successful participation in the elective, requires a minimum grade of 5.8	
Expected prior Knowledge	MUDE Preferably Track base module Structural Engineering Contact the responsible teacher to discuss the possibilities of taking this course without the preferred prerequisite knowledge.	
Academic Skills	Structural modelling, linear and nonlinear finite element analysis, deriving differential equations and boundary conditions, solving differential equations analytically and computationally, validating and interpreting finite element results.	
Literature & Study Materials	Handouts provided in the lectures	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations A calculator will be needed. It is allowed to use books, handouts, notes, old exams, computer, tablet, software and internet during the exam. Obviously, it is not allowed disturb others and to communicate with others during the exam. It is not allowed to use a phone. It is not allowed to use software with which one can communicate.	
Collegerama	No	

CIEM5304	CO2 Neutral Structures	5
Responsible Instructor	Prof. M. Veljkovic	
Module Manager	Prof. M. Veljkovic	
Instructor	Dr. P.C. Rem	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	Dr. H.M. Jonkers	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Contact Hours / Week x/x/x/x	4 hrs/week at first 4 weeks (lectures by instructors and guests), 2hrs/week remaining period (consultations), the last week 4 hours (presentation workshop)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The module consists of three parts. The introductory part will be provided by responsible instructors covering various structural expertise focusing on sustainability aspects of the built environment in which load bearing structure is made of concrete, steel, timber or as hybrid system (steel, biobased, concrete). Different views on interpreting basic terms used in policy documents and literature relevant to CO2-neutral structures/built environments will be discussed. In the second part, case studies of flagship projects in terms of CO2-neutral structures will be presented by architects and engineers from industry and related disciplines. In the third part of the module, students (in a group of max 4 students) will select one of the presented case studies and provide an alternative solution with the intention to improve solution addressing weaknesses of the presented solutions with a focus on CO2 neutrality.	
Study Goals	On completion of this module, the student will be able to: Critically assess and apply civil engineering technical process of generating and evaluating alternatives in a complex natural and societal environment [ILO 4]. Design, verify, detail, manage, assess, and decommission new and existing buildings, structures and infrastructure, whilst observing functional, economic, ethical, social and sustainability constraints [ILO 1,4,5,8]. Design potential routes for prefabrication, modular construction, recycling and reusing of building components [ILO 7] Develop strategies for sustainable use of materials in structures with the goal of zero or negative CO2 balance [ILO 8]	
Education Method	Classroom teaching, best practices from industry and other faculties/universities, project in a group of max 4 students	
Assessment	Short, one-week tasks during the course, to exercise various aspects of built environment with respect to sustainability. A written report, and a presentation at the final workshop with all teachers and guests present. The grading will be done using criteria for projects at MSc level. A successful participation in the elective requires a minimum grade of 5.8.	
Expected prior Knowledge	BSc Concrete Structures, Concrete and Steel Structures, Climate Impacts and Engineering, Construction Mechanics. MUDE + Program Base in the MSc programme Preferably also: Module A and B of the Structural Engineering track (and Construction Materials track)	
Academic Skills	Students will acquire new knowledge by self-studying and critically reviewing possible structural alternatives, considering the influence of civil engineering systems on the environment and vice versa. A holistic view on the lifetime of the built environment, including different scenarios for use. Students will gain experience in Community Engaged Learning, i.e. the module is based on partnership between academic and non-academic actors to learn with, from and for each other to contribute to sustainable progress which is scientifically based.	
Literature & Study Materials	Flagship reports and analysis, teaching material at brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Collegerama	No	

CIEM5305	Fire Safety Design	5
Responsible Instructor	Ir. G.J.P. Ravenshorst	
Module Manager	Ir. G.J.P. Ravenshorst	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	Prof. M. Veljkovic	
Co-responsible for assignments	M. Poliotti	
Contact Hours / Week x/x/x/x	6 x 45 minutes contact hours week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course will contain classroom lectures to explain the aspects that have to be regarded to make a fire safe design. The content will be delivered by experts on construction materials from TU Delft and experts from industry on fire physics, fire testing, fire safety measurements and fire safety engineers. The content will address: - Background of the fire safety regulations according to national and European laws. - Introduction into fire physics. Standard and natural fire curves. Thermal loading. Theoretical and numerical methods to predict the fire development and temperature distribution in compartments. - Explanation in the difference in Resistance to Fire and Reaction to fire, and testing methods to determine these properties. - Smoke control and smoke propagation. Escape scenarios and human behaviour. ASET and RSET. Active and passive fire safety measurements. - Response of construction materials steel, timber and concrete to thermal loading. Contribution of the construction material to the fire loading. - Design and verification rules for construction materials steel, timber and fire under thermal loading. - Fire Safety Engineering: Risk analysis to evaluate the effectiveness of fire safety measures.	
Study Goals	Besides the verification of every structure for the Ultimate Limit State (ULS) and Serviceability Limit State (SLS), it also has to be assessed for the accidental load case fire. This load case can be governing in the design, therefore every engineer should have knowledge how to handle the load case fire and how to verify a structure based on the material response on the high occurring temperatures for the different construction materials steel, timber and concrete. On successful completion of this module, the student will be able to: 1. Explain the fire physics with respect to the initiation and development of a fire in structures. [ILO 1,4] 2. To predict the temperature development of a compartment subjected to fire loading (ILO4) 3. Explain the behaviour of materials and structures under fire loading.(ILO 1,7) 4. apply the background of the fire safety regulations on national and European level.(ILO 3) 5. To identify active and passive fire safety measures and how they influence the safety of the structure.(LO 7) 6. To apply the design and verification rules for structural safety under fire conditions for the construction materials steel, timber and concrete, and their contribution to the fire load. (ILO 1,3,9)	
Education Method	Classroom lectures and self study. The consequences of fire safety on the structural design will be discussed with students and students will be challenged to apply the gained knowledge in a case study in the assignment.	
Assessment	Consultation hours for the assignment the students work on will be scheduled. The assessment will be in the form of an assignment (individual or group) where on the basis of a case study of a structure will have to design and verify components of steel, timber and concrete, predict the temperature development in a compartment of that structure and design passive and active fire safety measures. This will be presented by the students in a report and a presentation. The pass grade must be a minimum of 5.8	
Expected prior Knowledge	Bachelor of Civil Engineering Knowledge of Concrete and Steel Structures (BSc course level) Preferred prerequisite: CIEM5000 Track Base Structural Engineering Students are expected to be able to verify the capacity of a beam, columns and walls made of steel, timber and concrete, under bendings and axial load (including buckling verification). Additional preparatory materials can be provided by the responsible lecturer for student who do not meet the preferred prerequisite knowledge.	
Academic Skills	Critical and interdisciplinary thinking, problem-based learning and solving, scientific writing and presentation, application of theory to new situations.	
Literature & Study Materials	Course reader "Fire safety Design" Lecture presentations Additional readings on brightspace	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Colleggerama	No	

CIEM5306	Assessment of Existing Concrete Structures	5
Responsible Instructor	Prof.dr.ir. M.A.N. Hendriks	
Responsible Instructor	Ir. E.O.L. Lantsoght	
Module Manager	Ir. E.O.L. Lantsoght	
Instructor	Dr. Y. Yang	
Contact Hours / Week x/x/x/x	6hrs lectures	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In many countries the originally devised design life of a substantial part of the existing buildings and infrastructure has been reached or will be reached soon. These structures need to be assessed. Most often, existing structures are verified using simplified procedures based on the partial factor method which is commonly applied in design of new structures. Such assessments are often conservative and may lead to expensive renovation or unnecessary replacements of structures. Recently, the engineering practice and the research community started to acknowledge that an assessment of existing structures is fundamentally different from designing new structures. For existing structures, it pays off to consider more advanced load and capacity models, as well as the overall structural behaviour, and to align the reliability assessment of concrete structures with all the information we have. The result of the underlying reliability considerations is a different set of load factors for assessment, as well as a philosophy of increasing levels of approximation for assessment. More advanced models are then referred to as models with higher levels of approximation. In this elective you will learn about these types of models and associated considerations. On the one hand we have more information about an existing concrete structure than we have for a structure which is still in the design process. E.g., the structure has shown a certain capacity over its lifetime, we have more knowledge about the loading, or we have knowledge about the actual concrete strength properties. Monitoring data and results from proof test might further contribute to the knowledge of a structure. We also must deal with higher uncertainty. E.g., corrosion levels might be unknown, and the structural behaviour of structural vintage details might be hard to quantify. A proper assessment should take all this information into account. This elective will address the following topics: evolution of design codes and practices, assessment at various levels of approximation, and the link to reliability-based safety philosophies, modelling of the structures with vintage detailing (nonconforming design), determination material properties and the deterioration condition of structures, methodologies to deal with involved uncertainties, field testing, and in particular bridge load testing, assessment studies under multiple hazards, such as seismic action, structural impact of deteriorating mechanisms, global structural behaviour for assessment: transverse distribution, unintended composite action, contribution of non-structural components, decision-making strategies	
Course Contents Continuation	The course material of the elective will be applicable to both concrete buildings and concrete infrastructures. Most applications will be shown for concrete bridges. Since the assessment of existing concrete structures is an active research field and since codes for existing concrete structures are far from mature, this course will be updated from year to year to follow the most recent state of the research and practice. MUDE: The first lectures of this elective will focus on the topic of safety philosophy of existing concrete structures. The influence of various sources of uncertainty, with regard to the state of information and the degradation and deterioration, will also be discussed throughout this elective. Ethics: A discussion on the choices of the target reliability for concrete structures at different stages, namely, design, assessment, renovation is given. It will be made clear that such choice is not based on numbers but also on the ethics considerations. The topic of decision-making strategies for the assessment of existing structures will include the ethical considerations involved with this process. Design skills: This elective will lean on general design skills of structural engineering students, which can then be applied for the analysis of existing concrete structures.	
Study Goals	At the end of this course, you will be able to: 1. Describe and explain the safety philosophy of existing concrete structures [ILO 3] 2. Evaluate the capacity of existing concrete structures with and without deterioration [ILO 4] 3. Exemplify the evolution of concrete codes and design practices and its impact on assessment [ILO 1,4] 4. Apply decision-making strategies for the assessment of existing bridges [ILO 5,7,10] 5. Appraise available test methods that can provide additional information of the structure (material properties, boundary conditions, or the bearing capacity, etc.) [ILO 5,10]	
Education Method	Synchronous online and in-person lectures. The lectures will include discussions and smaller exercises. The students will work on case studies outside of the lecture hours.	
Assessment	3 assignments Assignment 1: Report on safety levels and development of codes Assignment 2: Calculation exercise on capacity with and without deterioration of a structural element Assignment 3: Case study of existing structure and proposed field testing method. Assignments 1 and 2 are individual. Assignment 3 is in groups of 3-4 students. Feedback of assignment 1 will first be given by a student peer and then by the instructor by filling out a rubric. Feedback on assignment 2 will be given by the instructor. Feedback of assignment 3 will be given by peer groups first and then by the instructor by filling out a rubric. 100% on oral exam A successful participation in the elective, requires a minimum grade of 5.8.	
Expected prior Knowledge	Knowledge of material properties of concrete, concrete mix design, mechanics of materials. Design of reinforced and prestressed	

	concrete structural members. Structural analysis and matrix analysis of structures. Understand the safety philosophies and theoretical background of Eurocode. This elective is also suited for participants who have not taken module B3.
Academic Skills	Critical and interdisciplinary thinking, problem-based learning and solving, information collection and synthesis, communication and responsiveness, application of theory to new situations using the relevant equations, making inferences of physical situations by analysing observations/data
Literature & Study Materials	Course reader published on Brightspace
Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	The oral exam is a closed-book exam.
Collegerama	No

CIEM5307	Forensic Structural Engineering	5
Responsible Instructor	Ir. K.C. Terwel	
Responsible Instructor	H. Alkisaai	
Module Manager	Ir. K.C. Terwel	
Instructor	Dr.ir. M. Korff	
Instructor	O. Copuroglu	
Instructor	Dr.ir. H.R. Schipper	
Contact Hours / Week x/x/x/x	4hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course partially consists of theory on structural damage, failure mechanisms, investigation methods and instruments, and legal aspects. The other part consists of cases to illustrate these aspects. Theory: &#9679; Structural damage: what is damage? Various types of failure mechanisms. Performance criteria and codes. How often does failure occur? What are technical causes? What are human and organisational causes? &#9679; Legal aspects: liability, Insurance, Court, expert witnesses &#9679; Delft Method for Forensic Investigations: steps of investigations, life cycle, Tree House of Failures, Ring of Trustworthiness &#9679; Instruments for investigation: for example microscopy &#9679; Damage in foundations and retaining structures: e.g. deterioration of timber piles, unequal settlements, failures related to deep excavations and retaining structures Cases with for example the following types of damage: &#9679; Damage in concrete: e.g. temporary structures, reinforcement in ridges &#9679; Damage in steel: e.g. connections and instability &#9679; Damage in timber: e.g. connections &#9679; Damage in finishing structures: e.g. leakage and connections &#9679; Damage in foundations: e.g. deterioration of timber piles, unequal settlements, building pits	
Study Goals	On completion of this module, the student will be able to: 1. Explain various types of failure and recognize important structural failure mechanisms in steel, timber, concrete, finishing structures and foundations. [ILO 1, 2, 4, 6, 8, 10] 2. Identify the role of human and organisational factors as contributing factors to failure. [ILO 1, 3, 4, 8] 3. Point out legal aspects that are important for liability after occurrence of damage. [ILO 3, 8] 4. Explain the Delft Method for Forensic Investigations and possible instruments for forensic investigations. Assess the usefulness of application of various instruments. [ILO 3, 8, 10] 5. Apply the various steps of Delft Method for Forensic Investigations in a forensic investigation process. [ILO 2, 6, 8, 9, 10] 6. Develop design measures to deal with important failure mechanisms. [ILO 1, 2, 4, 6, 8, 10]	
Education Method	&#9679; Workshops related to relevant aspects of Delft method for Forensic Structural Engineering. &#9679; (Guest) lectures with CSI moments where students are requested to think about possible causes. &#9679; Live instructions for the case study, to receive feedback Brightspace environment offers structured overview of tasks per week, in combination with instruction videos and study material.	
Assessment	A student compiles a portfolio throughout the course consisting of at least in class participation, a case-study report and a theory test. Feedback is provided throughout the process. An oral discussion on the portfolio of a student. The student passes the course if a minimum grade of 5.8 for the oral discussion was awarded.	
Expected prior Knowledge	This course has a maximum student participation of 40, because of the practical part (workshop) of the course. Prerequisite courses: Building Structures 1 and/or Concrete structures 2	
Academic Skills	Critical and interdisciplinary thinking, problem-based learning and solving, continuous investigation, information collection and synthesis, scientific writing and presentation. Making inferences of physical situations by analysing observations/data.	
Literature & Study Materials	TBD Possibly: - Why Buildings Fall Down (book) - Selected scientific papers on Forensic Structural Engineering (FSE) - SED, a book with multiple FSE examples (IABSE)	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CIEM5308	Parametric Design and Digital Fabrication	5
Responsible Instructor	Dr. M.A. Popescu	
Module Manager	Dr. M.A. Popescu	
Contact Hours / Week x/x/x/x	8hrs lectures/workshops 1-day symposium	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Parametric Design and Digital Fabrication will provide you with skills and (deepened) knowledge of using computational techniques in the design process within the paradigm of parametric design. This will be taught through tools as they are currently implemented in modern software. This includes 3D modelling (e.g., Rhinoceros 3D possibly coupled with visual programming plugins such as Grasshopper 3D) and coding (e.g., Python and COMPAS). We will be looking at principles of designing efficient structural forms using form-finding techniques and informed by fabrication constraints. To this end, relevant digital fabrication methods for construction will be explored using computational pipelines for production using available CNC tools (e.g., 3D printing, laser cutting, CNC knitting etc). This elective will offer you powerful toolsets to work on geometry modelling, optimisation, and evaluation of performance in structural design, as well as the natural extension of parametric models towards digital fabrication. These skills will be directly applied in an integrated building design and manufacturing exercise/assignment carried out in groups. The module will be taught and co-taught by experts from both TU Delft, other international research institutions, and industry who have detailed knowledge of emerging research fields as well as current practice.	
Study Goals	On completion of this module, the student will be able to: 1. Translate architectural or structural geometry into an accurate and systematic (parametric) model [ILO 1,2,3] 2. Effectively analyse and optimise design problems and geometrical non-standard structures taking fabrication constraints into account [ILO 3,4,8] 3. Design multiple alternatives for a structure and identifying the most suitable solution from an integrated and inter-disciplinary perspective [ILO 7,8]. 4. Manufacture a (scaled) structure or building using (available) digital fabrication methods [ILO 7,8].	
Education Method	Lectures, practical workshops (coding, digital fabrication), group work, presentation	
Assessment	Continuous feedback on assignment and during workshops Group assignment with clear justification of individual contributions, final presentation with Q&A (100%). A successful participation in the elective requires a minimum grade of 5.8.	
Elective	Yes	
Tags	Algorithmics Building & Spatial Development Building Engineering Design Diverse Group work Modelling Practicals Programming Project Projects Prototyping Software Sustainability	
Expected prior Knowledge	CEGM1000 MUDE (especially the Python part)	
Academic Skills	Critical, analytical, and algorithmic thinking Coding Information collection and synthesis Reasoning and arguing Presenting skills Reporting Collaboration in a group Practise hand-on and tinkering skills in (DigiConstruct)Lab	
Literature & Study Materials	Relevant literature will be added	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Colleggerama	No	

CIEM5309	Advanced Building Physics	5
Responsible Instructor	Dr.ir. W.H. van der Spoel	
Module Manager	Dr.ir. W.H. van der Spoel	
Instructor	Dr.ir. M.J. Tenpierik	
Instructor	Ir. E.R. van den Ham	
Instructor	Dr.ir. H.R. Schipper	
Instructor	Dr. E. Brembilla	
Contact Hours / Week x/x/x/x	6hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The module consists of Theory of heat, air and moisture (HAM) transfer in buildings and construction materials - Convective, radiative and conductive heat transfer - Dynamic thermal models - Basic numerical approach for heat transfer modelling - Natural ventilation and infiltration - Convective and diffusive moisture transfer - Moisture storage in materials - Basic numerical approach for moisture transfer modelling - Indoor thermal quality Theory of room acoustics and sound absorption Room acoustics - reverberation time - energy balance of a room - influence of sound absorption - sound pressure level inside - indoor acoustical quality Sound absorption - porous absorbers - panel absorbers - Helmholtz absorbers - quarter wave pipes Sound simulations - geometrical acoustics - Image Source Modelling - Ray tracing Theory of lighting - daylight and solar radiation properties (units, instruments, spectra, etc) - electric lighting types and properties (CCT, luminous efficacy, etc) - solar exposure studies, effectiveness of PV, urban farming - blinds and effective sun shading, complex fenestration systems - static and dynamic simulation methods and metrics (Daylight Factor, Climate-Based Daylight Modelling and related metrics) - non-visual effects of light on health and wellbeing - indoor visual quality and glare assessments Study assignments (choose 2 topics): Heat, air and mass transfer assignment You will analyse a problem in the area of building physics related to the heat/air/moisture (HAM) quality of a building component, installation, room or building. Matlab/Simulink will be used to solve the problem. Acoustics assignment You will analyse a problem in the area of building physics related to the acoustical quality of a building room. State-of-the-art modelling software will be used and possibly measurements will be carried out for comparison with modelling results. Advanced light analysis You will analyse a realistic design problem related to daylight, or solar radiation, using advanced simulation software (e.g. Radiance, Ladybug)	
Study Goals	On completion of this module, the student will be able to Theory of heat, air and moisture (HAM) transfer in buildings [ILO 2,9] explain theory on indoor thermal quality estimate and model surface heat transfer due to convection and radiation for various geometries and circumstances. calculate natural ventilation flows of simple geometries. apply analytical and simplified numerical methods to solve time-dependent building related heat, air and moisture transfer problems effectively and clearly communicate and present the outcomes of the assignment Theory of room acoustics [ILO 2,9] predict the acoustical quality of a (simple) room, and what type of absorbers best to be used; Effectively and clearly communicate and present the outcomes of the assignment Theory of lighting [ILO 2,9] assess the daylight performance of a room with both static (Daylight Factor, point-in-time illuminance) and dynamic (CBDM) methods	

	estimate the solar radiation received on a building facade and/or roof with simplified numerical methods, and consequently take appropriate design decisions about windows, shading and PV elements apply analytical methods to size the electric lighting system required to meet task illuminance levels in a simple room Effectively and clearly communicate and present the outcomes of the assignment Critically use advanced software to either [ILO 2,7] apply a numerical method for solving (time-dependent) heat and mass transfer carry out and acoustical quality analysis. analyse a realistic design problem related to daylight, artificial light or solar radiation
Education Method	Lectures, exercises in class, feedback on assignment progress, presentations.
Assessment	The assignments will be made in groups of 2 or 3 students. Each group writes two papers (conference style). One paper per chosen topic.
Expected prior Knowledge	100% assignment: the two papers will need to be handed in at the end of the elective and results need to be presented by each group. A successful participation in the elective, requires a minimum grade of 5.8. Theoretical basics of air, heat, and moisture transfer in buildings, room acoustics and lighting in buildings.
Academic Skills	Preferred prerequisite knowledge: CIEM5250 Building Engineering (SE-B5) Or AR1B022 Climate Design critical thinking problem-based learning and solving scientific writing and presentation application of theory to new situations
Literature & Study Materials	Theory of heat, air and moisture (HAM) transfer in buildings and construction materials: - Lecture notes and several readers on heat, air and moisture transfer which are available on BrightSpace - Van der Linden, Building Physics, chapter 5 on thermal comfort: https://klimapedia.nl/wp-content/uploads/2022/07/3-Basics-Thermal-Comfort-Ventilation-Solar-Control.pdf Theory of room acoustics and sound absorption: - Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Francisco. - Chapter 2 (page 27-34): Fundamentals - Chapter 3 (page 37-43): Psycho-acoustics and hearing - Chapter 4 (page 45 50): Introduction into measuring - Chapter 6 (page 69 115): room acoustics -Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. -Knowledge Base Building Physics, module A11- Speech intelligibility -Nijs, L. and Rychtáriková, M. (2011), Calculating the optimum reverberation time and absorption coefficient for good speech intelligibility in classroom design using U50, Acta Acustica united with Acustica 97 (1): 93-102. Only page 95 till 97 Theory of lighting - lecture notes available on Brightspace - Lighting Research Center, Illumination fundamentals, PDF available at weblink https://www.lrc.rpi.edu/resources/publications/pdf/illuminationfund.pdf - Reinhart C., Daylight Performance Predictions, Building Performance Simulation for Design and Operation, Chapter 7. Book available as e-book from the TU Delft library: https://tudelft.on.worldcat.org/oclc/1099466351
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

CIEM5310	Contact and Interface Mechanics for Engineering Structures	5
Responsible Instructor	Dr. K. Anupam	
Responsible Instructor	Ir. C. Kasbergen	
Module Manager	Dr. K. Anupam	
Instructor	Dr. V.L. Markine	
Instructor	Dr.ir. M.J.M.M. Steenbergen	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	M. Lukovi	
Instructor	Prof.dr.ir. R.P.B.J. Dollevoet	
Instructor	Dr. Z. Yang	
Instructor	Dr. H. Wang	
Instructor	Dr. M. Pavlovic	
Contact Hours / Week x/x/x/x	6hrs lectures/practicals	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, the students will learn the history of contact mechanics, and its relevance for different fields of modern structural engineering, followed by a basic discussion of fundamental aspects of contact mechanics. Then numerical modeling of contact mechanics will be discussed, taking material properties into account, via the Finite Element Method (FEM) and the Boundary Element Method (BEM). Next, the focus will shift to contact applications in structural engineering. For transport infrastructures, tire-pavement and wheel-rail interaction will be covered in depth. Aspects like surface conditions, degradation, noise will come into the picture and an experimental study with lab test-rig and deformable tire contact footprint will be presented. Applications in the fields of civil engineering structures such as concrete, steel and composite structures are presented. Finally, the remaining time in this course is used to work on a group assignment. This group assignment includes giving a presentation about the work done.	
Study Goals	On completion of this module, the student will be able to: 1. Explain fundamental concepts of contact mechanics. [ILO 1, 2, 6, 8] 2. Apply contact mechanics formulations/models for engineering problems. [ILO 1, 2, 3, 4, 5, 7, 8] 3. Implement analytical and numerical models to solve engineering level contact mechanics problems. [ILO 1, 2, 3, 4, 7, 8] 4. Analyse and interpret results of a given engineering problem. [ILO 2, 3, 4, 8, 9] 5. Develop methods to control interfaces in engineering structures from the perspective of lifecycle optimisation. [ILO 1, 3, 4, 8, 9, 10]	
Education Method	Classroom teaching; Laboratory and simulation experiments	
Assessment	Handout; Classroom interactions Assignment 100%: group report and presentations of group report; individual knowledge will be checked during the presentation A successful participation requires a minimum grade of 5.8.	
Expected prior Knowledge	Basic and advanced structural mechanics (from BSc and Program base module; MUDE; A1 or A2 or A3)	
Academic Skills	Critical and interdisciplinary thinking; Problem-based learning and their solutions; Inferences of physical systems applied across engineering structures; The application of theory in new situations (contact mechanics); Team-based work; Technical writing and presentation	
Literature & Study Materials	Main reference: Contact Mechanics; Kenneth Langstreth Johnson Advanced: Introduction to Contact Mechanics; Anthony C. Fischer-Cripps https://link.springer.com/book/10.1007/978-0-387-68188-7 Computational Contact Mechanics; Peter Wriggers Computational Contact and Impact Mechanics (Artikel 2003); Tod Laursen	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

CIEM5311	Loading and Life Time Prediction of Transportation Infrastructure		5
Responsible Instructor	Ir. C. Kasbergen		
Module Manager	Ir. C. Kasbergen		
Instructor	Dr. X. Liu		
Instructor	Dr. V.L. Markine		
Instructor	Prof.dr. Z. Li		
Instructor	Dr. A.A. Nunez Vicencio		
Instructor	A. Varveri		
Instructor	Dr. K. Anupam		
Instructor	Prof.dr.ir. R.P.B.J. Dollevoet		
Instructor	Dr. Z. Yang		
Instructor	Dr. H. Wang		
Contact Hours / Week x/x/x/x	6hrs lectures/workshops		
Education Period	1		
Start Education	1		
Exam Period	1 2		
Course Language	English		
Course Contents	The course starts with discussing the fundamentals of transportation infrastructures, like fatigue analysis, degradation mechanisms and life-time prediction techniques. Then the focus will go to natural harsh conditions influencing transport infrastructure. Here the stability of soil foundations, liquefaction phenomena, and the effects of extreme conditions like strong wind, very low and very high temperatures on the (long-term) performance of roads and railways are some of the topics that this module will focus on. Moreover, the impact of disasters like earthquakes or flooding of existing infrastructure will be discussed. In addition, the influence of both increasing traffic volumes and speed on the condition of transport infrastructures will be addressed. Finally, the course will present some countermeasures against the before-mentioned harsh conditions and disasters.		
Study Goals	On completion of this module, the student will be able to: 1. Quantify the individual and combined effects of traffic loadings, and environment on life time prediction of transportation infrastructure under extreme conditions. [ILO 1, 2, 3] 2. Use advanced experimental techniques to characterize, model and monitor mechanical response of various materials, in particular their fatigue behavior and lifetime prediction under extreme loads and environmental conditions. [ILO 1, 2,3] 3. Develop prevention, forecast, protection and maintenance strategies for transportation infrastructure and their supporting materials under extreme conditions. [ILO 1, 2, 3, 4, 8] 4. Identify and quantify the different physico-chemical degradation processes affecting transportation infrastructures using experimental data under extreme conditions. [ILO 1, 2, 3, 4] 5. Explain the extreme conditions transport infrastructure is exposed caused by nature and human involvement. [ILO 4, 5] 6. Identify, quantify and evaluate the countermeasures to be taken against harsh conditions and disasters. [ILO 2, 3, 4, 5] 7. Identify the consequences of extreme conditions to transportation infrastructure and propose tailored-based technological solutions for specific case studies/scenarios. [ILO 1, 2, 3, 4, 5, 7, 8] 8. Assess indices of vulnerability for infrastructure [ILO 3, 4]		
Education Method	Classroom teaching, workshops		
Assessment	Classroom interactions Written exam 100% A successful participation in the elective requires a minimum grade of 5.8		
Permitted Materials during Tests	Will be indicated during the course.		
Expected prior Knowledge	Basic and advanced structural mechanics (from BSc and Program base module); MUDE		
Academic Skills	Critical and interdisciplinary thinking; Problem-based learning and problem solving; inferences of transportation infrastructure systems under critical loading; the application of theory under harsh surrounding conditions		
Literature & Study Materials	Lecture slides, in class recommended reading material		
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations		
Permitted Materials during Exam	Will be indicated during the course.		
Colleggerama	No		

CIEM5312	Emerging Technologies for Transportation Infrastructure	5
Responsible Instructor	Prof.dr.ir. R.P.B.J. Dollevoet	
Module Manager	Prof.dr.ir. R.P.B.J. Dollevoet	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Prof.dr. Z. Li	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	Dr. H. Wang	
Contact Hours / Week x/x/x/x	6hrs lectures/practicals	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this elective, students will get acquainted with the latest technological solutions in the transportation industry facing increasing demands for passengers and freight transport, as well as challenges related to energy and climate change. The elective covers the state-of-the-art and potential technical breakthroughs that can revolutionize existing transportation infrastructure, with a focus on future resilient and green railway and road infrastructure. Backgrounds, principles, and implementations of the latest techniques and their limitations compared with the current transportation industry in terms of practicability, reliability, efficiency, and sustainability will be addressed. Leading specialists from the industry will be invited to give guest lectures to ensure the actuality of the problems discussed in this elective. Activities including laboratory and field tests, and excursions to transportation companies are planned. The following topics (not limited to) will be discussed: 1. Advanced topics on novel/sustainable designs and optimization, e.g., modular tracks, interface conditioning from the perspective of tribology, bio-mimics, smart materials, A-Train, Hyperloop, Maglev, Parcels/Pods. 2. Advanced experimental and numerical methods, e.g., nanomaterials, new ballast, discrete element method, molecular dynamics modelling. 3. Smart infrastructure, e.g., sensing, instrumentation, self-healing, platooning, MaaS equipment, digital twins, artificial passengers, cyber-security, solar roads, low-noise roads, robots for maintenance, cyber-security, railway markets, block-chain in transportation, privacy for passengers information to people in the platforms, IoT passengers as a sensor (mobile technologies, analysis of social networks, etc.), autonomous door to door transportation for passengers and freight/parcels - last mile problems. 4. Enhancing the reliability and resilience of transportation infrastructures and coping with the impact of climate change with emerging/advanced technologies for condition monitoring and decision making. 5. Energy efficiency and management, including hydrogen, fuel cells, batteries, maglev, friction modifier, friction energy reuse, automatic coupling, rolling pantograph, ATO, automated driving, ERTMS, charging station design, optimization of locations for catenary systems, and energy harvesting roads. 6. Circularity, including wheel-to wheel/LCA full loop of life cycle, construction, sustainability, CO2 and emission reduction, diesel line conversion (hybrid and hydrogen), modular design with circular materials. 7. Ethical implications of implementing the emerging technologies.	
Study Goals	On completion of this module, the student will be able to: LO1: Assess the function and implementation of novel designs and new technologies for transportation infrastructures. [ILO 3, 4, 8, 10] LO2: Apply advanced experimental and numerical methods to analyse and enhance the reliability and resilience of transportation infrastructures. [ILO 1, 2, 4, 5, 8] LO3: Explain the principles of design, operation and degradation of smart transportation infrastructures. [ILO 2, 3, 9] LO4: Apply the emerging technologies of condition monitoring and decision making to cope with the impact of climate change (including extreme weather) on transportation infrastructures. [ILO 1, 2, 3, 4, 5, 7, 8] LO5: Explain the energy management of transportation infrastructures and evaluate technologies to improve the energy efficiency. [ILO 1, 2, 9] LO6: Explain the principle, procedure and leading technologies for the circularity of transportation infrastructures. [ILO 1, 2, 9] LO7: Discuss the ethical implications of implementing the emerging technologies. [ILO 1, 2, 8, 9]	
Education Method	Classroom teaching; Laboratory and simulation experiments	
Assessment	Handout; Classroom interactions Written exam A successful participation in the elective requires a minimum grade of 5.8.	
Expected prior Knowledge	Basic structural mechanics (from BSc and Program base module);	
Academic Skills	Critical and interdisciplinary thinking; Problem-based learning and solving; Inferences of physical systems applied across engineering structures; the application of theory in new situations; Team-based work; Technical writing and presentation	
Literature & Study Materials	Lecture notes	
Judgement	See Assessment	
Collegerama	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Elctives Traffic & Transport Engineering (TTE)

CIEM6301	Railway Traffic Management	5
Responsible Instructor	E. Quaglietta	
Module Manager	E. Quaglietta	
Instructor	Prof.dr. R.M.P. Goverde	
Contact Hours / Week x/x/x/x	4hrs lectures	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Railway timetabling: Microscopic and macroscopic rail traffic simulation models Timetable design techniques and optimization models Timetable assessment using microscopic simulation Energy-efficient train operation Real-time railway traffic management: Driver Advisory Systems Disruption management Train delay prediction and analysis using a statistical train delay analysis tool Railway traffic management systems Conflict detection and resolution Simulation-based assessment of traffic management strategies This elective will include discussions about societal impacts of advanced tools and techniques for train service planning and management. Students will be aware of responsibilities and ethical implications in using automated decision support systems in rail traffic planning/management practice. Roles and responsibilities of railway stakeholders in planning / managing rail traffic will be tackled together with interaction processes during nominal and degraded operational conditions.	
Study Goals	On completion of this module, the student will be able to: LO1: Design railway timetables (CE ILO1) LO2: Assess rail traffic management methods (CE ILO4) LO3: Analyse train delays (CE ILO2) LO4: Apply rail traffic simulation (CE ILO2) LO5: Evaluate energy-efficient train operation (CE ILO3)	
Education Method	Blended learning mixing on-campus lectures, reading, videos, guest lectures and formative assignments	
Assessment	Formative group assignments on rail traffic simulation, scheduling and traffic data analysis with supervised practicum sessions. Formative assessment also includes pre- and/or post-lecture tests, on-line self-assessment, training with model exams with answers. Final digital written exam. The exam will include questions on lectures and formative group assignments. A minimum grade of 5.8 is required to successfully pass the exam.	
Expected prior Knowledge	Prior knowledge about planning and design of public transport systems is recommended, such as units of Module B3 of the MSc programme in Transport and Traffic Engineering (i.e. Public Transport Planning and Operations, Public Transport networks and modelling, Railway operations and control). In particular, blocking time theory is assumed working knowledge. Additional reading (one chapter, 3 hours) is required for those not familiar with blocking time theory. In addition, an introduction course on discrete optimization is recommended.	
Academic Skills	Problem formulation and problem solving Critical and analytical thinking Interpretation Writing technical reports Applying professionally-relevant tools and methods	
Literature & Study Materials	I.A. Hansen & J. Pachl (eds.), Railway Timetabling & Operations, Eurailpress, Hamburg, 2nd edition, 2014. Lecture slides of the course available on Brightspace	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Open book - digital exam	
Collegerama	No	

CIEM6302	Advanced Data Science for Traffic and Transportation Engineering	5
Responsible Instructor	P.K. Krishnakumari	
Module Manager	P.K. Krishnakumari	
Instructor	M. Rinaldi	
Contact Hours / Week x/x/x/x	5hrs workshops/practicals (see form of instruction for further info)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course is a project-based course, where the data and requirements are provided by companies (CO) and governmental organizations (GO). The topics for the project include but not limited to: - Transport Networks and Systems - Road Traffic Systems - Public Transport and Railway Systems - Road and Railway Engineering - Transport and Pollution - Transport and Equity With the 8 weeks course, the students will follow the common data pipeline data processing, analytics, and visualisation - and use the same tools to answer their assigned/chosen projects. In these weeks, they will learn to work together by exploring a single dataset in detail to extract useful insights for their clients/stakeholders. Students will also have an opportunity to discuss with established industry experts on the different projects. To aid in developing the project, different data topics are discussed every week complemented by lab exercises. The topics include: Data processing and visualization Data analytics feature engineering Ethics in Traffic and Transportation includes data ethics, societal relevance, relevance to the 17 SDGs Machine learning methods - Unsupervised, Supervised, and Reinforcement Validation and evaluation One important aspect of their project requirement would be to reflect and provide concrete steps or design on how their project could be relevant for the SDGs and how to scale up their prototype for production. We have a minimum requirement of 5 students for this course to be delivered. This is to ensure at least one project group can be formed.	
Study Goals	At the end of the course, the student should be able to: - Collect and translate the requirements from multidisciplinary stakeholders into a data science project - Identify, build and assess the core components of a data pipeline - Apply data science tools and techniques to solve real-world traffic and transport engineering problems - Reflect on the ethical and societal implications of a data-based solution - Effectively communicate the data solution to stakeholders - Implement industry-standard coding practices to real-world traffic and transport engineering problems.	
Education Method	Workshop + Lab sessions + Projects Workshop: 8 x 2 hours Lab sessions: 8 x 2 hours Group work: 8 x 10 hours Project feedback: 8 x 1 hour Self-study: 20 hours Total: 140 hours In the course, students will have on-campus workshops, corresponding lab sessions and group project sessions. Parallel to the workshops, students will work in groups on the industry project). They are presented with a specific problem by the industry that needs a data analysis, or solution. The project groups apply at least one ML technique and then build a deployable prototype. The tangible outputs of the project are a data story (in medium or similar platform), code base, prototype, and presentation to the industry. Direct group supervision, guidance and progress monitoring are provided by an academic supervisor and an industry supervisor. Project teams receive regular feedback on their performance and milestones.	
Assessment	The output of the project work (data story, code base, presentation) is assessed. The students receive a single grade per group, with some adjustments to account for individual contributions. A successful participation in the elective requires a minimum grade of 5.8. Projects have a repair option. A repair option is only allowed for students that receive a fail mark for their project (<5.8). The project resit mark will be capped to 5.8	
Expected prior Knowledge	Basic knowledge of Python	
Academic Skills	Interdisciplinary thinking; Problem-based learning; Data analytics; Communication	
Literature & Study Materials	TBD	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	

CIEM6303	Transitions, Sustainability & Innovation	5
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Module Manager	Dr.ir. A.J. van Binsbergen	
Contact Hours / Week x/x/x/x	4hrs lectures per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Transport and mobility is changing rapidly - partially as a consequence of technological and societal developments, partially also because our demands towards the transport system change. This course is about understanding this changing context and being able to develop and/or assess new ideas and concepts for transportation. This is done along the lines of sustainability (requirements), innovation (introducing new ideas) and transitions (radically changing the perspective on transport and mobility). Sustainability: on basis of (changing) demands in the fields of wellbeing ('people'), economic activities ('prosperity') and the environment ('planet') as well as anticipated developments in demography, economy and spatial development, we analyse the requirements and needs for future transport systems. The concepts will be discussed during lectures, but you'll work on these during the first part of an assignment as well: Identification of main developments and future aims and ambitions for a specific urbanised area, to be described in a base case, Business-as-usual (BAU) and two alternative (near or far) future scenarios. Innovation: introduction of innovation theory (steps, roles of actors, role of technologies, product life cycle, technology readiness levels, pilots, living labs, upscaling); tools for analysing and assessing 'new' ideas, what is really new, is a new concept really useful, what about interoperability and interconnectivity? Some concrete examples will be discussed, for instance MaaS, mobility hubs and synchro-modality. The topics will be introduced and discussed in class, and in part two of the assignment you're asked to identify, analyse and assess 'new' transport and mobility concepts. Transitions: introduction of transition theory and transition management (wicked problems, regimes and regime change; actors, co-evolution, self-organisation, adaptation), illustrated with some concrete examples. The approach will be discussed in class and in the third part of the assignment you're asked to identify or a potentially 'transitional' development yourself, to identify current regimes and actors, needed changes, barriers and ways to overcome these. During the fourth part of the assignment you're asked to develop a futuristic transport system based on (assumptions of) spatial and demographic development, complying to sustainability requirements. Analysing potential future use and effects under different future scenarios, developing innovation or transition pathways, identifying relevant regimes, key actors, practices and needed changes therein, identifying the necessity of co-evolution and the potential of self-organisation/co-makership. Overall assessment of the validity of the envisaged transport system.	
Study Goals	On completion of this module, the student will be able to: LO1. Create innovation pathways for passengers and logistics concepts answering to societal challenges: ILOs 1, 5, 8, 9, 10. LO2. Evaluate performance and (sustainability) impacts of innovative transport concepts in future contexts: ILOs 2, 3, 9, 10. LO3. Assess suitability of tools for evaluating innovative transport concepts in future contexts: ILOs 4, 6, 10. LO4. Analyse the expected functionality and (sustainability) performance of future transport concepts: ILOs 2, 5, 9, 10. LO5. Apply different standard methods and approaches for developing innovation pathways: ILOs 4, 7, 8, 10. LO6. Operationalise performance and (sustainability) effects indicators for (future) transport systems, including estimation methods and approaches: ILOs 5, 6, 10. LO7. Explain the complexity of sustainable transport system transitions and innovation pathways: ILOs 6, 7, 9, 10. LO8. Define (current and future) transport systems, their (envisaged, expected) characteristics, and main development challenges: ILOs 6, 9, 10. LO9. Effectively communicate results and opinions with stakeholders of diverse backgrounds: ILOs 6, 9	
Education Method	Mix of on-campus lectures, reading, videos, guest lectures and a summative assignment with intermediate feed-back	
Assessment	Formative in-class exercises / interactive sessions with direct feed-back. Group assignment (via report and presentation) Summative group assignment with supervisor assessment resulting in a group grade based for 85% on the report and 15% on the presentation, with the possibility of tuning individual grades via peer-assessment (varying max. 20% of the group grade). The minimum passing grade is 5.8. In case an individual grade is below 5.8 but higher than 4.5, an additional task can be given; when done satisfactory, the final grade can be improved to max. 5.8.	
Expected prior Knowledge	1. Basics of (transport) economics: basics of economic theory, economic rationale and behaviour (utility theory), costs & benefits (fixed, variable; direct, indirect), economic assessment (cost/benefit ratio), value of time. 2. Basic concepts of Transport Modelling: including both agent based modelling (as modelling technique) and activity based modelling (as modelling approach next to the traditional trip-based modelling). 3. Basic insights in societal goals such as (preconditions for) economic developments including accessibility, sustainability and social equity/inclusiveness, and terms such as reliability, robustness, resilience. 4. Insight in societal challenges and social, economic and ecological sustainability objectives. For participants lacking background knowledge in these fields, self-study materials sourced from the MSc Civil Engineering Transport Technology and Engineering (CE-TTE) track will be available.	
Academic Skills	Critical, interdisciplinary and ethical thinking, problem aware and solution oriented learning, information collection and synthesis, communication and responsiveness, scientific writing and presentation, application of theory to new situations.	
Literature & Study Materials	Lecture slides available on Brightspace, and: Book chapter: René Kemp, Jan Rotmans (2004) Managing the transition to sustainable mobility (Chapter 7 in System Innovation and the Transition to Sustainability, Edgar Elgar Kemp R., Avelino, F., Bressers, N. (2011) Transition management as a model for sustainable mobility in European Transport / Trasporti Europei No. 47, April 2011 Snelder, M., Wilmink, I., van der Gun, J., Bergveld, H.J., Hoseini, P., van Arem, B. (2019). Mobility impacts of automated driving and shared mobility: explorative model and case study of the province of North-Holland. European Journal of Transport and Infrastructure Research, vol. 19, n. 4. https://doi.org/10.18757/ejtr.2019.19.4.4282 Milakis, D., Snelder, M., Arem, B. van, Wee, B. van, & Homem de Almeida Correia, G. (2017). Development and transport implications of automated vehicles in the Netherlands: scenarios for 2030 and 2050. European Journal of Transport and Infrastructure Research, 17(1).	

Judgement	Lucas, K., Maat, K. & van Wee, GP., (2016) A method to evaluate equitable accessibility: combining ethical theories and accessibility-based approaches, in: Transportation. March, p. 1-18 18 p. See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

CIEM6304	eXtended Reality for Civil Engineering	5
Responsible Instructor	Y. Feng	
Module Manager	Y. Feng	
Instructor	A. Freeke	
Instructor	Dr.ir. D.C. Duives	
Instructor	Dr.ir. H. Farah	
Assistant	Ir. A.H.N. van Beek	
Contact Hours / Week x/x/x/x	6hrs lectures/workshops, last week 2hrs lectures/workshops	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course teaches students the fundamental knowledge and key skills required to apply XR - eXtended Reality (i.e., augmented, virtual, and mixed reality) to CE systems. It covers the core concepts of XR, tools, and techniques for XR development, and how to design and develop XR research applications (incl. data collection & analysis). Moreover, this course highlights the application of XR for CE systems, including transportation, structure, construction, and infrastructure. All students will obtain hands-on experience with multiple state-of-the-art XR technologies and learn how to apply XR to the CE domain. Enrollment information: Based on the current course, and the planned assignments, a minimum of 6 students will be required to enroll in the course. A maximum of 45 students is also in effect.	
Study Goals	On completion of this course, the student will be able to: LO1: Explain the conceptual and technological differences between Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR), and eXtended Reality (XR). [ILO 8] LO2: Use XR-related software to develop your own XR research application for investigating a selected topic of civil engineering. [ILO 1, 7, 8] LO3: Design and execution of a XR research plan for the selected CE topic. [ILO 1, 2, 8] LO4: Apply XR research application and suitable hardware to conduct the XR research plan for the selected CE topic. [ILO 1, 2, 8] LO5: effectively and clearly communicate and present the outcomes of the assignment. [ILO 9]	
Education Method	Theoretical lectures + technical lectures + practice workshops	
Assessment	Formative assessment: Peer feedback: each group will be asked to prepare a draft research proposal (including research question, experiment design, data collection plan, etc.) and a prototype of their XR application. Another group will be asked to give feedback on the research plan and the prototype XR application. Lecturers give direct formative feedback during the practice workshop. Summative assessment: Each group will present their final project in a poster presentation format during which the teacher will interview the groups. A successful participation in the elective, requires a minimum grade of 5.8.	
Expected prior Knowledge	Ability to program in python	
Academic Skills	Critical and interdisciplinary thinking, Problem-based learning and solving, Information collection and synthesis Scientific writing and presentation, making inferences by analyzing observations/data Communication and responsiveness Work in teams with diverse backgrounds, skills, and cultures	
Literature & Study Materials	Literature regarding the development and usage of XR for civil engineering related topics Lecture notes, slides, and tutorial documents will be available via BrightSpace	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Projects

CEGM3000	Multidisciplinary Project (MDP)	15
Responsible Instructor	Dr.ir. M.M. Rutten	
Module Manager	Dr.ir. M.M. Rutten	
Instructor	Y. de las Heras	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	In the Multidisciplinary Project (MDP) students investigate a problem for a client in a team that consists of students from multiple tracks within the Faculty of Civil Engineering and Geosciences and develop a solution strategy or other advice. The client can be external or faculty staff. Students from other faculties can be involved, yet permission from the module coordinator is required for this. The faculty encourages students to perform a MDP abroad and to acquire a project themselves by reaching out to external organisations. Project topics are also offered via the online learning platform Brightspace. After acquisition, the MDP consists of three phases: Phase 1: Preliminary investigation. Students start with a problem analysis resulting in a problem statement and an objective. Coupled to that, a project strategy will be formulated. This strategy includes which methods will be used, which contribution can different disciplines provide to the project, which steps have been passed through successively, which information is still necessary, where that information can be found etc. In addition, the organization of the group will be set up in this phase. This phase ends with a presentation of the project proposal to the client and supervisors. Phase 2: Design and/or research. During this phase students will execute the strategy developed under phase 1 will and adjust it if needed. The work may exist of research and/or of developing design alternatives or solution alternatives. Alternatives may be worked out for the complete problem or for sub-problems. Half-way this phase the progress is presented in a midterm presentation. Phase 3: Round-off. In the round-off, students deliver the project results to the client and supervisors. First, a draft report is discussed with the supervisor and client, whereupon the definite version is made. The participants evaluate the project, both regarding the content and regarding the project process. Also a public final presentation is given.	
Study Goals	After completion of this module the students are able: MDP-LO1: Communicate project plans, intermediate and final results with a client and other relevant stakeholders in written and oral form. (ILO: i) MDP-LO2: Initiate, devise and plan project, in order to meet requirements, set by a client. (ILO: j) MDP-LO3: Monitor and manage a project in a team considering the client and other stakeholders involved. (ILO: j) MDP-LO4: Analyze a complex problem in the field of civil engineering, environmental engineering and/or geosciences. (ILO: j) MDP-LO5: Select and apply appropriate research and design methods to address this complex problem. (ILO: j) MDP-LO6: Understand the effect of diversity in disciplines, talents, skills, characters and cultures on a group process and address this effectively in a group process. (ILO: k) MDP-LO7: Apply working methods that stimulate combining and integrating knowledge and methods from different disciplines. (ILO: k)	
Education Method	MDP starts with a kick-off in which the steps of the project and methods for complex problem analysis and multidisciplinary working will be introduced. During the project students have regularly discussions with their supervisors and client and peer review and intervision sessions with other groups. All these activities will be organized online to allow participation from abroad.	
Prerequisites	As of academic year 2023-24 Q2: students from must have obtained at least 45 ECTS. Mail the application form together with a study progress overview from each group member (all combined in one mail or one pdf) to studentoffice-esa-ceg@tudelft.nl. In the study progress overview EC's of parts of the modules can be used. This needs to be made clear by the student.	
Assessment	Formative assessment (feedback): Feedback from supervisors and client during presentations of project proposal and midterm. The draft report will be discussed with the group. Peer review on project proposals and draft reports will be organized. Summative assessment: Students complete the module successfully when passing the assessment that consists of: 1. Quality written report and oral presentation (10%) 2. Quality research / final design as presented in written and oral report (50%) 3. Group process and multi disciplinarity (40%).	
Special Information	As of academic year 2023-24: all students from cohort 22-23 and later performing a MDP will be rewarded with 15 ECTS. For cohort 21-22 and earlier it will be divided into 10 EC + 5 EC extracurricular. See TER/Annex of your cohort.	
Expected prior Knowledge	A large part of the first year of your program. See prerequisites.	
Academic Skills	During the project, students work on their skills regarding inquiry, collaboration, communication, integration of different perspectives, and reflection on personal functioning.	
Literature & Study Materials	Students will be offered handouts and web resources on problem analysis, working in diverse teams, project management, reporting a presenting. For problem analysis following material (or comparable resources) will be used together will dedicated material on Brightspace: https://ocw.tudelft.nl/courses/ngi-2/subjects/8-bonus-module-problem-demarcation-actor-analysis-skills-2/, (accessed 2-4-2021) For teamwork and project process following material (or comparable resources) will be used together will dedicated material on	

Brightspace:
 Schwartz, S. H. (2014). National culture as value orientations: Consequences of value differences and cultural distance. In Handbook of the Economics of Art and Culture (Vol. 2, pp. 547-586). Elsevier.
<https://www.belbin.com/about/belbin-team-roles>
<https://www.mbtionline.com/>
<https://scrumguides.org/>
<https://online-learning.tudelft.nl/project-management-skills/>

Students can use the resources below for report writing and presenting:
<http://eloquentscience.com/>
<https://faculty.washington.edu/heagerty/Courses/b572/public/StrunkWhite.pdf>
 Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine
 Publisher: Cambridge University Press, 2012
 ISBN: 978-0-521-16521-1

Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CIEM0120	Research Internship	10
Responsible Instructor	Dr.ing. M.Z. Voorendt	
Module Manager	Dr.ing. M.Z. Voorendt	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	A Research Internship (RI) is a civil engineering research or design project conducted in a professional environment. The aim of an RI is twofold. On the one hand, the RI is about formulating and conducting a research or design project directly related to some topical quest, raised by either an external party (such as private company, (non-)governmental body, scientific institute, pressure group, either in the Netherlands or abroad) or an internal party (research or advisory projects of any kind, for instance related to externally funded projects, private-public co-operation and so on). This basically means that the topic of an RI must be of evident societal, entrepreneurial or scientific relevance and that it needs to be conducted in the professional environment that raised the quest. On the other hand, the professional environment the concerned project is conducted in, must be reflected on. This means that students conducting an RI are required to evaluate the influence they experience from their working environment on their performance and deliverable. This has a broad notion. From an organisational perspective, this comprises at least supervision, guidance, availability of relevant information, technical and administrative facilities. Besides, the student is required to reflect on her/his role in and contribution to the organisation where the RI is conducted, as well as her/his personal development. It is important to note that conditions may apply, for instance regarding confidentiality, intellectual property and so on. It is the responsibility of the student to ensure that such conditions are agreed upon by all parties involved prior to the start of the project and in compliance with TU regulations (see TER).	
Study Goals	On completion of this course, the student will be able to Formulate a research or design question, aligned with the clients demands, as well as produce and conduct a plan of work evidently converging to an answer to that question; [ILO 7, 8, 9, 10] Integrate and apply academic knowledge together with engineering and design skills acquired in previously taken modules of the master programme and relevant previous education; [ILO 8, 10] Deliberate, communicate and report both progress and results with the client and other stakeholders; [ILO 9] Reflect on obtained results in regard to conclusiveness, relevance for the clients demands, ethical aspects; [ILO 7, 9, 10] Reflect on personal performance in the context of the clients organisation and in regard to personal development during the project. [ILO 7, 9, 10] Reflect on the formal and informal performance of the organization where the project has been conducted. [ILO 6, 9] The Research Internship, as it is about a research or design project conducted in a professional environment, links to practically all ILOs, but from a generic perspective it links primarily to ILOs 7-10.	
Education Method	Internship at an external or internal party	
Assessment	Formative assessment: Feedback is given by the university supervisor (1) on the project plan (including motivation) and (2) at least twice during the project, either in progress meetings or based on progress reports and (3) on the draft final report. Feedback is given also by the involved representative of the institution where the RI is conducted (the company supervisor). Summative assessment: - Engineering skills (25%) Application of theory, motivation of methods/techniques Experimental work, programming, calculations Ambition level, amount of work Usability of results for company - Academic approach (25%) Formulation of a goal and approach Argumentation Extension of knowledge and/or methods Achievement of the goal, reflection, judgement - Reflection component (20%) Reflection on personal objectives and future career Indication personal strengths, weaknesses and future personal development Reflection on the performance of the company Reflection on ethical and societal aspects of main project - Reporting (15%) Structure and consistency Language proficiency, style Use of literature, sources Lay-out, use of figures, tables, equations, etc. - Behavioural competences & skills (15%) Initiative, own contribution	

	Responsibility Communicative skills Independency For each of the above five criteria, a partial grade is set by the university supervisor (70%) and the company supervisor (30%). In the case of a substantial difference (more than 1 point on a 10 point scale), the responsible instructor is consulted. The final grade is the weighted average of the five partial grades. To pass, the rounded final grade must be at least 6.0 and all grades for the main five assessment categories are at least 5,0.
Expected prior Knowledge	MUDE, Programme Base and from the track the concerned student has selected the Track Base and Module A.
Academic Skills	The student has completed at least 45 EC's from the ISP, including partial results.
Literature & Study Materials	Analytical skills, report writing, project planning, communication
Judgement	Recommended: Elling et al (2011): Report Writing for Readers with Little Time. 1st edition, Noordhof Uitgevers.
	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	N/A
Collegerama	No

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Cross overs choose one

CEGM2000	Suspensions, Sludges and Soils	10
Responsible Instructor	Dr. C. Chassagne	
Module Manager	Dr. C. Chassagne	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr.ir. S.G.J. Heijman	
Instructor	O. Kirichek	
Instructor	Prof.dr.ir. J.B. van Lier	
Instructor	Dr. J. Gebert	
Contact Hours / Week x/x/x/x	2x4 uur = 8 uur in lecture room 2 x 8 uur = 16 uur practicum	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Sediments and sludges are mixtures of mineral clay, organic material and water. Sediment and slurries are dredged for port maintenance purposes. Sludges found in (waste)water treatment plants are stabilized and dewatered for resource recovery and/or incineration. Some of these slurries/sludges are used in engineered constructions like dikes. In this module, the biochemical and physical properties of sediments and slurries will be reviewed. The way to assess their properties will be studied from the aqueous phase to the solid phase. The module consists in lectures, group assignments and practical work (lab/in-situ). Learning Unit 1: From Suspensions to Slurries EC: 4 Weeks: 1-3 The microscopic/mesosopic properties of clays and organic matter will be reviewed from the point of view of colloid science. The flocculation of particles in combination with different environmental conditions will be studied and modelled. The properties of slurries in terms of settling thickening, rheology and consolidation will be measured and modelled, as function of composition and environmental conditions. List of key topics (Elements of Learning Unit): - Nature of clays (mineralogy) and clay properties - Nature of organic matter and sludge (origin, composition, stability) - Interfacial properties of colloidal particles (surface charge, Derjaguin Landau Verwey Overbeek (DLVO) theory, polymers), - Living organic matter interactions with clay and sludge - Flocculation, coagulation - Rheology of slurries - Settling, thickening, dewatering and consolidation Laboratory experiments and measurements to determine the following properties: - particle size distribution - surface charge - settling rate - coagulation/flocculation - rheological properties - conductimetry - capillary suction time - compressibility - biodegradation of sediment organic matter (SOM) - methane potential of sludges Learning Unit 2: From Slurries to Soils EC: 3 Weeks: 4 - 6 This unit addresses the changes that sediment properties undergo during post dredging treatment and ripening on land. Students will understand and be enabled to engineer the coupled processes of physical, biological and chemical ripening which transform sediments from saturated, soft, anaerobic material to an unsaturated, well-structured and aerated soil fit for use as earthen construction material. Students will understand the associated environmental impacts, for example, related to sediment contamination or degradation of their organic components and the benefits related to sparing excavation of natural soil resources for construction purposes. List of key topics (Elements of Learning Unit): - Mechanical/physical, chemical and biological ripening processes - Methods and criteria to assess physical/mechanical, biogeochemical and biological stability of sediments and soils - Carbon cycling in the sediment-to-soil process - Environmental behaviour of conditioned slurries - Beneficial use of conditioned slurries and sediment Laboratory experiments - geochemical processes during ripening - biodegradation of organic matter - redox potentials - basic soil physical/mechanical and geochemical properties Learning Unit 3: In-situ Characterization of Sediments EC: 3 Weeks : 6-8 The student will learn to use the methods and equipment to monitor the important properties of mud in a natural environment. These properties are assessed for mud in the water phase as well as on land. These methods will be connected to the knowledge acquired in Unit 1 and 2. List of key topics (Elements of Learning Unit): - Monitoring of suspended sediment (submersible laser-diffraction, camera, optical back-scattering equipment, Acoustic Doppler current profiler) - Acoustic methods used for navigation in ports and waterways - Determining in-situ density and flow properties of slurries - Electromagnetic and seismic methods for monitoring of soils and fluid waste repositories	
Study Goals	1. Understand the principal processes that determine behaviour of fine-grained sediments (mud) and sludge in the aqueous phase	

	and in the solid phase. 2. Formulate hypotheses regarding the effect of specific environmental conditions on the behaviour of sediments, sludges and soils. 3. Apply analysis and monitoring methods to study sediments, sludges and soils properties in different environmental conditions (in the laboratory and in-situ) 4. Use the corresponding model(s) to extract the relevant parameters from the raw data obtained from the specific characterization method (point 3) and be able to analyse the results. 5. Assess the properties of mud in relation to a specific engineering problem such as the improvement of water clarity, stabilization of dikes, impact of dredging plumes, navigability of fairways in mud-rich areas.
Education Method	For all units: Interactive lectures (videos, Q&A, discussion about recently published or seminal papers) Targeted self-study using scientific articles, reports or book chapters Laboratory (see description in Units) and field work. Field work will be 1 or 2 days to put theory in practice and will be organized with our partners. Field work is complementary to lab work. At module level: Group project: Case study (cross-over all units) Material and equipment: MUDNET laboratory / sanitary eng. laboratory / Field equipment (partnership) / Geo lab
Assessment	Formative assessment (feedback) Oral feed-back to students (contact hours) will be planned during the whole course, as teachers will be present for demonstrating the use of the equipment, the way to analyse results and the way to report these results. (see in particular LO1, LO2, LO5) Contact hours will also be used to clarify fundamental points that are unclear (in plenary lectures, books, exercises, articles, etc. that students will have to study). The course will be given in the spirit of flipped classroom . Students will be asked to read and comment one or two articles every week. These articles will then be discussed during plenary sessions. The experiments realized during the course of a unit will be formalized in the form of a small report. The design of this report will be discussed and assessed with the teacher, so that students will be prepared for the case study, whereby a full report should be delivered (and that students should design themselves, see summative assessment). During lively discussions, the students will be challenged by the teachers and fellow students to explain their choices and comments their results. The students will acquire the scientific critical mind and their out-of-the-box thinking that is expected from good engineers. Summative assessment Exam each student (60%) At this point, oral examinations are foreseen, which are possible due to the expected amount of students attended the CO module. If this number rises, it will be discussed to have a written exam. During this exam, each student will be asked about the theoretical aspects of the course. Group project report (40%) A case study will be performed and each group of students will deliver a common report on the work that is performed. The case studies will bridge the knowledge acquired in the course of all units. The groups will present the work to an audience and will have to reply to questions. Each student within the group will be challenged, and the final mark for each student will be a weighted coefficient between group report, presentation and individual attitude during the case study and individual response to questions. All subgrades must be > 5,8.
Expected prior Knowledge	N/A
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	All units : State-of-the-art articles (pdfs available on BS) Video recordings (to be recorded available on BS) Slides (pptx) corresponding to the given lectures Unit 1 New Lecture Notes have been made (available on BS), partially based on the book Introduction to Colloid Science application to sediment characterization https://books.open.tudelft.nl/home/catalog/book/16 and several leading publications exercises are being prepared (pdfs available on BS) Chassagne, Claire, Zeinab Safar, Zhirui Deng, Qing He, and Andy Manning. "Flocculation in estuaries: Modeling, laboratory and in-situ studies." In Sediment Transport-Recent Advances. IntechOpen, 2021. handouts for the use of equipment will be prepared and made available Unit 2 Blume et al. (2016): Scheffer/Schachtschabel Soil Science. https://link.springer.com/book/10.1007%2F978-3-642-30942-7 Weil & Brady (2016): The nature and properties of soils. Pons, L. J., & Zonneveld, I. S. (1965). Soil ripening and soil classification : initial soil formation of alluvial deposits with a classification of the resulting soils = Pedogenese initiale et classification des sols = Initiale Bodenbildung und Bodenklassifikation = Pedogenesis initial y clasificacion de los suelos. (Publication / International Institute for Land Reclamation and Improvement; No. 13). Veenman. https://edepot.wur.nl/60013 Journal papers on sediment ripening and beneficial use, exemplary journal: Journal of Soils and Sediments, Springer. https://www.springer.com/journal/11368 Lawrence et al. (2008). Biosolids Engineering and Management, Handbook of Environmental Engineering. ISBN 978-1-59745-174-1 Spinoza and Vesilind (2001). Sludge into biosolids. IWA publishing, London, UK. ISBN-10: 1900222086; ISBN-13: 978-1900222082 Published case studies on beneficial use of sediments: https://www.dredging.org/resources/ceda-publications-online/beneficial-use-of-sediments-case-studies Unit 3 https://library.seg.org/doi/book/10.1190/1.9781560801719 Shakeel, Ahmad, Alex Kirichek, and Claire Chassagne. "Rheology of Mud: An Overview for Ports and Waterways Applications." Sediment Transport-Recent Advances (2021). Draganov, D., Ma, X., Buisman, M., Kiers, T., Heller, K., & Kirichek, A. (2021). Non-intrusive characterization and monitoring of fluid mud: laboratory experiments with seismic techniques, distributed acoustic sensing (DAS), and distributed temperature sensing (DTS). In Sediment Transport-Recent Advances. IntechOpen.
Judgement	N/A

Permitted Materials during Exam	N/A
Collegerama	No

CEGM2001	Sustainable Cities	10
Responsible Instructor	Dr. H.M. Jonkers	
Module Manager	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	Dr. D.F.J. Schraven	
Contact Hours / Week x/x/x/x	approximately 12 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Sustainable cities addresses the need to transform current cities not only to climate mitigating- (substantially lower CO2 and environmental footprint), climate resilient-(adaptive to future climate changes) and circular- (minimal requirement of primary resources and minimal damage to environment) but simultaneously to desirable ones, meaning that they support overall wellbeing of its inhabitants. Some and specifically larger Dutch cities are currently characterized by the presence of aged and desolate city blocks, quarters and infrastructures with often co-occurring environmental and social problems which negatively affect wellbeing of inhabitants. Challenges to overcome are clearly manifold and solutions should therefore be integral in order to be both economically and functionally effective. The indicators that cover the four targeted sustainable (wellbeing) built environment performance criteria (climate mitigation, climate resilience, circularity and wellbeing) and addressed in this module are the following seven: 1. Quality (chemical composition) of outdoors air, water and/or soil (including presence of litter), 2. State of maintenance and quality of present buildings and infrastructure (service life-, circularity-, energy- and environmental performance), 3. Risk of heat stress occurrence (specifically in summertime), 4. Risk of flooding (specifically during intense rain showers), 5. Noise occurrence, 6. Recreational- and social interaction opportunities (quantity and quality of present public spaces) and 7. Quality and quantity of present nature (natural capital) comprising biodiversity and beneficial goods and services that nature can provide to city inhabitants. These seven living quality indicators are specifically addressed in this module as these are considered to be representative for liveability and desirability of sustainable cities. In this Cross Over module specific Dutch city quarters will be analysed both for the qualitative and quantitative state of these seven wellbeing indicators, followed by reviewing available and potential technical solutions for improvement of shortcomings related to individual indicators, and finalized by defining and designing integral solutions for often co-occurring problems. Specific focus of this module will be on the potential of nature-based solutions as these can typically provide integral solutions to multiple climate mitigating-, climate resilience- and wellbeing-related problems. E.g. the Dutch PBL (Netherlands Environmental Assessment Agency) states in their every two-years updated report that more public green (vegetation) in cities is required to increase wellbeing of its inhabitants, and that this can be achieved in balance and in concert with other urgent and space-requiring activities. Latter relate to the societal challenges for the need for more houses and residences, renewable energies, and climate resilience of- and circular building performance in cities. The module Sustainable Cities is practice oriented and case based. Each year specific city quarters will be designated for analysis. Although for practical reasons Dutch cities located close to Delft will be investigated, the problem of insufficient climate mitigation, climate resilience, circularity performance, and lack of wellbeing of inhabitants is relevant for metropolitan cities worldwide. Integral solutions found for Dutch cities are therefore also expected to be applicable to cities worldwide.	
Study Goals	After completion of this module the students are able to: CO-SC-LO1. List and analyse socio-environmental problems occurring in the built environment with specific focus on the air-, water- and soil compartments in city and infrastructural environments CO-SC-LO2. Formulate research and design questions related to socio-environmental problems in the built environment and conduct a multi-disciplinary project in a small group to find answers to the defined question, dealing with deadlines and requirements set in advance CO-SC-LO3. Use written and oral communication skills to effectively communicate theory, facts and results with respect to urban-specific health and socio-environmental problems with researchers and stakeholders in a multi-disciplinary context in order to achieve integral design solutions and strategies CO-SC-LO4. Work effectively in international, multicultural, inter-disciplinary teams CO-SC-LO5. Acquire new knowledge and skills to achieve multi-disciplinary and integral design solutions and strategies for multiple climate-, environmental- and wellbeing related problems occurring in the urban environment CO-SC-LO6. Quantify the environmental impact of linear and circular building practices using LCA (Life Cycle Assessment) techniques CO-SC-LO7. Define the socio-environmental-economic benefits of circular- and nature based building practices using SCBA (Societal Cost Benefit Assessment) related techniques CO-SC-LO8. Develop integral nature-based design solutions for multiple challenges incidents related to the need for climate mitigation and resilience, circularity performance of built structures, and wellbeing of inhabitants	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: - Lectures. Book chapters and handouts specifically relevant to each lecture should be studied beforehand. In addition to theory explanation, students will be engaged through exercises and quizzes. Small scale experimental demonstrations are given during the lectures in which students participate. - Workshop, hands-on, on campus. Students will learn how to build water-, air-, soil- or noise quality sensors and measuring systems. These will be subsequently used in a practical measuring campaign and obtained results will be analysed and reported. - Visits to designated city quarters is part of the assessment theme. Students will make an inventory of quality of living conditions based on the 7 sustainability performance indicators listed above, and report obtained results. - Case study portfolio. Students will prepare reports of tasks (individual and group tasks) and gather these in an individual case study portfolio. This portfolio will ultimately comprise reports from all assignments covering the module themes assess, review and design. Delivery of assignments will be biweekly formatively assessed. - Case study assignments. Students will form multi-disciplinary groups related to their background expertise during the design theme. Each group will work on a specific set of socio-environmental problems occurring in the designated city quarter and develop multidisciplinary integral solutions based on the knowledge and expertise obtained in the preceding assess and review themes. Groups will report findings to other groups in form of oral presentations. These outputs will be formatively assessed. Final integral solutions proposed by groups will be communicated written (in form of report) and orally (presentation). The final group assignment will be formatively assessed and added to the students individual case study portfolio.	
Assessment	Formative assessment (feedback): During the module, composed of the three themes Assess, Review and Design, tasks have to be delivered. Tasks comprise written reports and oral presentations, in part individual and in part in form of group assignments. The reports will be formatively assessed on completeness and quality by involved lecturers and in one specific case by students in form of a peer-review assignment and feedback will be provided to the students. All tasks will be assembled in an individual case study portfolio which will be summatively assessed for completeness and quality in the last week of the module. Summative assessment:	

	1. One Mobius (computer) individual examination in the last week of the module, covering the theory treated in the given lectures in the three themes (counting for 30% of final grade) 2. The individual case study portfolio will be summatively assessed for completeness and quality in the last week of the module (see Formative assessment, case study portfolio tasks) (counting for 70% of final grade) Minimal grade for both subgrades must be 5.8 to pass the module
Special Information	The Sustainable Cities Cross Over module is divided in the three consecutive and connected themes Assess, Review and Design. A short description of the three themes and its contents is as follows: Module theme 1 Assess: Quantitative assessment of city quarter quality covering 7 performance indicators. In the first theme assess the designated city blocks will be visited and quantitatively analysed for current state of quality based on the 7 indicators listed above: Topics: - Quality of built structures in relation to environmental footprint, circularity- and service life performance- Environmental quality in relation to air-, water- and soil quality and presence of litter and occurrence of noise - Build your own air-, water-, soil- or noise quality sensor (practical; including analysis and interpretation of obtained data and data uncertainty assessment) - Wellbeing quality in relation to risk of heat stress, risk of flooding, quantity and quality of public spaces (recreational- and social interaction opportunities) and quantity and quality of present nature - Methods and techniques for quantification of indicator values- Sustainability assessment methods: LCA (Life Cycle Assessment) and SCBA(Societal Cost Benefit Assessment) Module theme 2 Review: Methods for quality improvement of the built environment In the second theme review specific techniques and methods potentially suitable for mitigation or solving of problems related to the 7 sustainability indicators addressed in module unit 1 will be reviewed. Specific focus will be on green solutions, i.e. through application and implementation of vegetation in the built environment as a way to integrally tackle multiple problems simultaneously. Topics: - Methods and techniques for improvement of air-, water-, soil and noise quality- Circular and other sustainable building strategies for improvement of environmental performance of and wellbeing in city quarters - Eco-engineering solutions for improvement of liveability, climate resilience and climate adaptation of cities. Module theme 3 Design: Integral solutions for improved wellbeing in cities In the third theme design, integral solutions that can significantly mitigate problems, as identified in theme assess, through implementation of technologies reviewed in theme review, will be proposed and quantitatively evaluated. Specifically in theme design students from different MSc program and track backgrounds (EE, AES, CE, Architecture) will work in multidisciplinary teams to apply own background and new obtained knowledge in the assess and review themes to formulate and develop effective nature-based integral solutions to the multiple problems occurring in the analysed city quarters. Specific goal of theme review is to convert current non sustainable city quarters to climate resilient and desirable ones through application of integral sustainable solutions. Topics: - Integral building- and ecoengineering design solutions - Multidisciplinary creation and presentation of design solutions - Critical (qualitative and quantitative) review of design solutions including LCA and SCBA analysis
Expected prior Knowledge	N/A
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	Examples of books: Urban Climates (Oke, Mills and Christen 2017); The city of the future by BNA (Branchevereniging Nederlandse Architectenbureaus), publisher Blauwdruk (2019); Stad x Klimaat Het gebouw als watermachine by BNA and TUDelft (2020) (in Dutch, but English handouts will be provided to non-Dutch speaking students); TEEB Manual for Cities: Ecosystem Services in Urban Management (2011) (by The Economics of Ecosystems and Biodiversity (TEEB): Geneva; teebweb.org). These books comprise theory and review a number of practical cases concerning urban heat stress and integral ecoengineering solutions for additional socio-environmental problems occurring in the built environment. Additional specific literature in form of scientific papers and book chapters on analytical techniques will be provided via BrightSpace. Experimental material for the sensor workshop will be provided during the practical.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2002	Engineering for Global Development	10
Responsible Instructor	Dr.ir. M.W. Ertsen	
Module Manager	Dr.ir. M.W. Ertsen	
Instructor	Prof.dr.ir. D. van Halem	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr. E. Abraham	
Co-responsible for assignments	M.D.M. Palmeros Parada	
Contact Hours / Week x/x/x/x	12 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Focusing on engineering in the Global South, this module relates engineering applications (in global engineering) to a societal debate (what is (desirable) development), a theme spanning domains, programs and tracks within the CEG Faculty. Students learn about similarities and differences of techniques and methods from different domains. Building on the idea that engineering solutions useful and needed as they are are always positioned within a societal debate, students discover their own position. Students learn how to engage their own expertise within debates with other experts and other stakeholders on issues of development, political aspects of engineering, options for interventions, contexts, etcetera. Learning Unit 1: Engineering (and) Development 1 (Credits: 5) In a series of workshops, with discussions, guest lectures, design sessions, etcetera, students are introduced to themes and applications in the Global Engineering domain. After group discussions, individual students prepare documentation on their own position(s). Differences and similarities in debates and positions on technological development influencing engineering solutions are explored. The exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. These sessions are related to this list of key topics: Development theory (societal development, power relations, economic issues, globalization); Role of technology (engineering decisions, innovation ideas, history of technology); Co-creation and design interventions (participatory design, tinkering, values in design); Sustainable development goals (Differences and connections, development cooperation, role of states and UN institutes). Learning Unit 2: Engineering (and) Development 2 (Credits: 5) Teams of 5 to 7 students work on a real-life issue, relevant to at least a number of different stakeholders (design, planning, methods, etcetera), in the Global Engineering domain. The teams mobilize content from unit 1 in their project, in applying concepts and bringing their team work into unit 1. The projects are made available through the Delft Global network and will vary each year. Examples of possible projects include water provision in urban or rural settings, plastic pollution, affordable housing, access to energy. Team members will discuss the project with members of other teams, to facilitate reflecting on team results. List of key topics: Design teams working on a specific challenge; Team working sessions under guidance of a moderator; Exchange meetings between teams (discussions and workshops); Reporting on team results; Presenting team results in different formats. Please note that the exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. The projects are made available through the Delft Global network and will vary each year.	
Study Goals	After completion of this module the students are able: CO-EGD-1. Explain their definition of and position towards societal development CO-EGD-2. Recognize existing approaches on the role of technology in society CO-EGD-3. Recognize how different stakeholders link engineering solutions to (ideas on) societal development CO-EGD-4. Relate their own expertise to other fields of expertise when developing technologies CO-EGD-5. Activate their own expertise to design engineering solutions for given development issues CO-EGD-6. Specify their own role in processes of co-creation in engineering	
Education Method	Teaching methods involved in Unit 1: Interactive setups, different per session (discussions, role plays, etcetera); Exercises (from closed to more open); (Guest) lectures (a few) Teaching methods involved in Unit 2: Team training (discussion, designing, team roles, team task division); Reporting and presentation skills workshops; Exchange workshops	
Assessment	Formative assessment (feedback): Unit 1: Individual reports: reflection of the learning goals overall in relation to the session; Active participation in sessions (demonstrated in minutes and during sessions) Unit 2: Team: report (quality of content, societal relevance, team work process); Individual: presentations during exchange (quality of content, presentation skills); Individual: debates during exchange between teams Summative assessment: - Unit 1: Individual essay on a theme that relates to the sessions and study material (consistency of the argument, references) - Unit 2: Individual feedback essay on the team project (position in the team, reflection, writing skills) - The final grade is the average of the two grades. Each grade required to be at least 5.8.	
Special Information	Week 1 - week 8: 3 sessions of 4 hours each, with associated reading, group discussions and individual reflection. The exact setup of the sessions is decided on for each study year, as themes and topics are adapted to debates as they are encountered in relevant circles. The projects are made available through the Delft Global network and will vary each year. Week 9 - week 10: Self study	
Expected prior Knowledge	This course is open to all students from MSc programs.	
Academic Skills	Analysis of societal positions; reading texts; discussion skills; written and oral communication; team work	
Literature & Study Materials	The study material is collected each year, related to actual debates. Students read papers (about 5 to 7) and discuss these.	
Judgement	N/A	
Permitted Materials during Exam	N/A	
Collegerama	No	

CEGM2003	Data Science and Artificial Intelligence for Engineers	10
Responsible Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Responsible Instructor	Dr. H. Wang	
Responsible Instructor	Dr. R. Taormina	
Module Manager	Dr. R. Taormina	
Instructor	F.P. van der Meer	
Instructor	Dr. K. Anupam	
Instructor	Dr. O. Cats	
Instructor	P.K. Krishnakumari	
Instructor	Dr. D.V. Voskov	
Instructor	Dr. M.A. Schleiss	
Instructor	J.O. Colomes Gene	
Instructor	Dr. G. Rongier	
Contact Hours / Week x/x/x/x	5-7 blocks a week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Elective for MSc programmes AES, CE, ENVM	
Course Contents	Module Introduction Data Science (DS) and Artificial intelligence (AI) are poised to revolutionize all aspects of Civil Engineering, Environmental Engineering and Applied Earth Sciences. Our interdisciplinary module in Data Science and Artificial Intelligence for Engineers (DSAIE) will offer 2nd year Masters students the opportunity to learn these powerful tools and apply them from the very beginning. We will focus primarily on Probabilistic Machine Learning and Deep Learning methods, effective data-driven approaches to deal with the uncertainty and the complexity characterizing the challenges of the interconnected natural, living and built environments. During this module, the students will broaden their engineering understanding through collaboration with peers from diverse backgrounds, following instruction from experts across various programs, and applying what they learn on real-world case studies. The course will also provide plenty of opportunities to further develop their Python coding skills through hands-on exercises and workshops in areas such as collaborative coding, high performance computing, building automation and testing, and advanced data analytics. By the end of the module, the students will have acquired a practical and holistic understanding of DS/AI and its applications in their fields of choice, which will be extremely valuable for the students future careers. Module Units The proposed cross-over module is divided into three units: Unit 1. Data Science & Probabilistic Machine Learning (4 EC) Unit 2. Deep Learning (3 EC) Unit 3. Project (3 EC)	
Study Goals	After completion of this module, the students can: LO1. Explain core concepts in data science, machine learning and artificial intelligence, including the fundamentals of probabilistic and deterministic data-driven techniques. LO2. Translate a real-world problem related to the natural, the living and the built environment into a DS/AI application by following a multidisciplinary approach. LO3. Adapt suitable probabilistic and deterministic DS/AI techniques to solve real-world problems related to the natural, the living and the built environment. LO4. Implement DS/AI pipelines in Python using best coding practices for execution robustness, result reproducibility and validation. LO5. Present the results of a DS/AI analysis in oral and written formats, being able to i) motivate their choices, ii) defend their opinions, iii) cater to diverse audiences, and iv) critically evaluate feedback. LO6. Build workable DS/AI solutions for a multidisciplinary set of realistic problems in cooperation with peers of various backgrounds.	
Education Method	Unit 1 & Unit 2 In a typical lesson, the instructor will illustrate theoretical concepts to the class in a plenary session. When suitable, the instructor will alternate theory with exercises (individual or group) and coding walkthroughs/demos in Python. These will be based on selected datasets chosen by the instructor. While most of the lectures will be on campus, lectures for advanced topics will be delivered via recordings. Unit 3 Theory. Instructors can provide limited content related to the applications via recordings. Advanced topics of UNIT2 can be chosen in combination with the specific project selected. Applications. There will be 3 applications on very diverse datasets for students to apply the concepts of Unit 1 and Unit 2 while preparing for the final project. Students will work in groups of 5; instructors will be available with TAs for support in class. Each application will last for approximately 4 hours, as done in the MUDE. Projects. In week 4, project instructors will introduce case studies/datasets that students can exploit for developing their final projects. Contrary to applications,	

the proposed projects may not have clearly defined research questions. Students will form groups of around 5 to work together on their final projects. Students can also come up with their own dataset and case study, provided they find the support of an instructor that can act as a coach/mentor (open inquiry, see Figure 3). Groups will be formed in Week 5. Students will preferably link the self-study advanced topics of Unit 2 (Week 6) to the project they choose.

Coding Workshops.

The coding workshops will be usually delivered as a short lecture followed by hands on exercises (e.g., on notebooks, HPC terminals, ...) to master specific coding knowledge related to AI and Data Science. Instructors and TAs will be available in class for the entire duration of the workshops.

Practical Guide

Unit 1. Data Science & Probabilistic Machine Learning (4 EC) [LOS 1, 3, 4]
Week 1 to Week 3

In this Unit we will first introduce the module and reactivate prerequisite knowledge from the MUDE with respect to Supervised Regression. We will then explore Supervised Classification, the other major family of Supervised Machine Learning methods. We will then study the Bayesian perspective on model selection, before exploring tree-based methods and kernel based methods for regression and classification. After refreshing MUDE knowledge on clustering and PCA, we will explore more advanced topics in unsupervised machine learning and generative machine learning. Students will work on two different application sessions to reinforce their knowledge.

Unit 2. Deep Learning (3 EC) [LOS 1, 3, 4]
Week 4 to Week 6

This unit will mainly deal with Deep Learning methods, representing the state-of-the-art of AI-based applications in many fields, including CEG-related ones. We will focus on key topics such as representation learning, and the use of inductive biases and symmetries to overcome the curse of dimensionality. These will be introduced with respect to convolutional and recurrent neural networks, but we will explore other powerful and more recent approaches such as physics-based deep learning and transformers. Students will work on one application session to reinforce their knowledge.

In Week 6, students will select two advanced topics to study independently (i.e., videos, materials and notebook). These will be preferably linked to the projects they select in Week 4 (see Unit 3).

Unit 3. Project (3 EC) [LOS 2, 3, 4, 5, 6]
Week 4 to Week 10

Projects are an essential part of the module and constitute a dedicated unit. Students will work in groups on a selected CEG-related project to apply and reinforce the skills acquired in Unit 1 and Unit 2 while collaborating on real case studies. All available projects will be showcased by the responsible instructors in Week 4, and students have time until Week 5 to choose their project.

Potential case studies

Here follows a tentative list of potential case studies from which students can select their projects. Students can choose their projects regardless of their background. Students can also come up with their own project from scratch, provided they find a mentor by Week 5.

Environmental Engineering

Rainfall-runoff modelling using Deep Learning methods
Prediction of drinking water demands
AI-based modelling of water quantity and quality processes in urban water networks
Plastic detection using AI-based computer vision
Defect detection in sewer systems with AI-based computer vision

Civil Engineering

Surrogate modelling for flow through porous structures
Accelerate finite elements methods for structural mechanics with AI.
Defect detection of railway infrastructures using deep convolutional neural networks and on-board camera images.
Flood Risk Mapping with Deep Learning methods.
Unsupervised learning for rail health condition classification.
Unravelling the Dynamics of Urban Demographics
Incorporating safety and beauty perceptions into travel journey planners using street view images

Applied Earth Sciences

Rainfall nowcasting using Deep Learning methods.
Machine Learning for applications in Energy Transitions
One-shot learning for generative modeling of geological structures
Exploring the effect of different formulations of Gaussian processes on spatial predictions

Assessment

Formative assessment (feedback)

Theory

Computer-based quizzes with multiple answers for each topic, including coding exercises. These will be made available every week. A mock-up theory exam will be made available in Week 5.

Project

Formative feedback provided in class during applications 1-3 which serve preparatory activities for the project. Feedback from instructors and peers on project progress at the end of Week 7.

Summative assessment on module level

There will be two summative assessments: 1) an ANS written exam for the theoretical part, and 2) a group presentation with interview for project evaluation.

Students understanding of theoretical content/coding skills will also be assessed as part of the project learning line, specifically their ability to apply the theory and reflect on the meaning and limitations of produced results. This justifies the equal value of the two summative assessments (i.e., 50% each).

The project deliverables include the final presentation plus a code repository. The grade will reflect the originality and impact of the final solution, the quality of presentation, and the quality of code repository and documentation. Dedicated assessment tools for group work will be employed to facilitate marking of individual contributions.

All subgrades must be >5.8 to pass the module.

Special Information

Weekly Syllabus

10 ECs = 280 hours. We consider 28 hours per week divided in 7 blocks of 4 hours each. There will be around 50 hours of lectures, mainly focused on the first half of the module. The second half focuses on the project. Sufficient time is reserved for

self-study during the entire module. The students are free to organize their self-study as they want, but instructors and/or teaching assistants will be available in a classroom for explanations after lectures. When the project runs, there is a block reserved each week to interact with project instructors and TAs.

Week 1

Introduction to the course with overview.

Translating real-world problems to build data science and artificial intelligence solutions.

MUDE Recap: supervised regression, deterministic perspective on model selection: bias-variance trade-off, regularization, cross-validation, neural networks.

Coding Workshop #1: PyTorch

Data types: time series & sequences, tabular data, images & raster data, unstructured vs structured data

Probabilistic perspective on model selection: informative/uninformative priors, hyperpriors, Maximum a Posteriori (MAP) estimates, Bayesian model comparison, Empirical Bayes, Variational Bayes

Week 2

Supervised classification: classification with decision regions, decision theory for classification, classification with basis functions, logistic regression, classification with neural networks

Feature engineering, input selection and dealing with unbalanced data.

Coding Workshop #2: collaborative coding for DS/AI projects.

Frequentist and Bayesian kernel-based methods, instance-based learning

Application #1: Supervised Classification

Week 3

MUDE Recap: unsupervised machine learning, k-means and PCA

Inference and learning with discrete and continuous probabilistic latent variables, introduction to Generative Machine Learning, Probabilistic Principal Component Analysis (PPCA)

Application #2: Unsupervised & Generative Machine Learning

Week 4

From Probabilistic Machine Learning to Deep Learning: curse of dimensionality, representation learning, inductive biases & symmetries for data efficiency

Projects showcase

Convolutional Neural Networks: 1D/2D/3D convolutional layers, pooling, composition, feature extraction, translational invariance and equivariance, data augmentation, image regression and classification.

Regularization: early-stopping and dropout layers.

Pre-trained models and transfer learning.

Coding Workshop #3: AI on DelftBLUE

Application #3: Deep Learning

Week 5

Recurrent Neural Networks: sequential inductive bias, simple RNNs, gated recurrent neural networks (LSTMs and GRUs), time series data.

Intro to Physics-based deep learning: basics of neural ODE/PDE solvers, neural operators, physics informed neural networks, physics guided machine learning.

Other type of inductive biases: quick introduction on transformers, graph neural networks, bayesian deep learning.

Presentation of advanced topics that students can select (see below).

Project selection

Initial work on projects

Week 6

Advanced Topics

Project work

Week 7

Project work

Project preliminary presentations + feedback

Week 8

Project work

Week 9

Project work

Written exam

Week 10

Project work

Project submission and presentation

Expected prior Knowledge

CEGM1000 - Modelling, Uncertainty and Data for Engineers

Academic Skills

Analysis of complex (multi-cause) problems related to the built/natural/living environment; information collection and evaluation; multi-disciplinary and integral problem solving; project management; oral communication; teamwork; basic coding skills.

Literature & Study Materials

Unit 1 & Unit 2

Study material: lecturers slides, recorded videos/Collegerama, Jupyter notebooks, GitHub code repositories + documentation, datasets.

Literature:

Bishop, Christopher M. "Pattern Recognition and Machine Learning", 2006.

Murphy, Kevin P. "Probabilistic Machine Learning: An introduction.", 2021.(open source <https://probml.github.io/pml-book/book1.html>)

Murphy, Kevin P. "Probabilistic Machine Learning: Advanced Topics.", 2022. (open source <https://probml.github.io/pml-book/book2.html>)

Goodfellow, Ian, et al. Deep learning, 2016. (open source <https://www.deeplearningbook.org/>)

Molnar, Christoph. "Interpretable machine learning", 2020. (open source <https://christophm.github.io/interpretable-ml-book/>)

Wilke, Claus O. Fundamentals of data visualization: a primer on making informative and compelling figures, 2019. (open source <https://clauswilke.com/dataviz/index.html>)

Selected seminal papers in Data Science, Machine Learning, Deep Learning and Interpretable Machine Learning.

Select webpages and blogs.

Unit 3

Study material: datasets with description, code repositories, recorded videos (optional), all material of Unit 1 and Unit 2.

Literature: each instructor will provide a selected list of scientific papers and other publications (e.g., MSc Theses of previous

students) for the applications/projects they propose.

Coding Workshops & Computational Resources

The coding workshops will be partially based on high-quality, worldwide known, open-source Data Science and Computing workshops provided by The Carpentries (<https://carpentries.org/>), CodeRefinery (<https://coderefinery.org/>), and Aalto Scientific Computing (<http://scicomp.aalto.fi/training/>). The instructors will select, summarize, modify, and expand the material available from these resources to fit the scope of this Module. Students will be able to access the DelftBLUE supercomputer for the entire module (<https://www.tudelft.nl/en/dhpc>).

Judgement

See assessment

Permitted Materials during Exam

N/A

Collegerama

No

CEGM2004	Noise and Vibration: Generation, Propagation and Effect on Humans and Environment	10
Responsible Instructor	A. Tsouvalas	
Module Manager	A. Tsouvalas	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	approximately 4*2HC Theory and 1*2HC project with instructors per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	General description: Noise is sound considered unpleasant, loud or disruptive to hearing, which has become an increasing concern throughout the world. While urban noise (from transportation, construction, etc.) brings a negative influence on human health, underwater noise (from pile driving, seismic surveys, etc.) threatens the leaving space of marine mammals and fish. Similar to noise, vibration is undesirable in many cases because it causes damage to engineering structures and discomfort to people. From the viewpoint of physics, noise is vibrations transmitted through a medium, such as air or water. Therefore, effective control of noise and vibration can be possibly be achieved at the source, path, or receiver. In this module, students learn the fundamentals of noise and vibration, including structural vibrations, sound generation and transmission, perception of noise by inhabitants, conduct in-class/lab tests to obtain and analyse vibro-acoustic data, and use numerical tools to mitigate the adverse effects on the urban and marine environment. Overall structure of the module: The module structure consists of two interrelated themes; Theory and Project. The Theory theme starts first and students are exposed to fundamentals of noise and vibrations, important sources of noise in air and underwater and dynamics of structures. The coupling between the structure and the surrounding media, in which waves are transmitted, is central in the Theory theme. Vibro-acoustic modelling techniques are discussed and software (that the students use for the Project theme) is introduced. Besides modelling, students are also exposed to measurement setups and equipment regularly used to measure and quantify noise and vibration levels. The theme on Theory is followed by the Project theme in which students apply the theoretical concepts to realistic environmental noise and vibration problems. Case studies, in which levels of vibrations and/or noise are unacceptable, are selected by the instructors together with the relevant industry. Students form groups to tackle the environmental challenging case. Results of this investigation are reported by the group and presented at the end of the quarter. In the theme theory the mathematical background of acoustics and vibration is taught with an eye on real-life applications. Emphasis is placed on an in-depth analysis of structure-borne noise emission and its effective cancellation either at the source or at the propagation path. This unit is developed such that it includes illustrative applications of the different noise sources in the environment including the effects on people and animals. This unit is fundamental and interlinked to the theme project in which a practical implementation of the theoretical concepts is accomplished. In the theme Project, students apply the theoretical concepts to realistic environmental noise and vibration problems. The students are divided into groups. Groups are formed by 2-3 students following different MSc programmes in order to enrich the expertise within the groups and to be able to reach scientific depth in the final result of the project.	
Study Goals	After completion of this module, the students are able to: LO1 Formulate the governing equations for problems in which structural vibrations and sound generation and propagation are involved. LO2 Construct analytical formulae and numerical models to simulate structure-borne acoustic emission. LO3 Analyse the environmental noise and translate those measurements to (internationally) accepted noise metrics. LO4 Evaluate the impact of noise and vibrations on the environment and humans using metrics for noise and vibration perception. LO5 Design noise mitigation strategies to tackle urban and underwater noise pollution. LO6 Compose scientific reports on vibration- and noise-related topics using professional terminology.	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process. - Lectures. Pre-study material (book chapters and handouts) will be provided to students before the lectures. The course content will be described with real case studies, examples and interactive demonstrations with the active participation of the students. Students will be asked to refresh their knowledge at parts to avoid repetition of material covered earlier in the curriculum. - Hands-on assignments. While the students learn about the theory and mathematical descriptions related to the physics of noise and vibration, they should apply the outcomes to the case studies. The students should complete bi-weekly tasks in small assignments which will help them to execute the project. Delivery of bi-weekly assignments will be optional and formative feedback will be provided by the instructors. - Workshop. The students will learn the basic noise and vibration measurement equipment, use them, and analyse the collected data. The measurements sessions will be performed either in-class, in the lab facilities or, ideally, in the campus during the workshop. - Self-study: During weeks 1-5, CE students read selected topics related to acoustics and EnvEng, AES students topics that relate to structural dynamics. This way students get exposed systematically to theory from another discipline. The Project theme is organized by means of feedback/plenary sessions. The students form multi-disciplinary groups related to their backgrounds. Each group can choose one of the provided urban or underwater noise problems as a project topic under the supervision of the instructors so that all subjects are covered equally. Each group should develop innovative solutions based on the obtained knowledge in the preceding theory theme. Groups will present their process and results to other groups during bi-weekly meetings with the aim to initiate a live discussion and provide a formative assessment of the work-in-progress. Each group will prepare a detailed final report related to their project. These outputs (presentation and final report) will be summatively assessed.	
Computer Use	The software to be used by the students for the execution of the project will consist of: - available licensed software from TUD (e.g. Comsol Multiphysics) and - in-house software prepared by the teaching team.	
Practical Guide	Weeks 1 and 2: Introduction to noise and vibrations - Fundamentals of acoustics: acoustic metrics, sound wave propagation. - Fundamentals of structural vibrations: vibrations of beams and plates, coupling to the acoustic domain and sound radiation. - In-class examples with analytical solutions, homework.	

	Week 3: Noise and vibration sources and impact on environment (distribution of projects and group forming at the end of week 3) - Noise and vibration sources in the urban and marine environment (traffic, rail, underwater and air noise) - Underwater noise and vibration sources, defining the acoustic problems. - Audibility range of species, levels of injury, etc. - WHO reports, MSFD, biological research, noise metrics, international standards. - Levels of vibrations acceptable in buildings. - Impact of traffic, rail and air-induced noise and vibrations on human life - Underwater noise impact on marine life: Noise from shipping, pile driving, seismic surveys, explosions and wind. - In-class examples with analytical solutions, homework. Weeks 4 and 5: Propagation of waves and modelling - Brief information related to ray theory, mode theory, finite elements, and boundary elements (air/water/soil layered waveguides). - Why do we need simulations? How to choose among the different simulation techniques? - Model validation and interpretation of the modelling results. - Generation of noise maps. - In-class numerical examples (homework related to simulations with FE or inhouse software; the software to be used will be determined by the instructors depending on the type of problem analyzed). Week 6: Sound and vibration measurements - Noise measurement equipment: sound level meters, hydrophones. - Lab-scale (or in-class) demonstration of working principles and data analysis. - Analysis of noise and vibration data, converting pressure to relevant metrics for the noise impact (homework related to the analysis of measurement data). - Tire-pavement measurement methods. - Rail-wheel measurement methods. Weeks 7 and 8: Noise mitigation and vibration suppression - Noise mitigation and vibration suppression strategies, smart mitigation strategies (acoustic black holes, metamaterials), examples for the noise mitigation related to pile-driving. - Measurements in lab/field. - In-class discussion of case studies: noise mitigation and vibration suppression. Project - Conducting a project regarding either urban or underwater noise: Each group will conduct a project (distributed in the end of the 3rd week of the module) with the focus being placed in one of the subjects listed below: 1. Structural-borne noise and vibrations in urban environments: traffic noise including modelling of tire-pavement interaction, railway noise including modelling of rail-wheel interaction, noise from civil construction in urban environments. 2. Structural-borne noise and vibrations in offshore environments: seismic surveys offshore, noise and vibrations from installation of foundations offshore, shipping noise, drilling. To accomplish the project, students are required to conduct simulation, analyse the results of noise and vibration data (data sets will be provided by the instructors), design a strategy of noise mitigation and/or vibration suppression, and compose a final report.
Assessment	Formative assessment (feedback): Students will discuss their results and formative feedback will be provided by the instructors and based on plenary sessions during the quarter on a weekly basis. Formative assessment of weekly assignments regarding theory unit. Summative assessment: The module is assessed by means of the project work with a report and an in-class presentation.
Expected prior Knowledge	N/A
Connection to other modules	The CO module brings together engineers of various backgrounds and challenges them with a problem in which both the source of noise and vibrations, the emission of waves and the mitigation of noise and vibrations are tackled in a holistic manner. This requires teams consisting of students of various expertise which is mobilized during the execution of the project. Given the focus of the module, there is some overlap with the following parts of the individual programmes: - Overlap with the AEE track in the EnvEng programme on the fundamentals of acoustics and measurement of noise; - Overlap with the SE track in the CE programme on fundamentals of structural vibrations and measurements; - Overlap with AES programme on elements of wave propagation. Despite the overlap though, the students here will need to go beyond their comfort zone, tackle a multi-disciplinary problem, and work together to find a suitable remedy to reduce the disturbance to inhabitants. They will also deepen their knowledge further by tackling not only the individual elements dealt with earlier in their programmes but also their complex interaction.
Academic Skills	Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.
Literature & Study Materials	Theory: Mathematical description and physics related to noise and vibration, analysis of practical problems of structure-borne noise and vibrations. Selected chapters of the following books: [1] Allan. D Piece, Acoustics: An Introduction to Its Physical Principles and Applications, 2019, Springer. [2] Steven L. Garrett, Understanding Acoustics: An Experimentalists View of Sound and Vibration, 2020, Springer [3] M. Möser, G. Müller, Handbook of Engineering Acoustics, 2013, Springer [4] D. Thompson, Railway Noise and Vibration, Mechanism, Modelling and Means of Control, 2010, Elsevier https://www.sciencedirect.com/book/9780080451473/railwaynoise-and-vibration. [5] F. Fahy, P. Gardonio, Sound and Structural Vibration: Radiation, Transmission and Response, 2008, Academic Press. [6] Robert Bernard and Roger L. Wayson, An Introduction to Tire-Pavement Noise of Asphalt Pavements, 2007. [7] TRBs National Cooperative Highway Research Program (NCHRP) Report 630: Measuring Tire-Pavement Noise at the Source examines a Suggested Procedure for Measuring Tire-Pavement Noise at the Source Using the On-board Sound Intensity (OBSI) Method. Additional specific literature (if any) in form of scientific papers, other book chapters, or public reports will be provided in class. Project: Conducting a project regarding either urban or underwater noise Books: same as theme Theory above. Additional materials including the manuals of the measurement equipment and the papers reviewing the existing studies of the specific topics on urban or underwater acoustics.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	Yes

CEGM2005	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE)	10
Responsible Instructor	E. Ragno	
Module Manager	E. Ragno	
Instructor	Ir. F.C. Vossepoel	
Instructor	G.F. Nane	
Instructor	Dr.ir. O. Morales Napoles	
Instructor	Dr.ir. R.C. Lanzafame	
Contact Hours / Week x/x/x/x	approximately 4 hours lectures, 4 hours workshop per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Probabilistic Modelling of real-world phenomena through ObseRvations and Elicitation (MORE) is a methodological type module oriented and structured around advanced topics in probability and statistics applied to engineering problems. Rainfall, water flowing into rivers, cars and trucks crossing bridges, ocean waves, and earthquakes are all examples of phenomena of which we wish to quantify the magnitudes, frequencies, uncertainties, and risks to make our daily life safer and more manageable. Such phenomena are characterized by some degree of randomness, meaning that their future outcomes are unknown. Hence, how do we design sewer systems to deal with precipitation in urban areas, estimate the loads that will compromise the structural integrity of a bridge or protect coastlines from flooding? In this module, students will learn methods of probability and statistics to infer characteristics of random and uncertain phenomena around us when observations (data) and physics-based numerical models are available, by using (non)stationary extreme-value analysis, dependence modelling in higher dimensions and data-assimilation techniques. When data pose challenges or are simply missing, students will learn about structured expert judgment methods to quantify uncertainty by using expert opinion. The module will provide students with a theoretical understanding of different probabilistic techniques so that they can discern which method(s) is(are) most suitable for analysing the problem at hand. At the same time, students will learn how to translate their theoretical knowledge into computer codes. As part of this course, students will apply the methods learned in an interdisciplinary group project around specific themes, that will vary on a yearly basis. Example of projects themes are (i) shoreline protection in the face of sea level rise; (ii) subsidence as a results of ground water extraction; (iii) water management assessment under urbanization and climate change; (iv) reliability of aging infrastructure under climate change. Students will work in groups, about 4 members each, and based will come up with specific research questions for their theme that they will answer using methods learned. The total 10 EC will be divided as follows: 6 EC in Unit 1 - Theory and Example applications and 4 EC in Unit2 - Group Project.	
Study Goals	After completion of this module the students will be able to: LO1 Discuss the concepts of non-stationary extreme value analysis, multivariate probability distributions, multivariate uncertainty quantification, data assimilation, and expert judgment. LO2 - Apply the concepts of non-stationary extreme value analysis, multivariate probability distributions, multivariate uncertainty quantification, data assimilation, and expert judgment to problems in civil engineering, applied earth sciences and environmental engineering. LO3 - Analyse and discuss the engineering applications (for example climate change, extreme discharge, induced seismicity, maximum bending moments, traffic loads, infrastructure design, pollution health effects) that accompany the techniques in probability and statistics taught in this module with an emphasis on the added value of uncertainty quantification. LO4 Write computer codes using open-source, e.g., Python, or commercial, e.g., Matlab, programming languages to translate key methods and techniques presented in the module and to carry out analysis related to specific track-related problems. LO5 Make use of existing open-source code and software to carry out analysis using more advanced methods which require higher-level coding skills. LO6 - Design and carry out an interdisciplinary research project of interest in science or engineering that incorporates methods taught in this module within a guided inquiry-based learning framework. LO7 - Work effectively in international, multicultural, inter-disciplinary team	
Education Method	Theory and example applications Lectures: the module will offer a mix of classical style lectures from an instructor, collaborative group sessions and educational videos. Dedicated classroom time will be provided each week for the lectures and collaborative sessions with students and instructors. Hands-on learning via computer practicum and think-pair-share activities will be implemented in class. The students will benefit from the interaction with their peers and from the need to re-interpreting theoretical concepts in consecutive executable actions. At the same time, teacher-center lectures are necessary to show the students the theory and rationale underlying the statistical methods covered in class. An example application of these activities can be related to extreme value analysis. Students will be provided with a time series for a general variable (for example maximum yearly wave height in the North Sea for CE, maximum PM2.5 concentrations in Amsterdam for EE, or expected probability of earthquake occurrence in AES) and they are asked to extrapolate a sub-sample of the observations containing only extremes. First, they will do this individually and then they will share ideas with their partner. After gathering students ideas, the instructor will explain how extremes can be sampled based on current methods in extreme value theory. Students will then apply such methods into executable coding scripts to sample extremes from a time series of their interest. Applications: Students will work on weekly assignments related to practical applications of the theory learned in class. Such assignments will derive from the three programs so that the students can appreciate the interdisciplinary character of statistical methods. The applications proposed will be related to a specific track in such a way that also students that do not come from that track can understand them, as exemplified in the Module Content section. Students are encouraged to follow the think-pair-share approach. In this way, they will first try to solve the problem individually and then compare their work with their peers for feedback. Dedicated classroom time will be made available for peer feedback and collaboration.	

Group project

A guided inquiry-based learning approach will be followed, similar to that implemented in the MUDE module (Q1 and Q2 of the first year MSc programme). This module builds on the project-related skills developed in MUDE by allowing students to identify their own topic of interest to further analyse it using at least two methods covered in class. Students progress will be followed by the instructors starting from week #4, in which an introduction on the remaining topics covered in the module will be given to facilitate the project definition. Guidance documents for students (e.g., effective group work strategies, formulation of research questions, report writing) will be provided. Students will have all the material from the theory section. However, students will be encouraged to carry out a literature review to get familiar with how the scientific community has approached similar cases. Teaching methods in Unit 2 are largely focused on guidance with individual groups during the Q&A sessions in weeks 7 and 8. For example, some students may need help solving coding issues, whereas others may need advice extending the methods covered in Unit 1.

Practical Guide

Theory and example applications:

In this unit, students will learn theoretical concepts related to advanced techniques in probability and statistics. The unit content is spread over a period of 6 weeks. In each week, one topic will be treated comprehensively, from theory to practical applications.

The following topics will be covered:

(w1) Extreme value theory: Power law distributions, rank ordering statistics (including tail index estimation - e.g., Hills method), Goodness of Fit diagnosis (Kolmogorov-Smirnov, Anderson-Darling etc.), Heavy tailed distributions; Review of extreme value theory peak over threshold, block maxima; Non-stationary extreme value theory. Example assignment: data analyses of (extreme) observations for inferring, e.g., design values.

(w2) Data Assimilation: maximum a posteriori probability methods, ensemble methods, variational methods and particle methods. Example assignment: quantification of subsidence due to fluid extractions.

(w3) Expert Judgment: Cooks method for uncertainty quantification, Elicitation of probabilistic dependence (multivariate uncertainty). Example assignment: Elicitation of future hydro-meteorological variables due to climate change, or, Assessment of uncertainty regarding mortality and morbidity due to PM10 and PM2.5 through Cookes method.

(w4) Bivariate distribution using copulas: Introduction to copulas compared to traditional bivariate distribution; Covariance, product moment correlation coefficient, rank correlation coefficients (Spearman's rho, Kendall's tau), tail dependence coefficient, estimation (maximum likelihood, moments); One parameter copulas: Gaussian copula, Gumbel copula, Frank copula, rotations of copulas. Examples of 2 parameter copulas (t-copula), and 3 parameter copulas goodness of fit diagnostic tools: semi correlations. Example assignment: modelling correlation between PM2.5 (air pollutant) and health effects (e.g., daily mortality) in the context of regression analysis for long and short term effects.

(w5) Multivariate probability via Bayesian Networks: Advanced topics on discrete Bayesian networks; Non-Parametric Bayesian Networks; Dynamic Bayesian Networks. Example assignment: risk and reliability assessment, e.g., Calculation of flooding probabilities, stochastic modelling of vehicle loads in bridges, calculation of maximum bending moments in bridges, health risks, others.

(w6) Multivariate probability using vine-copulas: multivariate distribution via pair-copula construction. Example assignment: multivariate probabilistic design of coastal structures subjected to multiple forces, e.g., wind speed, wave height, wave periods.

Group project:

Students work on a research topic within the proposed themes that they choose themselves where at least two techniques covered during the module will be applied. During week #4, the topics covered in weeks #5 and #6 will be introduced briefly so that students can better identify potential projects. The students will propose a project topic beginning in week #5, which is reviewed by the instructors and finalized by the end of week #6, including a work plan (analysis methods and anticipated results) and definition of the project data set. Weeks #7-8 include a number of Q&A sessions by the instructors for checking students progress and discuss possible common questions. While there will be some review of material, these weeks are dedicated to finalizing the group project. A project is performed in groups of 2 or 3 students (depending on the size of the class), preferably from different programs.

Assessment

Formative assessment (feedback):

Formative assessments on common errors identified in weekly assignments and during projects progress meetings will be provided to the class. Group and individual formative assessments will be provided when necessary.

Peer review based on weekly assignments will be implemented. This in particular for the programming components of each week.

At the end of week #8 and before the final deadline for project report submission, the students will have the chance to present their partial results to their peers and receive peer feedback. The students will receive a feedback form to fill in during the group presentation. Feedback will be on the structure of the presentation, relevance of the project topic, and clarity of the methods implemented.

Summative assessment:

Theory and applications

Summative assessment will consist of an individual oral exam covering both theoretical aspects and applications of the statistical methods taught in class. The oral exam will build upon the analyses carried out in the report and expand to the rest of the topics (theory) discussed in the module. The progress (depth and extent that each student can reach) of the oral exam will depend on the ability of each student to answer to questions. The depth and extent that each student can reach during the oral exam will determine the individual grade. The oral examination is 50% of the total grade. As stated earlier, this grade will be individual and hence may differ from the group project grade per student. LOs 1,2,3 will be tested.

Group project

Summative assessment consists of a written report summarizing project setup and results. The purpose of the group project is to LOs 2,4,5,6,7 will be tested. The group project is 50% of the final grade.

All subgrades must be ≥5.8

Special Information

The methods are first discussed theoretically and then exemplified via applications. Specific applications will be drawn from applications related to the three programs in the faculty. The applications proposed will be related to a specific track in such a way that also students that do not come from that track can understand them. For example, a nonstationary statistical model of sea level height will be developed to include the signal of sea level rise. This example is relevant to coastal engineers but also to earth system scientists. At the same time, environmental engineers can understand the relevance of it without too much in-depth pre-knowledge of coastal dynamics and climate change.

Students will write their own code to solve simple problems using the fundamentals of each method. In-house open-source Matlab- and python-based software will be used for more advanced problems. Example applications will be provided by members from the faculty, some of which will also be linked to the MUDE example applications and project proposals to ensure a coherent learning line in probability and statistics in the new master curriculum.

During week #1 a general overview of all topics in the module will be presented, including project-related examples, to allow students time to choose the methods to implement in their final project. Starting from week #4, topics covered in weeks #5 and #6 will be introduced briefly so that students can better identify potential projects.

Expected prior Knowledge

MUDE CEGM1000

Connection to other modules	Overlap with other parts of the programmes. In the MUDE, the students will get familiar with basic concepts of probability and statistics (e.g., axioms, discrete and continuous random variables, central limit theory, bivariate distributions, Bayes theorem). These basic concepts represent a necessary pre-knowledge for understanding the topics covered in this module. We expect some overlap in extreme value analysis, and bivariate probability distributions, but this would be propaedeutic to the related more advanced topics as nonstationary extreme value analysis, and Bayesian Networks and Vine-copulas respectively. Similarly, students will be introduced to data assimilation techniques that link the data, uncertainty and modeling learning lines in the MUDE, which will also be extended in the MORE. We foresee a small overlap in HOS regarding reliability-based design approach that includes extreme value analysis. Some overlap with CO modules Data Science and Artificial Intelligence for Engineers and Monitoring of Structural Health and Geohazards mostly with respect to Bayes rule (Bayesian updating) and other data assimilation techniques is also expected.
Academic Skills	Analysis of complex problems, communicate scientific knowledge and findings in inter-disciplinary context, critical thinking, project management, team work, written and oral communication
Literature & Study Materials	Theory and applications Group project Students will receive lecture slides, references to relevant scientific literatures, text book, MOOC, recordings of past lectures on related courses, and open source software developed over the year by the instructors. All the material will be accessible via Brightspace. Example literature: Coles, Stuart, et al. An introduction to statistical modeling of extreme values. Vol. 208. London: Springer, 2001. Genest, C. and Favre, A. (2007). Everything You Always Wanted to Know about Copula Modeling but Were Afraid to Ask. J. Hydrol. Eng. 12, SPECIAL ISSUE: Copulas in Hydrology, 347368. Morales Nápoles, O., Worm, D., van den Haak, P., Hanea, A., Courage, W., Miraglia, S.: Reader for course: Introduction to Bayesian Networks, TNO-060-DTM-2013-01115, TNO, Delft, The Netherlands, 2013. Sections: 1, 2. Anca Hanea, Oswaldo Morales Napoles, Dan Ababei, Non-parametric Bayesian networks: Improving theory and reviewing applications, Reliability Engineering & System Safety, Volume 144, December 2015, Pages 265-284 MOOC material for Expert Judgment: https://online-learning.tudelft.nl/courses/decision-making-under-uncertainty-introduction-to-structured-expert-judgment/ Software: Treatment of Uncertainty: OpenTURNS, https://openturns.github.io/. Open source. Extreme value ProNEVA matlab based. Available for distribution. Discrete BNs. Netica & Hugin (commercial but a limited free version). Powerful graphical user interface Matlab & Python (open source) options less friendly graphically but easy to implement coding. Continuous BNs UniNet (commercial but a limited free version). Powerful graphical user interface BANSHEE. Matlab version available in SoftwareX (own development). Python (open source) version under development available by the end of 2021. Vine-copulas R-vine. Open source Expert Judgment ANDURIL matlab based. Available in SoftwareX (own development) ANDURYL python based. Available in SoftwareX (own development) Data Assimilation Python-based Jupyter scripts
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2006	Subsurface Storage for Energy, Water and Climate Applications	10
Responsible Instructor	Dr. H. Hajibeygi	
Module Manager	Prof.dr.ir. M. Bakker	
Module Manager	Dr. H. Hajibeygi	
Instructor	Dr. K.H.A.A. Wolf	
Instructor	Dr.ir. J.M. Bloemendal	
Instructor	Dr. B.M. van Breukelen	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Prof.dr. S. Geiger	
Contact Hours / Week x/x/x/x	Diverse: lectures, computer practicals and lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The subsurface is increasingly used to store fluids associated with energy and for climate change mitigation. There are many potential technologies, and each has its place. In this module, the key technologies which are available and under-development are detailed, including key aspects of flow of fluids in the subsurface. These concepts are utilised to evaluate the technologies, such that we can efficiently use the subsurface. Students will move from broad concepts and how to assess the applicability (Learning Unit 1), through theoretical aspects of how the systems work (Learning Unit 2) to evaluation and design (Learning Unit 3).	
Study Goals	After completion of this module the students are able: CO-SS-LO1. Explain how earth underground resources can be used and sustained in an environmentally responsible manner for cyclic storage of fluids and energy (ILO: 4, 10). CO-SS-LO2. Design a conceptual framework for a given storage project, considering the geology and geomechanics of the host rock and physics of fluids under dynamics (ILO: 1, 3, 4, 8, 11). CO-SS-LO3. Evaluate the performance of storage systems, including the influence of their design parameters and nearby systems (ILO: 4, 5, 6, 7, 9). CO-SS-LO4. Evaluate geochemical processes in order to evaluate the environmental performance of systems (ILO: 4, 5, 6, 7, 9). CO-SS-LO5. Weigh the various advantages and disadvantages of subsurface storage options against each other and propose an optimal solution (ILO: 3, 6, 9, 10). CO-SS-LO6. Communicate their findings clearly in an appropriate comprehensive manner for various technical and non-technical stakeholders (ILO: 2).	
Education Method	Teaching and learning that encourages students to take responsibility for their learning process: Learning Unit 1: Subsurface storage technology - Lectures. Book chapters and handouts specifically relevant to each lecture should be studied beforehand. In addition to theory explanation, students will be engaged through exercises and quizzes. Learning Unit 2: Heat, solute and compressible fluid transport - Lectures on the transport of dissolved substances and heat in groundwater. Small scale experimental demonstrations are given during the lectures in which students participate. - Based on the scientific contents of the lectures, practical sessions will be performed to model the fundamental concepts behind successful storage of fluids in the subsurface across scales. These are mostly in the form of weekly computer modelling exercises using, for example, MODFLOW, MT3D, SEAWAT, DARTS, DARSim, or MRST. Python scripts will be used to generate input files, read output files, and analyse the results. - Group experiments in the lab will be performed to quantify physics of the stored fluids, e.g. H₂ and CO₂. Learning Unit 3: Storage design and evaluation project - Project based learning. Students will form multi-disciplinary groups. Each group will work on a specific energy scenario and conceptualise, design and evaluate a storage system based on their scenario. Groups will report findings on a weekly basis to other groups in form of oral presentations. These presentations will focus on the key messages to different stakeholders. These outputs will be formatively assessed. The final designs and evaluations will be communicated in writing (in form of report) and orally (via a presentation).	
Practical Guide	Learning Unit 1: Subsurface storage technology (1 ECTS) Link to learning objectives of the module: CO-SS: LO 1, LO 5 The various technologies related to subsurface storage and the evaluation of subsurface storage are detailed, including the current state of the art and future research directions. Theoretical and quantitative methods to gain insights into technical and environmental performance are taught. Evaluation methodologies, such as exergy analysis and multi-criteria analysis, are outlined. Topics: - Context of subsurface storage: - o global overview of Climate & Energy status - o from production to storage to mitigation. - Overview of the technologies and advancements - o Energy storage in porous media & cavities (reservoirs and salt caverns): - o Borehole thermal energy storage - o Aquifer thermal energy storage - o Hydrogen storage - o Compressed air energy storage - o Supercritical CO₂ energy storage - o Novel storage topics - o CO₂ storage - o Water storage (managed aquifer recharge (MAR)) - Assessment methods - o Exergy Analyses - o Multi-criteria analysis	

- o Environmental performance
- o Geomechanics

Learning Unit 2: Heat, solute and compressible fluid transport (6 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4

The theoretical aspects of heat, solute and compressible flow in the subsurface are presented, including the flow of heat via conduction, advection and convection, the transport of dissolved solutes and the transport of compressible fluids (e.g. CO₂ and H₂). Reactivity and the effect on transport, leading towards changes in properties or pollution are considered. The coupled behaviour of processes is detailed, e.g. the effect of temperature on physical properties and rate constants.

Topics:

- Solute transport in groundwater
- Heat transport in the surface and groundwater
- Coupled groundwater flow and concentration/heat transport
- o density and viscosity dependency on salinity and temperature
- Reactive transport of contaminants
- o Effect of temperature on rate constants.
- o Mobilization, adsorption and resorption of trace elements due to flow/concentration/temperature
- Compressible fluid flow
- o CO₂
- o H₂
- o Laboratory experiment (geochemistry): measuring CO₂ dissolution, pH change
- Subsurface flow
- o Pore scale to km scale modelling
- Simulation methods
- o Assignments and exercises using various open-source simulators: e.g. MODFLOW, MT3D, SEAWAT, PHT3D and/or PHAST, DARTS, DARSim, MRST
- Applications include seawater intrusion, managed aquifer recharge, aquifer thermal energy storage, CO₂ storage, contaminant plumes.

Learning Unit 3: Storage design and evaluation project (3 ECTS)

Link to learning objectives of the module:

CO-SS: LO 2, LO 3, LO 4, LO 5, LO 6

Students are presented with an energy scenario where they must evaluate the most appropriate storage concept and present a detailed design. The design must include predictions of performance, an evaluation of environmental impact and include analysis of uncertainties. The students must demonstrate the ability to select appropriate theory, modelling concepts and data, perform numerical analysis and critically evaluate the results to form conclusions. They must clearly and precisely communicate their results to their peers.

Topics:

- Scenario evaluation
- Application of theory and tools from other two units
- Critical evaluation of results
- Presentation of project via written technical report and presentation for different stakeholders (focus on non-technical users in the final presentation)

Assessment

Formative assessment (feedback)

Learning Units 1 and 2

During the first two units weekly tasks have to be delivered. Tasks comprise written reports based on laboratory experiments, computer exercises and theoretical knowledge. In addition, class quizzes will be given.

The tasks will be formatively assessed on completeness and quality by involved lecturers. These build towards: (i) the portfolio, and (ii) the activities in Learning Unit 3.

Learning Unit 3

Groups will give weekly presentations on their progress and considering different stakeholders. The presentations will be formatively assessed on completeness and quality by involved lecturers, and peer feedback will be solicited. In addition, formative peer feedback within the group will encourage equal contribution.

Summative assessment (grading)

Portfolio of individual learning activities at the end of Learning Units 1&2

50% of final module grade.

Assessment based on:

Individual cumulative grade. Combination of (approximately) weekly assignments defined during lectures and workshops of Units 1 and 2. The students will have received formative comments after each assignment and can upgrade the assignments for the final grade.

Integrated module assessment at the end of Learning Unit 3

50% of final module grade.

Assessment based on:

Group report

Group presentation at the end of the project

Answering questions after the presentation

A 1-2 page(s) individual report on critical evaluation of all group studies (also to stimulate reading others group reports and actively ask questions during presentations and evaluate based on capacity, efficiency, safety, exergy, public perceptions,).

Both subgrades must be >=5.8.

Expected prior Knowledge

For all MSc students CE, ENV, AES.

Academic Skills

Analysis of complex (multi-cause) problems, information collection and evaluation, multi-disciplinary and integral problem solving, project management, written and oral communication, teamwork.

Literature & Study Materials

Learning Unit 1: Subsurface storage technology

Books:

- o The city of the future by BNA (Branchevereniging Nederlandse Architectenbureaus), publisher Blauwdruk (2019); and Stad x Klimaat Het gebouw als watermachine by BNA and TU Delft (2020) (in Dutch). The study is interactive available at <https://www.bna.nl/kennis/bna-onderzoek/stad-x-klimaat> which can be automatically translated by Chrome Browser to English, for international students. These books comprise and review a number of practical cases concerning integral ecoengineering solutions for socio-environmental problems occurring in the built environment.
 - o Sustainable energy without the hot air by David Mackay: provides an overview of alternative energy options.
 - o Carbon Dioxide Capture and Storage, by Intergovernmental Panel on Climate Change (UNEP): this open-access report provides guidelines for climate-driven actions related to subsurface storage of greenhouse gases, especially CO₂.
- Specific literature in form of scientific papers and book chapters on diverse subjects of subsurface storage (energy and climate) related to techno-economical, experimental, modelling, computational and analytical techniques will be provided via

	BrightSpace. Experimental material for the CO₂/H₂ storage practical, as well as for the sensor workshop will be provided during the practical. Learning Unit 2: Heat, solute and compressible fluid transport Manuals of the open-source software and the associated packages for the modelling software used. Assignments will be developed in the form of Jupyter Notebooks so that they are fully reproducible and documented. Selected papers on the modeling of ATEs, MAR and storage of compressible fluids. Learning Unit 3: Storage design and evaluation project Resources from Learning Units 1 and 2. Additional specific literature in form of scientific papers and book chapters on multi-criteria analysis and scenario analysis will be provided via BrightSpace
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2007	Resilient Deltas	10
Responsible Instructor	Dr.ir. M.M. Rutten	
Module Manager	Dr.ir. M.M. Rutten	
Instructor	Dr. J.E.A. Storms	
Instructor	Dr. R.E.M. Riva	
Instructor	Dr. D.S. van Maren	
Instructor	Dr. A.P. Siebesma	
Instructor	Prof.dr. C.A. Katsman	
Contact Hours / Week x/x/x/x	approximately 7 hours/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This topical cross-over deals with the societal challenge of making deltas resilient to climate change, subsidence, unprecedented economic growth and urbanisation. Future increases in sea level rise, droughts, extreme precipitation and flooding can have adverse effects on the physics and dynamics of deltas, salt intrusions and groundwater, which requires new delta interventions to be developed. The focus is on deltas globally, using the Dutch Delta as a nearby example and laboratory. The module is split in three themes: Theme 1 (3EC) - Physics of the coupled delta system under climate change and human interventions - provides students with the scientific understanding of the interaction of atmospheric, fluvial and marine processes that comprise the delta system, their response to human interventions, and their sensitivity to climate change (LO 1,2,3) Theme 2 (3EC) - Climate change and scenarios gives students hands-on experience on the use of statistical and dynamical downscaling techniques and long-term observational datasets, which they need to analyze to understand the impact of natural and man-made climate change on the delta system. Students will study and discuss different methods to construct climate scenarios, contrasting approaches developed in various countries (for example Switzerland, UK and the Netherlands) and learn how to apply them to different regions/ Focus will be on the specific climate variables: (extreme) precipitation/ droughts and sea level rise. (LO 1, 2, 4, 5) Theme 3 (4EC) - Adaptation pathways for deltas - guides students to develop pathways towards new delta technologies (flood defence and water treatment) based on climate model/data analysis using the Dynamic Adaptive Policy Pathway method among other analytical tools, and guided by SCRUM project management (LO 2, 4, 5, 6, 7, 8) Weekly Syllabus The module takes an inquiry-based approach, in which students start in week 1 by exploring a delta system of their choice and interest. They will create a conceptual map of their current understanding of that delta system including the interlinkages or feedbacks between processes (a concept map). They will do this as part of a multidisciplinary team. In week 1, one of the final deliverables of this module a proposal for an adaptation pathway for a hypothetical client/stakeholder - will be explained and discussed with the students. Student teams will serve as advisors for one delta and stakeholders for another delta. The stakeholders are responsible for climate adaptation of a delta, the advisors are hired to develop pathways. Together they will agree on a product that the advisors will deliver to the stakeholders. This product should at least contain an analysis of the local climate change effects on the main delta systems, and proposals for solutions and adaptation pathways. As examples, they might focus on flooding, salt intrusion of beach nourishments in their chosen delta system under climate change. During Weeks 28, the students will take an iterative approach towards their deliverable, enriching their conceptual understanding of the delta system in each iteration with knowledge and skills from Unit 1 & 2, which they put in practise in weekly computer labs. In the computer labs, students may develop python notebooks in which climate projected data obtained through dynamical downscaling with high resolution models are evaluated against observations. Alternatively, students can develop statistical downscaling techniques by combining global climate model data with local observations. They will also be trained in obtaining useful trends from long observational data sets of the present climate and try to attribute these to global warming. Teamwork in Unit 3 towards the deliverable will be supported with SCRUM project management. That product will be developed in three iterative cycles ("sprints"), ending in week 4, week 8 and week 10 (the final assessment).	
Study Goals	1. Explain the interaction of physical processes that take place in different components of the (coupled) delta system (ocean, rivers, land, atmosphere) 2. Explain the influence of natural variability, climate change and socio-economic development on the coupled delta system 3. Explain mechanisms and adaptation timescales associated with historic human interventions and implications for adapting to climate change 4. Construct and interpret climate change scenarios including the use of dynamical and statistical downscaling techniques based on climate model output. 5. Analyse observed and modelled trends in sea level rise, rainfall and flooding, droughts, sediment transport, subsidence and socio-economic development 6. Acquire new knowledge on green-gray solutions for mitigating impacts of human interventions and climate change, e.g. beach nourishment, river training, building with nature. 7. Develop pathways towards engineering solutions for future delta challenges using model scenarios, including uncertainties 8. Conduct a project in a multidisciplinary team	
Education Method	Unit 1: Physics of the coupled delta system under climate change and human interventions Series of lectures on fundamentals and case studies (3 hrs per week) Mini-lectures (1 hr per week) Quizzes / theory test in week 1/2 Unit 2: Climate change and scenarios Series of lectures (2 hrs per week) Computer labs (2 hrs per week) Quizzes / theory test in week 1/2 Unit 3: Adaptation pathways for deltas Discussions, supervised team work, and in-class paper reviews (2 hrs per week) Workshops in system thinking and stakeholder analysis Computer labs (2 hrs per week) Field trip to the Rhine Meuse Delta to be considered	
Literature and Study	Unit 1: Physics of the coupled delta system under climate change and human interventions	

Materials

Hoitink, A. J., Nittrouer, J. A., Passalacqua, P., Shaw, J. B., Langendoen, E. J., Huismans, Y., & van Maren, D. S. (2020). Resilience of river deltas in the Anthropocene. *Journal of Geophysical Research: Earth Surface*, 125(3), e2019JF005201.

Leuven, J. R., Pierik, H. J., Vegt, M. V. D., Bouma, T. J., & Kleinhans, M. G. (2019). Sea-level-rise-induced threats depend on the size of tide-influenced estuaries worldwide. *Nature climate change*, 9(12), 986-992.

Kirwan, M. L., Temmerman, S., Skeehean, E. E., Guntenspergen, G. R., & Fagherazzi, S. (2016). Overestimation of marsh vulnerability to sea level rise. *Nature Climate Change*, 6(3), 253-260.

Van Maren, D. S., Oost, A. P., Wang, Z. B., & Vos, P. C. (2016). The effect of land reclamations and sediment extraction on the suspended sediment concentration in the Ems Estuary. *Marine Geology*, 376, 147-157.

van Maren, D.S., J.G.W. Beemster, Z.B. Wang, Z.H. Khan, R.A Schrijvershof, A.J.F. Hoitink. Tidal amplification and river capture in response to land reclamation in the Ganges-Brahmaputra delta. *Catena*, 2022

Syvitski, J. P., Kettner, A. J., Overeem, I., Hutton, E. W., Hannon, M. T., Brakenridge, G. R., & Nicholls, R. J. (2009). Sinking deltas due to human activities. *Nature Geoscience*, 2(10), 681-686.

Syvitski, J. P., Vorosmarty, C. J., Kettner, A. J., & Green, P. (2005). Impact of humans on the flux of terrestrial sediment to the global coastal ocean. *science*, 308(5720), 376-380.

Talke, S.A. and Jay, D.A. Changing Tides: The Role of Natural and Anthropogenic Factors. *Annu. Rev.Mar. Sci.* 2020. 12:12151

Wang, Z. B., Elias, E. P., van der Spek, A. J., & Lodder, Q. J. (2018). Sediment budget and morphological development of the Dutch Wadden Sea: impact of accelerated sea-level rise and subsidence until 2100. *Netherlands Journal of Geosciences*, 97(3), 183-214.

Winterwerp JC, Wang ZB, van Braeckel A, van Holland G, Kösters F. Man-induced regime shifts in small estuariesII: a comparison of rivers. *Ocean Dyn.* 63:1293306, 2013.

Unit 2: Climate change and scenarios

van den Hurk, B. et al. Drivers of mean climate change around the Netherlands derived from CMIP5. *Clim. Dynam.* 42, 16831697 (2013).

Swiss Climate Change Scenarios CH2011 (C2SM, MeteoSwiss, ETH, NCCR Climate, and OcCC, 2011).

van den Hurk, B. et al. New climate change scenarios for the Netherlands. *Wat. Sci. Technol.* 56, 2733 (2007).

Jenkins, G. J. et al. UK Climate Projections: Briefing Report (Met Office Hadley Centre, 2009).

Frederikse, T., et al. (2020). The causes of sea-level rise since 1900. *Nature*, 584(7821), 393-397. <https://doi.org/10.1038/s41586-020-2591-3>

Tamisiea, M.E., and J.X. Mitrovica (2011). The moving boundaries of sea level change: Understanding the origins of geographic variability. *Oceanography* 24(2):2439, <https://doi.org/10.5670/oceanog.2011.25>

Slangen, A. B. A., Katsman, C. A., Van de Wal, R. S. W., Vermeersen, L. L. A., & Riva, R. E. M. (2012). Towards regional projections of twenty-first century sea-level change based on IPCC SRES scenarios. *Climate dynamics*, 38(5), 1191-1209. <https://doi.org/10.1007/s00382-011-1057-6>

Slangen, A. B. A., Carson, M., Katsman, C. A., Van de Wal, R. S. W., Köhl, A., Vermeersen, L. L. A., & Stammer, D. (2014). Projecting twenty-first century regional sea-level changes. *Climatic Change*, 124(1), 317-332. <https://link.springer.com/article/10.1007/s10584-014-1080-9>

IPCC 5AR Ch 13.1/13.6 [to be replaced by appropriate 6AR chapter this summer]

Katsman, C. A., Sterl, A., Beersma, J. J., Van den Brink, H. W., Church, J. A., Hazeleger, W., ... & Weisse, R. (2011). Exploring high-end scenarios for local sea level rise to develop flood protection strategies for a low-lying delta the Netherlands as an example. *Climatic change*, 109(3), 617-645. <https://link.springer.com/article/10.1007/s10584-011-0037-5>

Practical Guide

Week/Unit

- Physics of Delta System / Climate
- Climate Change / Scenarios
- Adaptation pathways

1

- Deltas in the Anthropocene. An introduction to the range of systems and their response impacted by climate change and human interventions
- An introduction to climate science: climate sensitivity
- Diverse team formation, SCRUM Introduction.
 - Exploration and discussion of generic adaptation pathways and deliverables

2

- Sediment supply from rivers and from coastal seas; sediment type; changes in sediment supply (role of climate, land use & reservoirs)
- Our future climate - Representative Concentration Pathway (RCP) scenarios, climate modelling and downscaling techniques
- Selection of deltas and roles (advisor / stakeholder)
 - Systems thinking workshop & stakeholder Analysis workshop

3

- Tidal amplification response to human interventions (closure dams, channel deepening) b) Sea level rise: causes, processes + observations (altimetry / tide gauges Dutch coast)
- Discussion product deliverables and planning Sprint 1
 - Presentations Conceptual Model and Stakeholder Analysis

4

- Changes in salt intrusion and fresh water availability in response to channel deepening, reservoir construction, and channel reorganization
- Uncertainty in climate model and scenarios, climate change versus natural variability, and attribution of climate change
- Resilience debates

5

- Decline of water quality and ecological value in response to climate change and human interventions
- Atmospheric influences on deltas: Storms, surges, precipitation and river discharge
- Review and Retrospective Sprint 1 and Planning Sprint 2

6

- Subsidence and compaction of deltas
- Atmospheric influences on deltas: Projections of Extremes : Flooding and Droughts
- Discussion on evaluation of strategies

7

- Coastal erosion by natural processes, human interventions, and climate change
- Sea level rise: global and regional projections. High-end scenarios with skewed ice sheet contributions, links to extreme events / coastal safety.

8

- Morphologic response to sea level rise
- Compound effects of sea level rise, storm surges + precipitations / river discharge c) Review and Retrospective Sprint 2

	10 a) Adaptation to sea level rise c) Presentations of adaptation pathways to client/stakeholder, supported by data products consisting of e.g. downscaled climate model output for different scenarios, observational data analyses.
Assessment	Formative assessment (feedback): The formative assessment for Theme 1 and 2 is arranged in the form of quizzes and assignments with feedback. For Theme 3 the conceptual model and stakeholder analysis are formative assessed in week 3. In week 5 and 8 the SCRUM artefacts Review and Retrospective are used to give students feedback on content and process. Summative assessment: The summative assessment consists of: Written exam on content Unit 1 - 3 (50%) Proposal for an adaptation pathway / solution for a delta including analysis of the local climate change effects on the delta system, delivered in either a written/oral/(other form) presentation (50%)
Expected prior Knowledge	Students that enter the module are expected to have in-depth understanding of one of the subsystems of deltas (e.g. coastal area, rivers, the subsurface, the atmosphere etc.). All Master tracks within the programs Civil Engineering, Environmental Engineering and Applied Earth Sciences can participate. Students from other programs need to contact the module manager.
Academic Skills	The professional and personal skills are included in the learning objectives. Specifically students work on their metacognitive skills (learning objective 6) and their collaboration, management and communication skills (learning objective 8)
Literature & Study Materials	(Continuation) Unit 3: Adaptation pathways for deltas For pathway systems thinking and rapid modelling the following material (or comparable resources) will be used together will dedicated material on Brightspace: Haasnoot, M., Brown, S., Scussolini, P., Jimenez, J. A., Vafeidis, A. T., & Nicholls, R. J. (2019). Generic adaptation pathways for coastal archetypes under uncertain sea-level rise. <i>Environmental Research Communications</i>, 1(7), 071006. Wright, D., & Meadows, D. H. (2008). <i>Thinking in systems</i>. Earthscan. https://ocw.tudelft.nl/courses/ngi-2/subjects/8-bonus-module-problem-demarcation-actor-analysis-skills-2/, (accessed 2-4-2021) Goldreich, P., Mahajan, S., & Phinney, S. (1999). Order-of-magnitude physics: Understanding the world with dimensional analysis, educated guesswork, and white lies. University of Cambridge. Relevant part available via https://web.stanford.edu/class/ee353/SanjoyMahajanIntro-01-1.pdf (accessed 2-4-2021) Van Berchum, E. C., van Ledden, M., Timmermans, J. S., Kwakkel, J. H., & Jonkman, S. N. (2020). Rapid flood risk screening model for compound flood events in Beira, Mozambique. <i>Natural Hazards and Earth System Sciences</i>, 20(10), 2633-2646. Jafino, B. A., Kwakkel, J. H., & Taebi, B. (2021). Enabling assessment of distributive justice through models for climate change planning: A review of recent advances and a research agenda. <i>Wiley Interdisciplinary Reviews: Climate Change</i>, e721. For teamwork and project process the following material (or comparable resources) will be used together will dedicated material on Brightspace: · Schwartz, S. H. (2014). National culture as value orientations: Consequences of value differences and cultural distance. In <i>Handbook of the Economics of Art and Culture</i> (Vol. 2, pp. 547-586). Elsevier. · https://www.mbtionline.com/ · https://scrumguides.org/ (All accessed 2-4-2021)
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CEGM2008	Monitoring of Structural Health and Geohazards	10
Responsible Instructor	Dr. E. Lourens	
Module Manager	Dr. E. Lourens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr.ir. M.J.M.M. Steenbergen	
Instructor	Dr.ir. D.S. Draganov	
Instructor	Ir. F.J. van Leijen	
Instructor	Dr. Y. Yang	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	Dr. A. Barnhoorn	
Instructor	Dr. K. Anupam	
Instructor	C. Shen	
Instructor	Dr. H. Wang	
Contact Hours / Week x/x/x/x	12-14 hours per week (3-4 lectures, 3-4 workshops of 2*45 min)	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The cross-over will be of the methodological type, oriented around processes and techniques that are shared across multiple domains. Specifically, students from CE and AES will apply predominantly new methods related to the monitoring of civil structures, rock- and soil masses to problems originating in their own as well as other fields of study. The module consists of two units: 1. Theory & Techniques 6 EC 2. Case Studies: Advanced monitoring of structural health and geohazards 4 EC Unit 1: Theory & Techniques [LO1, LO2, LO3, LO4, LO5] The first unit provides the students with knowledge on the theory and techniques required to answer the questions what, when, how, why and where to monitor?. Parts of this unit will form natural extensions to what is taught in the MUDE, including brief recapitulations. Weekly syllabus Week 1: Introduction Degradation & failure mechanisms Relation loading &#8596; materials &#8596; mechanisms &#8596; time- & spatial scale Sensors and data acquisition Advanced signal processing (beyond MUDE) Week 2: Inversion Inversion and quality assessment (including recap MUDE) Prediction (Best Linear Unbiased Prediction, Kriging, Gaussian process Regression) Inversion for sub-surface (layered) structures Week 3: System identification and model updating System identification: EMA/OMA Deterministic (FE) model updating Bayesian inference in the context of structural health and geohazards monitoring (Bayesian inversion) Week 4: Machine learning perspectives Feature engineering Introduction to image segmentation, deep learning and convolutional neural networks (including recap MUDE) Anomaly detection: clustering techniques extended from MUDE, hypothesis testing (time series analysis) Week 5: Exam Written exam (30%) Unit 2: Case studies: Advanced monitoring of structural health and geohazards [LO3, LO4, LO5, LO6] Case studies covering a wide range of application domains will be presented, where a division is made between CE applications (bridge, railway, pavement, and wind turbine monitoring, stability of dykes and embankments), AES applications (soil mass movements and landslides, land subsidence, failure & fault reactivation due to gas extraction, short-term cyclic storage of fluids or resource mining), and applications relating to both fields of study (imaging of structures, rocks and the near surface, monitoring of acoustic emission and seismic activities). Each case study will be developed & supervised by the relevant, contributing research group. Students will work in groups of 2 on these case studies, 1 from CE and 1 from AES. Each case study comprises 2 ECTS. Insofar possible, each group will have representatives from both MSc programmes. The last 4 weeks of the module are reserved solely for the case studies. Weekly syllabus Weeks 6-10: Introduction of the case studies If required: theory specific to the case study & monitoring approach used State-of-the-art review per case study	
Study Goals	[LO1] Identify mechanisms of failure and functionality loss for applications in the geo- and civil engineering domain. [LO2] Discern techniques to inspect and measure the health or integrity of civil structures, rocks, and soil masses. [LO3] Design monitoring systems to detect patterns and anomalies in civil structures, rocks and soil masses and assess the value of information obtained with these systems. [LO4] Post-process monitoring data to extract structural health information. [LO5] Apply numerical and statistical inverse analysis to identify unknowns and update numerical and/or analytical models based on the monitoring data.	

Education Method	[LO6] Integrate monitoring data and physical modelling to support decision-making. Unit 1: The content in unit 1 will be delivered through theoretical teaching sessions, flipped classrooms, live demonstrations in the classroom/laboratory, and computer workshops. Unit 2: For the case studies, the students will be split into sets of groups, where the aim is to have multiple groups working on the same case study, and directly supervised by the relevant research group. For each case study, there will first be theoretical teaching sessions to introduce the case study and point the students towards literature/articles relevant to the case study. This will be followed by regular, supervised practical sessions (per case study), where the groups can work on their assignment and ask questions.
Assessment	Formative assessment (feedback): Unit 1: Small group assignments related to the computer workshops. Unit 2: Short presentations to expert panels, partly peer-graded. During the presentations, societal aspects related to assessment decision-making should also be addressed. Summative assessment: 1. Written exam in week 5 (after Unit 1), before students start the case studies (30%) 2. Assignment portfolio based on group projects (70%) - Case study 1 (35%) - Case study 2 (35%)
Special Information	Case study example 1 [CE]: Structural health monitoring of bridges Students receive a data set containing accelerations and strains measured on a bridge. They are asked to 1) perform operational modal analysis to identify the dynamic properties of the structure, 2) update a (given) simplified model of the bridge using the identified properties, 3) design features that can be used to monitor the global vibrational response of the bridge, 4) implement an algorithm for detecting anomalies in the response, and 5) suggest improvements to the monitoring system (locations/types of sensors, data acquisition). The last task is included for all case studies where it is not practically feasible for students to collect the data themselves. Case study example 2 [AES]: Monitoring of changes in subsurface structures Students receive seismic reflection data, a subsurface fault model, and seismicity data from a subsurface reservoir (e.g. from Groningen) and are asked to 1) perform an assessment of the seismicity locations, 2) perform fault slip processes modelling, 3) assess what the role of subsurface properties has on the outcomes of these analyses, including the role of possible changes in production during continuous production, and 4) devise an improved monitoring plan to improve the quantification of subsurface seismicity properties.
Expected prior Knowledge	N/A
Connection to other modules	Overlap with other parts of the MSc programmes: CEGM1000: Data acquisition, signal processing, Bayesian inference & time series analysis is taught in the MUDE as well. For all three these topics, the theory will be briefly refreshed, and contextualized for monitoring applications. CIEM5110 Structural Mechanics and Dynamics: Also contains content on data acquisition, signal processing, and experimental modal analysis. AESB403B Induced Seismicity: Possible overlap with case studies (Groningen data may be used in this B module as well. The focus is very different, however. AESB404B Raw Material Exploration and Data Acquisition Techniques: Contains one lecture on the seismic exploration method with active and passive sources.
Academic Skills	Project management, teamwork, written and oral communication, information collection and evaluation.
Literature & Study Materials	The complete list for unit 1 still has to be compiled. For the case studies, relevant readings for the specific case study will be collected.
Judgement	N/A
Permitted Materials during Exam	N/A
Collegerama	No

CIEM0210	Engineering Management Systems	10
Responsible Instructor	Ir. R. Binnekamp	
Responsible Instructor	Prof.dr.ir. A.R.M. Wolfert	
Module Manager	Prof.dr.ir. A.R.M. Wolfert	
Instructor	Dr.ir. G.A. van Nederveen	
Instructor	Ir. J.P.G. Ramler	
Contact Hours / Week x/x/x/x	8hrs lectures	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The goal of this crossover (CO) for CE (and/or ODE) students is to learn about/ develop different management systems to support the civil engineering management processes over the entire system service life cycle. Engineering management systems are a group of information concepts, models and methods for managerial decision to support for effective and efficient engineering management: i.e., project-, planning-, organization-, supply chain- and/or operations & maintenance management systems. In this module the student will construct three engineering management systems : 1) OMS -organization 2) PMS planning 3) IMS information within a design-, build-, maintain- and operate context of a service provider or contractor. This module allows CE students to broaden their systems engineering horizon within a management context. Rather than only focusing on what to engineer, this crossover focuses especially on how to manage & organize the engineering asset management processes within the context of an international operating service provider/contractor. Such an organisation delivers a certain type of service quality (e.g. availability, safety, capacity etc.) by a set of enabling engineering assets (e.g. road, tunnel, bridge etc.). Participation in this crossover will lead to CE students that are better prepared for real-life asset management practices resulting in open design &#950;-professionals rather than in specialized &#946;-engineers. This crossover is based on the principles of Open Design Learning concept, which is a reflective, creative and engaging learning approach that opens human development and unlocks new knowledge and design solutions in a constructive manner. A CO student will receive 10 ECTS, which consists of a development of three engineering management systems and conspection on performance improvement over current practice: i.e., 1) OMS (organization management system) 2) PMS (planning management system) and 3) IMS (information management systems) that are all interrelated. For the conspection part, students together with a reflective professional from the SPoI determines a need to support management decisions. The student creates three different management system that supports this need using the state-of-the-art engineering asset management concepts (i.e., a open glass-box computer model using e.g., 3C planning, Preferendus/ IMAP etc.).	
Study Goals	It is an excellent opportunity for civil engineering students to apply their systems engineering design knowledge and skills within an engineering management context, particularly within that of a service provider/contractor of interest (SPoI). This module builds /complements the Engineering Systems Design unit of the Programme Base. This module extends the multi-objective engineering design optimisation approach into a problem-solving context for engineering management decision making. On completion of this module, the student will be able to: LO1: Develop engineering management systems and conspect on performance improvement over current practice; LO2: Deal with open design learning and intercultural aspects within a (international) service provider organization; LO3: Apply research and/or development (R&D) methods to prepare for a MSc graduation thesis project proposal; LO4: Demonstrate and develop the personal outcome of the aforementioned learning goals, by internalizing the openings/learning achievements in an original Open Design Learning (ODL) response.	
Education Method	Students will be introduced to the following concepts in the first two weeks of this module: 1) Your SPoI and the QoS management systems within its socio-eco purpose context 2) Strategic Asset Management Planning 3) Information management systems These concepts are linked to the management systems to be created and are available online. - Student has to do the following: 1) organize a committed reflective practitioner within an (international) SPoI (the domain buddy), and 2) write a concise and motivated project development plan: i.e., the group submits a motivation for SPoI and relevant potential for engineering management systems improvement linked to the needs of the SPoI. This will be commended ultimately at the end of week 2. - Practical work meetings with both an academic supervisor and with a reflective practitioner of the service provider (SPoI)	
Assessment	Masterclasses (midterm presentations of and feedback on work-in-progress) ODL response (an individual/group response) to be delivered at the end of week 10. The pass grade should be a minimum of 5.8 for each of the ODL response.	
Expected prior Knowledge	Successfully completed the Programme base module Interdisciplinary design and mechanics and MUDE.	
Academic Skills	Systems thinking, Multi-objective optimization modelling, Reflective practitioning, Creating an ODL proposal and response, Multi-disciplinary skills	
Literature & Study Materials	In the Syllabus of this course one can find an extensive list of source material and further readings for this module (as also partly attached as an Appendix or see Brightspace). The most relevant literature is: 1) Binnekamp, Wolfert (2023) Open Design Systems, DUT Open Textbook series 2) Kammouh, O., Nogal, M., Binnekamp, R., & Wolfert, A. R. M. R. (2021). Multi-system intervention optimization for interdependent infrastructure (3C concept). Automation in Construction, 127, 1-11. [103698]. https://doi.org/10.1016/j.autcon.2021.103698 3) Wolfert, R., Nederveen, S. V., & Binnekamp, R. (2022). Fit for Purpose Building Information Modelling and Systems Integration (BIMSI) for Better Construction Projects Management. The Journal of Modern Project Management, 10(1), 174187. https://doi.org/10.19255/JMPM029011	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Master Thesis preparation

CIEM0400	Master Thesis preparation	5
Responsible Instructor	Dr.ir. A. Gavriilidou	
Responsible Instructor	Dr.ir. W. Daamen	
Responsible Instructor	Dr. J.M. Vleugel	
Responsible Instructor	Dr.ir. S.C. Calvert	
Responsible Instructor	Dr.ir. Y. Yuan	
Module Manager	Dr.ir. D.C. Duives	
Contact Hours / Week x/x/x/x	2hrs lectures & 2hrs writing lab	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4 5	
Course Language	English	
Course Contents	The module develops a comprehensive work plan for a MSc thesis project over the course of 7 weeks. During each week, a specific section of the work plan is highlighted in the lecture. Besides lectures, each week, a writing lab is hosted, which features personal formative feedback on their current plans from peers of their own track. Special attention is paid to the evaluation of research risks (i.e. Human participation in research, data protection and physical safety risks), professional & research ethics, Open science and FAIR data.	
Study Goals	Upon the successful completion of the module, the student is able to: LO1 formulate research gap, scope and question(s) LO2 develop a conceptual framework LO3 develop a literature review LO4 develop a research methodology and data collection plan LO5 evaluate the risks of a research plan (HREC, DMP, Lab Servant). LO6 formulate a (personal) standpoint on professional and research ethics w.r.t. a research plan	
Education Method	Lectures, writing labs, writing reports	
Prerequisites	60 EC	
Assessment	Formative assessment of workplan by the Responsible supervisor in week 4. Formative assessment of the ethics portfolio by peers and instructors in week 1 and week 5. Formative assessment of the safety risks assessment, HREC and DMP by peers and Responsible supervisor in week 8. Obligatory physical attendance during the ethics debate in week 6. Summative assessment (Pass/Fail) of the comprehensive MSc thesis project work plan incl. appendices (ethics statement, safety assessment, HREC assessment, DMP) at the end of week 8.	
Enrolment / Application	Enroll for CIEM0400 on brightspace. Accordingly, also enroll for the group of a specific quarter.	
Special Information	1. This course is given on campus, there is no opportunity to follow this course digitally while overseas 2. This course is given for the first time in Q2 of the university year 2023-2024. After that, it runs every quarter.	
Expected prior Knowledge	1. Student must have completed at least 60 EC. 2. Student must have finalized and handed in Assignment 1-4 of their ethics portfolio (Ethics learning line). 3. Student must have already: a. identified a MSc thesis project topic b. arranged a responsible supervisor for this MSc thesis project topic (c. for in-company graduations - arranged a contact person at the company you want to perform your MSc thesis project.)	
Academic Skills	We suggest students to follow this module in the quarter before they start their MSc thesis project. Critical and analytical thinking Information collection and synthesis Interpretation Judgemental skills Reasoning and arguing Moral judgement Scientific writing and presentation	
Literature & Study Materials	N/A	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

Year	2023/2024
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Master Thesis

CIEM0500	Master Thesis	30
Responsible Instructor	Ir. T.J. Zitman	
Contact Hours / Week x/x/x/x	N/A	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	The master thesis project is all about some topical (track-related) civil engineering quest with innovative potential, the answer to which requires integration of acquired knowledge and skills as well as creativity and critical thinking in combination with societal and scientific awareness. It builds on preparatory work conducted in the Thesis Preparation (CIEM0400). The project may be related to ongoing work, either in house or at a private company, a research institute or a (non-)governmental organisation, provided prior approval of the concerned track co-ordinator. Conditions may apply, for instance regarding confidentiality, intellectual property and so on. It is the responsibility of the student to ensure that such conditions are agreed upon with all parties involved prior to the start of the project and in compliance with TU regulations (see ...). Commonly, but not by definition, this is dealt with already along with the formulation of a project proposal in the Thesis Preparation (CIEM0400). The primary deliverables are - a written report (the thesis) and - an oral presentation and defence (this is a public event, to be scheduled on campus, unless exceptional circumstances dictate otherwise)	
Study Goals	1. Formulate a research or a design question and conduct a project to answer that question [ILO 7]. 2. Integrate knowledge and skills acquired during studies [ILO 4,10]. 3. Analyse, model and explain the behaviour of the civil engineering system addressed in the project to arrive at a sound and responsible answer to the research or design question [ILO 1,2]. 4. Reflect in a scientifically sound way on obtained results, against the background of societal and scientific challenges as well as ethics [ILO 3,5,8,10]. 5. Compose a structured and consistent technical report [ILO 9]. 6. Orally present and defend project execution and results [ILO 9].	
Education Method	The Master Thesis project is conducted independently on the basis of the results of the Thesis Preparation (CIEM0400), comprising primarily a proposal, a planning and a literature survey. Guidance is provided by a daily supervisor (or supervisory team), who is also a member of the graduation committee (see assessment below). It is the responsibility of the student to maintain contact with the supervisor during the project.	
Assessment	Formative: Next to feedback given by the daily supervisor or the supervisory team, the graduation committee (see assessment below) provides formative feedback in at least three meetings (that need to be included in the project plan): 1. a kick-off meeting at the start of the project; 2. a mid-term (when the project has progressed to the value of about 15 EC); 3. a green-light meeting, prior to which the student needs to hand in a final draft report and during which the student must give a draft final presentation (i.e. a draft of the presentation the student will give as part of the thesis defence). If the graduation committee concludes that the work done thus far by the student and including the response of the student to questions the committee has posed during the meeting, is sufficient to graduate (with a grade of at least 6.0), green-light is given. This means that the student may proceed finalising the report and prepare the final presentation and defence, using final feedback from the committee. Besides, a date can be set for the final presentation and the student can initiate the administrative procedure required for graduation (see BrightSpace). The final report needs to be provided to all graduation committee members at least five work days prior to the date set for the presentation and the defence. If the committee concludes that green-light cannot be given yet, it explains the student which improvements are at least required and a new meeting is scheduled. Such a postponement of green-light may adversely affect the final grade. (also see relevant TER/RGBE articles). Summative: The Master Thesis is assessed by a graduation committee as described in the TER. The grade is based upon the final thesis and the final presentation as a weighted average of five partial grades concerning: 1. Scientific approach (25%) - theoretical profundity - scientific argumentation - quality of experimental or design work - creativity 2. Quality of result/product (25%) - scientific reflection and judgement - utilisation of result or product - extension or generation of methods - amount of work 3. Behavioural Competencies (20%) - initiative and/or own contribution - independency - communicative skills - time planning 4. Quality of written presentation (15%) - structure and consistency - acknowledgement, references, quotes	

	- use of graphics 5. Quality of oral presentation and defence (15%) - clarity and structure of the presentation - quality of presentation material - answering questions To graduate, this weighted average must be at least 6.0 and only one of the partial grades may be less than 5.8, but not less than 5.0.
Expected prior Knowledge	Students may start the Master Thesis project only after they have completed the Thesis Preparation (CIEM0400).
Academic Skills	Critical thinking, problem-based learning and solving, information collection and synthesis, communication, scientific writing, presentation, time management, application of theory to new situations, drafting research questions, reflection on ethics and learning skills.
Literature & Study Materials	N/A
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/faculties/ceg-student-portal/education/education-information/educational-rules-and-regulations/
Permitted Materials during Exam	N/A
Collegerama	No

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